NON-SEMISIMPLE χ-CROSSED QUANTUM GROUPS

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Abstract. We introduce the notion of χ-crossed ribbon quasi-Hopf algebras for a crossed module $\chi\colon G \to H$, and show that their representations form χ-crossed ribbon categories introduced in [GRW26], with grading by G and action by H. For the projection $\chi\colon \mathbb{C}/2\mathbb{Z} \to \mathbb{C}/\mathbb{Z}$ with trivial action, we construct an explicit non-semisimple example based on quantum $\mathfrak{sl}_2$ at even roots of unity. The example also gives rise to a non-semisimple χ-graded monoidal 2-category with objects indexed by $\mathbb{C}/\mathbb{Z}$. Our construction arises from twisted-local modules over an algebra in the category of weight modules for the unrolled quantum $\mathfrak{sl}_2$. Each graded component is a finite abelian category, and the resulting category admits a non-trivial $\mathbb{C}/\mathbb{Z}$-action. Restricting to degrees 0 and 1 recovers the factorisable small quantum group of [CGR17]. We further give explicit formulae for a family of graded coends as free modules. This example provides algebraic input for our forthcoming paper [GGPRW], which recovers the non-semisimple 3-manifold invariant of [CGP12].

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1. Introduction

1.1. **Background and main results.** Since the foundational developments of quantum groups in the 1980s, Hopf algebras and their representations have played a central role in quantum algebra and quantum topology [Ka95, Ma95, Ku90, RT91, TV92, Tu94]. Over the years, a number of variants of Hopf algebras have been introduced for different purposes. Motivated by homotopical refinements of quantum invariants for 3-manifolds, one is naturally led to *crossed ribbon Hopf group-coalgebras* [Tu00, Vi00b, Tu10]. In recent years, there has been sustained interest in studying non-semisimple Hopf algebras, driven in part by their rich applications to quantum-topological invariants and topological field theories (TFTs) [CGP12, BCGP14, DGPR17, DGPR18].

In this paper, we generalise crossed ribbon Hopf group-coalgebras to *χ-crossed ribbon quasi-Hopf algebras* for a crossed module χ, whose representation categories are *χ-crossed ribbon categories* as introduced by the authors in [GRW26]. As an explicit example, we construct a family of χ-crossed ribbon quasi-Hopf algebras based on quantum $\mathfrak{sl}_2$ at even roots of unity, where $\chi\colon \mathbb{C}/2\mathbb{Z} \to \mathbb{C}/\mathbb{Z}$ is the projection with trivial action. When restricted to degrees 0 and 1, our example reduces to the factorisable small quantum group of [CGR17].

Our construction arises via reconstruction from a χ-crossed ribbon category obtained by twisted-local modules as introduced in [GRW26, Sec. 5]. Our example carries a nontrivial $\mathbb{C}/\mathbb{Z}$-action, and each of its graded components is finite as an abelian category. This guarantees the existence of a family of graded coends, for which we give explicit formulae as free modules. Recall that for a crossed module $\chi\colon G \to H$, any χ-crossed ribbon category canonically carries both G-crossed and $\mathrm{Im}(\chi)$-crossed ribbon structures [GRW26, Prop. 4.9]. In particular, our example yields a $\mathbb{C}/\mathbb{Z}$-crossed braided extension for the representation category of the ribbon quasi-Hopf algebra described in [CGR17, Sec. 4.2]. Following the idea of [Cu16, JPR20] that every G-crossed braided category gives rise to a monoidal 2-category, we thus also obtain a non-semisimple monoidal 2-category with a continuum of objects.

A *crossed module* is a group homomorphism $\chi\colon G \to H$ equipped with an action of H on G satisfying some conditions. We write $h \triangleright g$ for the action of $h \in H$ on $g \in G$. We introduce in Section 2 the notion of a *χ-crossed ribbon quasi-Hopf algebra*, which is a family of unital associative algebras $U = \{U_g\}_{g\in G}$ over a ground field $\mathbb{K}$ together with

- a family of unital algebra maps called *comultiplications* $\Delta = \{\Delta_{a,b}\colon U_{ab} \to U_a \otimes U_b\}_{a,b\in G}$;
- a unital algebra map called *counit* $\varepsilon = \{\varepsilon\colon U_1 \to \mathbb{K}\}$;
- a family of invertible elements called *coassociators* $\{\Phi_{a,b,c} \in U_a \otimes U_b \otimes U_c\}_{a,b,c\in G}$;
- a family of linear isomorphisms called *antipodes* $S = \{S_a\colon U_a \to U_{a^{-1}}\}_{a\in G}$;
- a family of *evaluation elements* $\{\boldsymbol{\alpha}_a \in U_a\}_{a\in G}$;
- a family of *coevaluation elements* $\{\boldsymbol{\beta}_a \in U_a\}_{a\in G}$;
- a family of algebra isomorphisms $\{\gamma_{h,a}\colon U_a \to U_{h\triangleright a}\}_{(h,a)\in H\times G}$;
- a family of invertible elements $\{\Omega^h_{a,b} \in U_a \otimes U_b\}_{(h,a,b)\in H\times G\times G}$;

satisfying compatibility conditions (H1)–(H6) and (C1)–(C7), and further equipped with

- a family of *χ-crossed R-matrices*

$$R = \{R_{a,b} = \sum_i (R'_i)_a \otimes (R''_i)_b \in U_a \otimes U_b\}_{(a,b)\in G^{\times 2}} , \tag{1.1}$$

- a family of *χ-crossed ribbon elements*

$$v = \{v_a \in U_a\}_{a\in G} , \tag{1.2}$$

satisfying compatibility conditions (R1)–(R4) and (RE1)–(RE4).

Remark 1.1. If $G = H = \{*\}$, we recover the notion of ribbon quasi-Hopf algebras [Dr91]. If $\chi = \mathrm{id} : G \to G$ with conjugation action, and if the invertible elements $\{\Omega^g_{a,b}\}$ are all trivial,

we recover the notion of ribbon quasi-Hopf G-coalgebras [Gu16, Def. 3.4]; if in addition, the coassociators, as well as the evaluation and coevaluation elements are all trivial, then we recover the notion of crossed ribbon Hopf G-coalgebras [Tu10, Vi00b].

The category of finite-dimensional U-modules is χ-crossed ribbon:

- the comultiplications, counit, and coassociators induce the G-graded monoidal structure;
- the antipodes, along with the evaluation and coevaluation elements induce rigidity;
- the algebra isomorphisms $\{\gamma_{(h,a)}\}$ produce a strict H-action on $\mathrm{Rep}_{\mathrm{f.d.}}U$, and the invertible elements $\{\Omega^h_{a,b}\}$ encode the monoidality of each h-action;
- the χ-crossed R-matrices and χ-crossed ribbon elements give rise to the χ-crossed braided and ribbon structures, respectively.

Conversely, we present a series of reconstruction results showing how the Hopf-algebraic data above can be recovered from the corresponding categorical structures. First, we show that every degreewise finite $\mathbb{K}$-linear abelian G-graded monoidal category admitting a normalised *quasi-fibre functor* is equivalent, as a G-graded monoidal category, to the category of finite-dimensional representations of a family of finite-dimensional algebras $U = \{U_g\}_{g\in G}$, equipped with comultiplications, a counit, and coassociators (Proposition 2.4). We further show that compatible categorical H-actions, χ-crossed braidings, and χ-crossed ribbon structures give rise to the corresponding algebraic data γ, Ω, R, and v, respectively (Propositions 2.9, 2.11, and 2.13).

Remark 1.2. For conceptual completeness, we note that the reconstruction results above miss one piece: the antipodes and the evaluation and coevaluation elements can be reconstructed from a suitable rigid G-graded monoidal category. We do not pursue this in the present paper. For a finite-dimensional quasi-bialgebra H, it was shown in [Sch02] that H admits a quasi-Hopf structure if and only if its category of finite-dimensional representations is rigid.

In Section 3 we study graded coends in a χ-crossed ribbon category, which form an *H-crossed Hopf G-algebra* object in the neutral component (Theorem 3.6).

Following the reconstruction strategy outlined above, we first present a χ-crossed ribbon category constructed via *twisted-local modules* (Section 4), and then reconstruct the corresponding χ-crossed ribbon quasi-Hopf algebra from it.

Fix a primitive $2p$-th root of unity $q = e^{\frac{\pi i}{p}}$ in $\mathbb{C}$, where $p \geq 2$ is an integer. Let $\overline{U}_q^H\mathfrak{sl}_2$ denote the *unrolled quantum* $\mathfrak{sl}_2$, and let $\mathcal{C} \coloneqq \mathrm{Rep}_{w.t.}\overline{U}_q^H\mathfrak{sl}_2$ be the balanced category of its weight modules (Section 4.2). Note that objects in $\mathcal{C}$ may be infinite-dimensional, as we allow arbitrary direct sums. Hence $\mathcal{C}$ is not rigid. For each $k \in \mathbb{Z}$, there is a one-dimensional $\overline{U}_q^H\mathfrak{sl}_2$-module on which H acts by the scalar $2pk$ and both E and F act by 0. We denote this module by $\mathbb{C}_k$. Following [CGR17], we consider the following commutative *simple-current algebra* object in $\mathcal{C}$:

$$A = \bigoplus_{k\in\mathbb{Z}} \mathbb{C}_k \ . \tag{1.3}$$

Denote by ${}_A\mathcal{C}$ the category of left A-modules in $\mathcal{C}$, and by ${}_A\mathcal{C}^{f.g.}$ its full subcategory of finitely generated A-modules. Consider the crossed module $\pi\colon \mathbb{C}/2\mathbb{Z} \to \mathbb{C}/\mathbb{Z}$, where π is the projection map, and $\mathbb{C}/\mathbb{Z}$ acts on $\mathbb{C}/2\mathbb{Z}$ trivially.

Theorem 1.3 (Thm. 4.10)**.** *${}_A\mathcal{C}^{f.g.}$ can be endowed with the structure of a π-crossed ribbon category.*

In Proposition 5.13, we further give formulas for graded coends in ${}_A\mathcal{C}^{\mathrm{f.g.}}$ as free A-modules.

For every $x \in \mathbb{C}$, and for all positive integers $p-1 \geq k \geq l \geq 0$, recall the notation:

$$q^x := e^{\frac{x\pi i}{p}} , \quad \{x\} := q^x - q^{-x} , \quad [k] := \frac{\{k\}}{\{1\}} , \quad [k]! := \prod_{j=1}^{k} [j] , \quad \begin{bmatrix} k \\ l \end{bmatrix} := \frac{[k]!}{[l]![k-l]!} . \tag{1.4}$$

In Section 5, we carry out the reconstruction and obtain the following π-crossed ribbon quasi-Hopf algebra over $\mathbb{C}$, arising from quantum $\mathfrak{sl}_2$.

For every $\alpha \in \mathbb{C}/2\mathbb{Z}$, we define a unital algebra $\overline{\mathfrak{U}}_\alpha$ generated by

$$\{E, F, K^{\pm 1}\} \tag{1.5}$$

with the relations

$$\begin{aligned} E^p = F^p = K^p - q^{p\overline{\alpha}} = 0 , \qquad KEK^{-1} = q^2 E , \\ KFK^{-1} = q^{-2} F , \qquad [E, F] = \frac{K - K^{-1}}{q - q^{-1}} , \end{aligned} \tag{1.6}$$

where $\overline{\alpha}$ is an arbitrary representative for α in $\mathbb{C}$.

We define a counit $\varepsilon \colon \overline{\mathfrak{U}}_0 \to \mathbb{C}$ by

$$\varepsilon(K^{\pm 1}) = 1 , \qquad \varepsilon(E) = \varepsilon(F) = 0 . \tag{1.7}$$

Let $\pi_1 \colon \mathbb{C}/2p\mathbb{Z} \to \mathbb{C}/2\mathbb{Z}$ be the projection. For each $a \in \mathbb{C}/2p\mathbb{Z}$ with $\pi_1(a) = \alpha$, there is an idempotent $e_a \in \overline{\mathfrak{U}}_\alpha$ defined via

$$e_a = \frac{1}{p} \sum_{l=0}^{p-1} q^{-al} K^l . \tag{1.8}$$

Let $\pi_3 \colon \mathbb{C} \to \mathbb{C}/2\mathbb{Z}$ be the projection homomorphism. For any set-theoretic section

$$\sigma \colon \mathbb{C}/2\mathbb{Z} \to \mathbb{C} \tag{1.9}$$

of π_3 with $\sigma(0) = 0$, we define a family of coproducts Δ^σ on $\overline{\mathfrak{U}} = \{\overline{\mathfrak{U}}_\alpha\}_{\alpha \in \mathbb{C}/2\mathbb{Z}}$:

$$\begin{aligned} \Delta^\sigma_{\alpha,\beta}(E) &= q^{-\sigma(\alpha)} \mathbb{1}_\alpha \otimes E + E \otimes K , \\ \Delta^\sigma_{\alpha,\beta}(F) &= q^{\sigma(\alpha)} K^{-1} \otimes F + F \otimes \mathbb{1}_\beta , \\ \Delta^\sigma_{\alpha,\beta}(K^{\pm 1}) &= K^{\pm 1} \otimes K^{\pm 1} , \end{aligned} \tag{1.10}$$

with the coassociators given by

$$\Phi^\sigma_{\alpha,\beta,\gamma} = \sum_{\substack{\pi_1(a)=\alpha \\ \pi_1(b)=\beta \\ \pi_1(c)=\gamma}} q^{c\iota_{\alpha,\beta}} e_a \otimes e_b \otimes e_c = \mathbb{1}_\alpha \otimes \mathbb{1}_\beta \otimes K^{\iota_{\alpha,\beta}} . \tag{1.11}$$

where $\iota_{\alpha,\beta} := \frac{\sigma(\alpha)+\sigma(\beta)-\sigma(\alpha+\beta)}{2} \in \mathbb{Z}$. We also define a family of antipodes S^σ by

$$S^\sigma_\alpha(E) = -q^{-\sigma(\alpha)} EK^{-1} , \quad S^\sigma_\alpha(F) = -q^{\sigma(\alpha)} KF , \quad S^\sigma_\alpha(K^{\pm 1}) = K^{\mp 1} , \tag{1.12}$$

with the evaluation and coevaluation elements being

$$\boldsymbol{\alpha}^\sigma_\alpha = \mathbb{1}_\alpha , \qquad \boldsymbol{\beta}^\sigma_\alpha = K^{\frac{\sigma(\alpha)+\sigma(-\alpha)}{2}} . \tag{1.13}$$

We denote the resulting quasi-Hopf $\mathbb{C}/2\mathbb{Z}$-coalgebra by $\overline{\mathfrak{U}}^\sigma$ (Section 5.2).

Remark 1.4. $\overline{\mathfrak{U}}^\sigma_0$ is the small quantum $\mathfrak{sl}_2$ in the sense of [Ka95, Definition 6.5.6]. Denoting by $t = -\sigma(1)$, $\overline{\mathfrak{U}}^\sigma_0 \oplus \overline{\mathfrak{U}}^\sigma_1$ coincides with the quasi-Hopf-algebra $\overline{U}^{(\Phi)}_q \mathfrak{sl}_2$ constructed in [CGR17, Sec. 4.2].

Remark 1.5. It is known that the category of finite-dimensional weight modules over the semi-restricted quantum group $\overline{U}_q\mathfrak{sl}_2$ is not braided [CGP14, Sec. 2.1]. In [Ha18, Sec. 5.1], this category was realised as the representation category of a Hopf $\mathbb{C}/2\mathbb{Z}$-coalgebra, whose component algebras are precisely the $\overline{\mathfrak{U}}_\alpha$ defined above, equipped with the coproducts and antipodes inherited from the standard Hopf algebra structure on $\overline{U}_q\mathfrak{sl}_2$. Thus, this Hopf $\mathbb{C}/2\mathbb{Z}$-coalgebra is not ribbon, and it is not known whether it admits a $\mathbb{C}/2\mathbb{Z}$-crossed ribbon structure. In our construction, we modify the Hopf structure so that the resulting quasi-Hopf $\mathbb{C}/2\mathbb{Z}$-coalgebra admits a π-crossed ribbon structure, as described below.

Choosing a set-theoretic section $\sigma'\colon \mathbb{C}/2p\mathbb{Z} \to \mathbb{C}$ for the projection $\pi_2\colon \mathbb{C} \to \mathbb{C}/2p\mathbb{Z}$, we define a $\mathbb{C}/\mathbb{Z}$-action on $\{\overline{\mathfrak{U}}_\alpha\}_{\alpha\in\mathbb{C}/2\mathbb{Z}}$ with the following data (Section 5.3):

- for any $h \in \mathbb{C}/\mathbb{Z}$, $\alpha \in \mathbb{C}/2\mathbb{Z}$, the algebra isomorphism $\gamma_{h,\alpha}\colon \overline{\mathfrak{U}}_\alpha \to \overline{\mathfrak{U}}_\alpha$ is given by
$$E \mapsto \sum_{\substack{a\in\mathbb{C}/2p\mathbb{Z}\\ \pi_1(a)=\alpha}} q^{2p\kappa_+(a)h}Ee_a\ , \qquad F \mapsto \sum_{\substack{a\in\mathbb{C}/2p\mathbb{Z}\\ \pi_1(a)=\alpha}} q^{2p\kappa_-(a)h}Fe_a\ , \qquad K^{\pm1} \mapsto K^{\pm1}\ , \tag{1.14}$$
where $\kappa_\pm(a) = \frac{\sigma'(a)\pm2-\sigma'(a\pm2)}{2p}$;
- for any $h \in \mathbb{C}/\mathbb{Z}$, $\alpha, \beta \in \mathbb{C}/2\mathbb{Z}$,
$$\Omega^h_{\alpha,\beta} = \sum_{\substack{\pi_1(a)=\alpha\\ \pi_1(b)=\beta}} q^{-2p\kappa(a,b)h}e_a \otimes e_b\ , \tag{1.15}$$
where $\kappa(a,b) = \frac{\sigma'(a)+\sigma'(b)-\sigma'(a+b)}{2p}$.

We further introduce a family of π-crossed R-matrices and π-crossed ribbon elements (Section 5.4). For any $\alpha, \beta \in \mathbb{C}/2\mathbb{Z}$, we define
$$R_{\alpha,\beta} = \sum_{n=0}^{p-1} \sum_{\substack{a,b\in\mathbb{C}/2p\mathbb{Z}\\ \pi_1(a)=\alpha\\ \pi_1(b)=\beta}} \frac{(q-q^{-1})^n}{[n]!} q^{\frac{n(n-1)+\zeta_a^n(\zeta_b^0-2n)}{2}} q^{\frac{\sigma(\alpha)\sigma(\beta)}{2}} q^{(\zeta_b^{-n}+n)\sigma(\alpha)} (E^n e_a \otimes F^n e_b)\ , \tag{1.16}$$
$$v_\alpha = \sum_{n=0}^{p-1} \sum_{\substack{a\in\mathbb{C}/2p\mathbb{Z}\\ \pi_1(a)=\alpha}} \frac{(q^{-1}-q)^n}{[n]!} q^{\frac{-n(n+3)}{2}} q^{-\frac{(\sigma'(a))^2}{2}} q^{a(p-n-1)} F^n E^n e_a\ , \tag{1.17}$$
where for any $c \in \mathbb{C}/2p\mathbb{Z}$ and any integer $1-p \le k \le p-1$, $\zeta_c^k := \sigma'(c+2k) - \sigma(\pi_1(c))$ is an even integer depending on the choices of σ and σ'.

Remark 1.6. $R_{0,0}$ coincides with the R-matrix for the small quantum $\mathfrak{sl}_2$ as in [Ka95, Theorem 9.7.1].

We denote the resulting π-crossed ribbon quasi-Hopf algebra by ${}^{\sigma'}\overline{\mathfrak{U}}^{\sigma}$, and its category of finite-dimensional weight modules by $\mathrm{Rep}_{\mathrm{f.d.}}{}^{\sigma'}\overline{\mathfrak{U}}^{\sigma}$.

Theorem 1.7 (Thm. 5.10)**.** $\mathrm{Rep}_{\mathrm{f.d.}}{}^{\sigma'}\overline{\mathfrak{U}}^{\sigma} \cong {}_A\mathcal{C}^{f.g.}$ *as π-crossed ribbon categories.*

As outlined in [GRW26, Sec. 1.3] and [Ce26, Ex. 4.19], the π-crossed ribbon category
$$\mathcal{D} = \bigoplus_{\alpha\in\mathbb{C}/2\mathbb{Z}} \mathcal{D}_\alpha := \bigoplus_{\alpha\in\mathbb{C}/2\mathbb{Z}} \mathrm{Rep}_{\mathrm{f.d.}}{}^{\sigma'}\overline{\mathfrak{U}}^{\sigma}_\alpha \tag{1.18}$$
gives rise to a *π-graded monoidal 2-category* D in the sense of [Ce26, Sec. 4.13] (after strictification). Explicitly, following [Cu16, Sec. 6.1], the structure of D is described as follows.

The objects of D are the elements of $\mathbb{C}/\mathbb{Z}$, and its Hom-categories are

$$\mathsf{D}(h,k) = \mathcal{D}_\alpha \oplus \mathcal{D}_{\alpha+1} \ , \qquad \pi(\alpha) = k - h \ . \tag{1.19}$$

For composable 1-morphisms $U\colon h \to k$ and $V\colon k \to l$, their composition is given by

$$V \circ U \coloneqq V \otimes U \ , \tag{1.20}$$

with the identity 1-morphism at each h given by the tensor unit $\mathbf{1}$ of $\mathcal{D}$. The associator and unitors for composition are inherited from the monoidal structure of $\mathcal{D}$.

The monoidal product $\boxtimes$ on D has unit object $0 \in \mathbb{C}/\mathbb{Z}$ and is defined on objects by

$$h \boxtimes k \coloneqq h + k \ . \tag{1.21}$$

For 1-morphisms $U\colon h \to h'$ and $V\colon k \to k'$, we have

$$U \boxtimes V \coloneqq U \otimes {}^h V\colon h + k \longrightarrow h' + k' \ , \tag{1.22}$$

where ${}^h(-)$ denotes the $\mathbb{C}/\mathbb{Z}$-action on $\mathcal{D}$. The monoidal product of 2-morphisms is defined analogously.

For composable 1-morphisms $U\colon h \to h'$, $U'\colon h' \to h''$, $V\colon k \to k'$, and $V'\colon k' \to k''$, the *tensorator*

$$(U' \boxtimes V') \circ (U \boxtimes V) \longrightarrow (U' \circ U) \boxtimes (V' \circ V) \tag{1.23}$$

is given by

$$U' \otimes {}^{h'}V' \otimes U \otimes {}^h V \xrightarrow{\mathrm{id}\otimes C^{-1}_{U,{}^h V'}\otimes \mathrm{id}} U' \otimes U \otimes {}^h V' \otimes {}^h V \xrightarrow{\mathrm{id}\otimes h_{V',V}} (U' \otimes U) \otimes {}^h(V' \otimes V) \ , \tag{1.24}$$

where C denotes the π-crossed braiding in $\mathcal{D}$, $h_{-,-}$ denotes the monoidal structure of the action ${}^h(-)$, and we supress associators of $\mathcal{D}$ for readability. For $h, k \in \mathbb{C}/\mathbb{Z}$, the identity comparison is given by

$$\mathrm{id}_{h+k} = \mathbb{1} \xrightarrow{\sim} \mathbb{1} \otimes \mathbb{1} \xrightarrow{\mathrm{id}\otimes h_{\mathbb{1}}} \mathbb{1} \otimes {}^h\mathbb{1} = \mathrm{id}_h \boxtimes \mathrm{id}_k \ , \tag{1.25}$$

where $h_{\mathbb{1}}$ denotes the unit structure isomorphism of ${}^h(-)$.

The *monoidal associator* of D has identity components at objects. For $U\colon h \to h'$, $V\colon k \to k'$, and $W\colon l \to l'$, the component

$$U \boxtimes (V \boxtimes W) \longrightarrow (U \boxtimes V) \boxtimes W \tag{1.26}$$

is given by

$$U \otimes {}^h(V \otimes {}^k W) \xrightarrow{\mathrm{id}\otimes (h_{V,{}^k W})^{-1}} U \otimes \left({}^h V \otimes {}^{h+k} W\right) \xrightarrow{\sim} (U \otimes {}^h V) \otimes {}^{h+k} W \ . \tag{1.27}$$

The left and right monoidal unitors also have identity components at objects, and on 1-morphisms they are given by

$$\mathbb{1} \boxtimes U = \mathbb{1} \otimes U \xrightarrow{\sim} U \ , \qquad U \boxtimes \mathbb{1} = U \otimes {}^h\mathbb{1} \xrightarrow{\mathrm{id}\otimes h_{\mathbb{1}}^{-1}} U \otimes \mathbf{1} \xrightarrow{\sim} U \ . \tag{1.28}$$

Finally, the *pentagonator* and the three *unit modifications* are given by the canonical coherence isomorphisms induced by the unitors of $\mathcal{D}$ and the unit structure isomorphisms $h_{\mathbb{1}}$.

In the present paper, we do not verify that these data satisfy all the coherence axioms of a fully weak monoidal 2-category.

Remark 1.8. The 2-category D is not locally semisimple, since its endomorphism category $\mathsf{D}(0,0) = \mathcal{D}_0 \oplus \mathcal{D}_1$ is the non-semisimple representation category of the quasi-Hopf-algebra $\overline{U}_q^{(\Phi)}\mathfrak{sl}_2$ constructed in [CGR17, Sec. 4.2].

1.2. **Potential applications to quantum topology.** A major advance in quantum topology is the Reshetikhin–Turaev construction of link invariants from ribbon categories [RT91, Tu94]. In [Tu00, Tu10], this framework was extended to G-links and G-tangles—tangles equipped with a flat connection in a principal G-bundle over their complements. The algebraic input for this refined theory is a G-crossed ribbon category.

In parallel, an alternative approach to defining invariants of tangles with flat G-connections was proposed and developed in [KR02a, KR10, GP13, BGPR18] via the notion of *holonomy braidings*. While a concrete non-semisimple example based on quantum $\mathfrak{sl}_2$ was considered within their framework, the holonomy braiding there could only be determined up to multiplication by a root of unity in every degree.

In the present work, we construct a family of non-semisimple G-crossed ribbon categories for $G = \mathbb{C}/2\mathbb{Z}$, which we propose as algebraic input for the aforementioned first approach to G-tangle invariants. We emphasise that explicit construction of non-semisimple G-crossed ribbon categories remains open and challenging in the case of non-abelian groups. In particular, the case $G = \mathrm{PSL}(2,\mathbb{C})$ is of special interest due to its relevance to the volume conjecture [Mc21].

Another important motivation for studying link invariants is their role in constructing 3-manifold invariants and 3-dimensional TFTs. In recent years, there has been growing interest in non-semisimple extensions of the Reshetikhin–Turaev framework, motivated by the search for finer topological invariants. An important example is the theory developed in [CGP12, BCGP14], which can be viewed as a non-semisimple and homotopical refinement of the Reshetikhin–Turaev theory. In particular, it can be understood as a homotopy quantum field theory (HQFT), for which a general construction from semisimple G-crossed ribbon categories was developed in [TV14]. However, the connection is not immediate, as the algebraic data underlying [CGP12, BCGP14] do not take such form. In [GGPRW], we bridge this gap by developing a non-semisimple refinement of the HQFT construction in [TV14] and showing that, when applied to the example constructed in the present paper, it recovers the invariants of [CGP12]. We expect this framework to provide a unified perspective on several existing constructions in quantum topology:

- Restricting the HQFT of [GGPRW] to bordisms equipped with trivial maps to the Eilenberg–MacLane space $K(\mathbb{C}/2\mathbb{Z},1)$ should recover the non-semisimple ordinary 3d TFT of [DGGPR19].
- The construction of [GGPRW] should extend to a theory assigning categories to circles, equivalent to the extended theory of [DeR17].
- Restricting this extended theory to the $(\emptyset, S^1)$ part, namely the category of surfaces with one boundary component and 3-dimensional bordisms between them, should recover the theory of [DMW24].

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AI disclosure. The initial draft of this paper, including all computations, was completed in May 2025 without the use of AI. During the final stages of this work, we used AI to verify the computations, which helped us identify a computational error regarding the ribbon elements (1.17) and a mistake in axiom (RE4) for the definition of a χ-crossed ribbon quasi-Hopf algebra.

2. Crossed-module crossed quasi-Hopf algebras

Fix a crossed module χ. In this section, after a brief review of *χ-crossed ribbon categories* as studied in [GRW26], we introduce the notion of a *χ-crossed ribbon quasi-Hopf algebra* (Definitions 2.3, 2.8, 2.10, 2.12). By construction, the finite-dimensional representation category of such a notion forms a χ-crossed ribbon category. This framework extends the theory of *crossed Hopf G-coalgebras* from [Vi00b, Tu10] in the following three respects:

- the group acting on the algebras differs from the grading group for comultiplications;
- the action of a group element is a coalgebra morphism only up to conjugation;
- the comultiplications are coassociative only up to conjugation by a counital 3-cocycle.

We also provide reconstruction results for a χ-crossed ribbon quasi-Hopf algebra from a χ-crossed ribbon cateogry (Proposition 2.4, 2.9, 2.11, 2.13).

2.1. χ-Crossed ribbon categories.

A *crossed module* is a group homomorphism $\chi\colon G \to H$ equipped with an action of H on G by group automorphisms, denoted $h \triangleright g$. Writing χ_g for $\chi(g)$, this data must satisfy the compatibility conditions

$$h\chi_g h^{-1} = \chi_{h\triangleright g}\ , \qquad gg'g^{-1} = \chi_g \triangleright g'\ , \tag{2.1}$$

for all $g, g' \in G$ and $h \in H$.

Fix a ground field $\Bbbk$. Let $\mathcal{C}$ be a $\Bbbk$-linear additive monoidal category. $\mathcal{C}$ is called *G-graded monoidal* if there exists a family of full subcategories $\{\mathcal{C}_g\}_{g\in G}$ in $\mathcal{C}$ such that

- $\mathcal{C} = \bigoplus_{g\in G} \mathcal{C}_g$;
- the tensor unit $\mathbb{1}$ lies in $\mathcal{C}_{1_G}$;
- for any $a, b \in G$ and any $X \in \mathcal{C}_a$, $Y \in \mathcal{C}_b$, one has $X \otimes Y \in \mathcal{C}_{ab}$.

Throughout this paper, we assume that $\mathcal{C}$ is *faithfully* G-graded, i.e. $\mathcal{C}_g \neq 0$ for all $g \in G$. Denote by $\mathcal{C}_{\mathrm{hom}}$ the full subcategory in $\mathcal{C}$ consisting of homogeneous objects. The degree of $X \in \mathcal{C}_{\mathrm{hom}}$ is denoted by $|X| \in G$. If $\mathcal{C}$ is moreover rigid, then for every $a \in G$ and every object $X \in \mathcal{C}_a$, its left and right duals satisfy $X^*, {}^*X \in \mathcal{C}_{a^{-1}}$.

Let $\underline{H}$ denote the discrete monoidal category whose objects are group elements of H. Let $\mathrm{Aut}_\otimes(\mathcal{C})$ denote the groupoid whose objects are monoidal $\Bbbk$-linear autoequivalences of $\mathcal{C}$ and whose morphisms are monoidal natural isomorphisms. $\mathrm{Aut}_\otimes(\mathcal{C})$ is strictly monoidal with the tensor product on objects given by composition of functors, and on morphisms given by horizontal composition of natural transformations.

An *H-crossed action* on a G-graded tensor category $\mathcal{C}$ is a monoidal functor

$$\varphi\colon \underline{H} \longrightarrow \mathrm{Aut}_\otimes(\mathcal{C}) \tag{2.2}$$

such that for any $(h, g) \in H \times G$, the functor $\varphi(h)$ maps $\mathcal{C}_g$ into $\mathcal{C}_{h\triangleright g}$.

Remark 2.1. Since φ is strictly monoidal in all examples considered in this paper, we assume this to be the case throughout unless otherwise stated.

For each $h \in H$, we assume that the monoidal endofunctor ${}^h(-) \coloneqq \varphi(h)$ is strictly unital, and we write

$$h_{X,Y}\colon {}^hX \otimes {}^hY \xrightarrow{\cong} {}^h(X \otimes Y) \tag{2.3}$$

for its structure isomorphisms. Writing λ, ρ, α for the left unitor, right unitor, and associator in $\mathcal{C}$, the following diagrams commute for all $h, h' \in H$ and $X, Y, Z \in \mathcal{C}$:

$$\begin{array}{ccc} {}^{h'h}X \otimes {}^{h'h}Y & \xrightarrow{{}^{h'h}_{X,Y}} & {}^{h'h}(X \otimes Y) \\ {\scriptstyle {}^{h'}_{{}^{h}X,{}^{h}Y}}\downarrow & \nearrow_{{}^{h'}_{X,Y}} & \\ {}^{h'}({}^{h}X \otimes {}^{h}Y) & & \end{array} \tag{2.4a}$$

$$\begin{array}{ccccc} {}^{h}X \otimes \mathbb{1} = {}^{h}X \otimes {}^{h}\mathbb{1} & \xrightarrow{{}^{h}_{X,\mathbb{1}}} & {}^{h}(X \otimes \mathbb{1}) & & \\ {\scriptstyle \rho_{{}^{h}X}}\downarrow & \swarrow_{{}^{h}\rho_X} & & & \\ {}^{h}X & \nwarrow_{{}^{h}\lambda_M} & & & \\ {\scriptstyle \lambda_{{}^{h}X}}\uparrow & & & & \\ \mathbb{1} \otimes {}^{h}X = {}^{h}\mathbb{1} \otimes {}^{h}X & \xrightarrow[{}^{h}_{\mathbb{1},X}]{} & {}^{h}(\mathbb{1} \otimes X) & & \end{array} \tag{2.4b}$$

$$\begin{array}{ccc} {}^{h}X \otimes ({}^{h}Y \otimes {}^{h}Z) & \xrightarrow{\alpha_{{}^{h}X,{}^{h}Y,{}^{h}Z}} & ({}^{h}X \otimes {}^{h}Y) \otimes {}^{h}Z \\ {\scriptstyle \mathrm{id}\otimes {}^{h}_{Y,Z}}\downarrow & & {}^{h}_{X,Y}\otimes\mathrm{id}\downarrow \\ {}^{h}X \otimes {}^{h}(Y \otimes Z) & & {}^{h}(X \otimes Y) \otimes {}^{h}Z \\ {\scriptstyle {}^{h}_{X,Y\otimes Z}}\downarrow & & {}^{h}_{X\otimes Y,Z}\downarrow \\ {}^{h}(X \otimes (Y \otimes Z)) & \xrightarrow{{}^{h}\alpha_{X,Y,Z}} & {}^{h}((X \otimes Y) \otimes Z) \end{array} \tag{2.4c}$$

Remark 2.2. If $\mathcal{C}$ is rigid, then for any $h \in H$, $X \in \mathcal{C}$, the objects $({}^{h}X)^*$ and ${}^{h}(X^*)$ are both dual to ${}^{h}X$. Hence there is a unique isomorphism $\nu_{h,X}$ between them, compatible with the (co)evaluation maps. Explicitly, $\nu_{h,X}$ is the composition

$$\begin{aligned} &{}^{h}(X^*) \xrightarrow{\sim} {}^{h}(X^*) \otimes \mathbb{1} \xrightarrow{\mathrm{id}\otimes\mathrm{coev}_{{}^{h}X}} {}^{h}(X^*) \otimes \big({}^{h}X \otimes ({}^{h}X)^*\big) \xrightarrow{\sim} \big({}^{h}(X^*) \otimes {}^{h}X\big) \otimes ({}^{h}X)^* \\ &\xrightarrow{{}^{h}_{X^*,X}\otimes\mathrm{id}} {}^{h}(X^* \otimes X) \otimes ({}^{h}X)^* \xrightarrow{{}^{h}\mathrm{ev}_X\otimes\mathrm{id}} {}^{h}\mathbb{1} \otimes ({}^{h}X)^* \xrightarrow{\sim} ({}^{h}X)^* \ . \end{aligned} \tag{2.5}$$

There is a corresponding isomorphism for right duals, but we will not need it.

Let $\mathcal{C}$ be an H-crossed G-graded monoidal category. A *χ-crossed braiding* on $\mathcal{C}$ is a family of isomorphisms

$$\{C_{X,Y} \colon X \otimes Y \to {}^{\chi_{|X|}}Y \otimes X\}_{X,Y \in \mathcal{C}_{\mathrm{hom}}} \tag{2.6}$$

such that

- for any morphisms $f \colon X \to X'$, $g \colon Y \to Y'$ between homogeneous objects, the following diagram commutes:

$$\begin{array}{ccc} X \otimes Y & \xrightarrow{C_{X,Y}} & {}^{\chi_{|X|}}Y \otimes X \\ {\scriptstyle f\otimes g}\downarrow & & {\scriptstyle {}^{\chi_{|X|}}g\otimes f}\downarrow \\ X' \otimes Y' & \xrightarrow{C_{X',Y'}} & {}^{\chi_{|X|}}Y' \otimes X' \end{array} \tag{B1}$$

- for any $h \in H$, $a, b, c \in G$, and any homogeneous objects $X, Y, Z \in \mathcal{C}$ of degrees a, b, c, respectively, the following diagrams commute:

$$\begin{array}{ccccc}
(X\otimes Y)\otimes Z & \xrightarrow{C_{X\otimes Y,Z}} & {}^{\chi_{ab}}Z\otimes(X\otimes Y) & \xrightarrow{\alpha_{{}^{\chi_{ab}}Z,X,Y}} & ({}^{\chi_{ab}}Z\otimes X)\otimes Y \\
\uparrow{\scriptstyle \alpha_{X,Y,Z}} & & & & \\
X\otimes(Y\otimes Z) & & & & \uparrow{\scriptstyle C_{X,{}^{\chi_b}Z}\otimes\mathrm{id}} \\
\downarrow{\scriptstyle \mathrm{id}\otimes C_{Y,Z}} & & & & \\
X\otimes({}^{\chi_b}Z\otimes Y) & & \xrightarrow{\qquad\alpha_{X,{}^{\chi_b}Z,Y}\qquad} & & (X\otimes {}^{\chi_b}Z)\otimes Y
\end{array} \tag{B2}$$

$$\begin{array}{ccccc}
X\otimes(Y\otimes Z) & \xrightarrow{C_{X,Y\otimes Z}} & {}^{\chi_a}(Y\otimes Z)\otimes X & \xleftarrow{{}^{\chi_a}Y,Z\otimes\mathrm{id}} & ({}^{\chi_a}Y\otimes {}^{\chi_a}Z)\otimes X \\
\downarrow{\scriptstyle \alpha_{X,Y,Z}} & & & & {\scriptstyle \alpha_{{}^{\chi_a}Y,{}^{\chi_a}Z,X}}\uparrow \\
(X\otimes Y)\otimes Z & & & & {}^{\chi_a}Y\otimes({}^{\chi_a}Z\otimes X) \\
\downarrow{\scriptstyle C_{X,Y}\otimes\mathrm{id}} & & & & {\scriptstyle \mathrm{id}\otimes C_{X,Z}}\uparrow \\
({}^{\chi_a}Y\otimes X)\otimes Z & & \xrightarrow{\qquad\alpha^{-1}_{{}^{\chi_a}Y,X,Z}\qquad} & & {}^{\chi_a}Y\otimes(X\otimes Z)
\end{array} \tag{B3}$$

$$\begin{array}{ccc}
{}^{h}X\otimes {}^{h}Y & \xrightarrow{C_{{}^{h}X,{}^{h}Y}} & {}^{\chi_{h\triangleright a}h}Y\otimes {}^{h}X \\
\downarrow{\scriptstyle h_{X,Y}} & & {\scriptstyle h\chi_a Y,X}\downarrow \\
{}^{h}(X\otimes Y) & \xrightarrow{{}^{h}C_{X,Y}} & {}^{h}({}^{\chi_a}Y\otimes X)
\end{array} \tag{B4}$$

A χ-crossed braided category $\mathcal{C}$ is called *χ-crossed balanced* if there exists a family of χ-crossed balancing isomorphisms

$$\{\vartheta_X\colon X\to {}^{\chi_X}X\}_{X\in\mathcal{C}_{\mathrm{hom}}} \tag{2.7}$$

satisfying the following conditions:

- for any morphisms $f\colon X\to Y$ between homogeneous objects, we have
$$\vartheta_Y\circ f=({}^{\chi_{|X|}}f)\circ\vartheta_X\ ; \tag{BL1}$$
- for any $a,b\in G$ and any $X,Y\in\mathcal{C}$ of degrees a and b, the following diagram commutes:
$$\begin{array}{ccccc}
X\otimes Y & & \xrightarrow{\qquad\vartheta_{X\otimes Y}\qquad} & & {}^{\chi_{ab}}(X\otimes Y) \\
\downarrow{\scriptstyle \vartheta_X\otimes\vartheta_Y} & & & & {\scriptstyle (\chi_{ab})_{X,Y}}\uparrow \\
{}^{\chi_a}X\otimes {}^{\chi_b}Y & \xrightarrow{C_{{}^{\chi_a}X,{}^{\chi_b}Y}} & {}^{\chi_{ab}}Y\otimes {}^{\chi_a}X & \xrightarrow{C_{{}^{\chi_{ab}}Y,{}^{\chi_a}X}} & {}^{\chi_{ab}}X\otimes {}^{\chi_{ab}}Y
\end{array} \tag{BL2}$$
- for any $h\in H$ and any $X\in\mathcal{C}_{\mathrm{hom}}$, we have
$${}^{h}(\vartheta_X)=\vartheta_{{}^{h}X}\ . \tag{BL3}$$

If $\mathcal{C}$ is moreover rigid, and for any $X\in\mathcal{C}_{\mathrm{hom}}$, we have

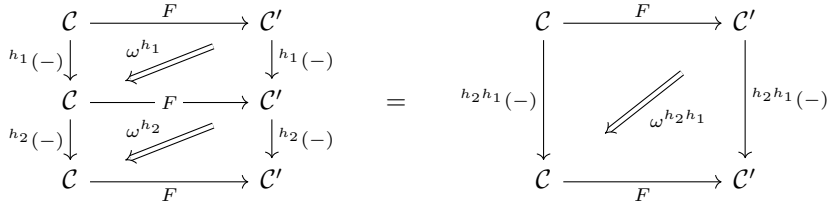
$$(\vartheta_X)^*\circ\nu_{\chi_{|X|},X}=\vartheta_{{}^{\chi_{|X|}}(X^*)}\ , \tag{RB1}$$

where ν is as in (2.5), then $\mathcal{C}$ is called *χ-crossed ribbon*.

Let $\mathcal{C}$ and $\mathcal{C}'$ be two χ-crossed braided categories. A *χ-crossed braided functor* $\mathcal{C}\to\mathcal{C}'$ is a pair (F,ω), where

- $F=(F,F_{-,-},F_{\mathbb{1}})$ is a degree preserving $\mathbb{K}$-linear monoidal functor:
$$F\colon\mathcal{C}\to\mathcal{C}'\ ,\qquad F_{-,-}\colon F(-)\otimes F(-)\Rightarrow F(-\otimes-)\ ,\qquad F_{\mathbb{1}}\colon\mathbb{1}'\to F(\mathbb{1})\ ; \tag{2.8}$$
- $\omega=(\omega^h)_{h\in H}$ is a family of monoidal natural isomorphisms
$$\omega^h\colon {}^{h}F(-)\Longrightarrow F({}^{h}(-))\ , \tag{2.9}$$

such that for all $h_1, h_2 \in H$, $a, b \in G$, and $X \in \mathcal{C}_a$, $Y \in \mathcal{C}_b$, we have the following identity and commutative diagram:



(2.10a)

$$\begin{array}{ccccc} F(X \otimes_{\mathcal{C}} Y) & \xrightarrow{F^{-1}_{X,Y}} & F(X) \otimes_{\mathcal{C}'} F(Y) & \xrightarrow{C'_{F(X),F(Y)}} & {}^{\chi_a}F(Y) \otimes_{\mathcal{C}'} F(X) \\ \downarrow{\scriptstyle F(C_{X,Y})} & & & & \downarrow{\scriptstyle \omega_Y^{\chi_a} \otimes_{\mathcal{C}'} \mathrm{id}} \\ F({}^{\chi_a}Y \otimes_{\mathcal{C}} X) & & \xrightarrow{F^{-1}_{{}^{\chi_a}Y,X}} & & F({}^{\chi_a}Y) \otimes_{\mathcal{C}'} F(X) \end{array} \tag{2.10b}$$

where C and C' are the χ-crossed braidings in $\mathcal{C}$ and $\mathcal{C}'$, respectively.

If $\mathcal{C}$ and $\mathcal{C}'$ are further χ-crossed balanced/ribbon, then a χ-crossed braided functor (F, ω) is said to be *χ-crossed balanced/ribbon* if, for all $a \in G$ and $X \in \mathcal{C}_a$, the following diagram commutes,

$$\begin{array}{ccc} F(X) & \xrightarrow{F\vartheta_X} & F({}^{\chi_a}X) \\ {\scriptstyle \vartheta'_{F(X)}}\downarrow & \nearrow_{\omega_X^{\chi_a}} & \\ {}^{\chi_a}F(X) & & \end{array} \tag{2.11}$$

where ϑ and ϑ' are the χ-crossed balancing isomorphisms in $\mathcal{C}$ and $\mathcal{C}'$, respectively. If F is an equivalence of categories, then $\mathcal{C}$ and $\mathcal{C}'$ are said to be equivalent as χ-crossed balanced/ribbon categories.

2.2. **Quasi-Hopf group-coalgebras.** Fix a ground field $\mathbb{K}$ and a group G. We start with a family of associative unital algebras $U = \{U_g\}_{g \in G}$. Consider the category of (not necessarily finite-dimensional) left U-modules

$$\operatorname{Rep} U \coloneqq \bigoplus_{g \in G} \operatorname{Rep} U_g \ . \tag{2.12}$$

$\operatorname{Rep} U$ is a G-graded $\mathbb{K}$-linear additive category. To endow it with a G-graded monoidal structure, one needs a family of G-graded comultiplications on U. This motivates the following definition.

Definition 2.3. A *quasi-bialgebra with G-graded comultiplications* is a family of associative unital algebras $\{U_g\}_{g \in G}$ with

- a family of unital algebra maps called *comultiplications* $\Delta = \{\Delta_{a,b} \colon U_{ab} \to U_a \otimes U_b\}_{a,b \in G}$;
- a unital algebra map called *counit* $\varepsilon = \{\varepsilon \colon U_1 \to \mathbb{C}\}$;
- a family of invertible elements called *coassociators* $\{\Phi_{a,b,c} \in U_a \otimes U_b \otimes U_c\}_{a,b,c,\in G}$,

such that for any $a, b, c, d \in G$, $x \in U_{abc}$,

$$(\varepsilon \otimes \mathrm{id}_{U_a}) \circ \Delta_{1,a} = \mathrm{id}_{U_a} = (\mathrm{id}_{U_a} \otimes \varepsilon) \circ \Delta_{a,1} \ ; \tag{H1}$$

$$(\Delta_{a,b} \otimes \mathrm{id}_{U_c}) \circ \Delta_{ab,c}(x) \cdot \Phi_{a,b,c} = \Phi_{a,b,c} \cdot (\mathrm{id}_{U_a} \otimes \Delta_{b,c}) \circ \Delta_{a,bc}(x) \ ; \tag{H2}$$

$$\begin{gathered} \Big(\Phi_{a,b,c} \otimes \mathbb{1}_d\Big)\Big((\mathrm{id}_{U_a} \otimes \Delta_{b,c} \otimes \mathrm{id}_{U_d})\Phi_{a,bc,d}\Big)\Big(\mathbb{1}_a \otimes \Phi_{b,c,d}\Big) = \\ \Big((\Delta_{a,b} \otimes \mathrm{id}_{U_c} \otimes \mathrm{id}_{U_d})\Phi_{ab,c,d}\Big)\Big((\mathrm{id}_{U_a} \otimes \mathrm{id}_{U_b} \otimes \Delta_{c,d})\Phi_{a,b,cd}\Big) \ ; \end{gathered} \tag{H3}$$

$$(\mathrm{id}_{U_a} \otimes \varepsilon \otimes \mathrm{id}_{U_b})\Phi_{a,1,b} = (\mathrm{id}_{U_a} \otimes \mathrm{id}_{U_b} \otimes \varepsilon)\Phi_{a,b,1} = (\varepsilon \otimes \mathrm{id}_{U_a} \otimes \mathrm{id}_{U_b})\Phi_{1,a,b} = \mathbb{1}_a \otimes \mathbb{1}_b \ , \tag{H4}$$

where $\mathbb{1}_g$ is the unit of U_g.

For a quasi-bialgebra with G-graded comultiplications $U = \{U_g\}_{g\in G}$, there is a G-graded monoidal structure on $\operatorname{Rep} U$. The tensor unit is $\mathbb{K}$ with U_1 acting on it via the counit ε. If $M \in \operatorname{Rep} U_a$ and $N \in \operatorname{Rep} U_b$, then $M \otimes N \in \operatorname{Rep} U_{ab}$ on which $x \in U_{ab}$ acts through $\Delta_{a,b}(x)$. This monoidal structure need not be strict. The left- and right unitors are trivial by (H1), while the associator may be nontrivial by (H2). More precisely, writing $\Phi_{a,b,c} = \sum_i x_i \otimes y_i \otimes z_i \in U_a \otimes U_b \otimes U_c$, for any $M \in \operatorname{Rep} U_a$, $N \in \operatorname{Rep} U_b$, $K \in \operatorname{Rep} U_c$, the associator $\alpha_{M,N,K}$ is given by

$$\begin{aligned} \alpha_{M,N,K} \colon M \otimes (N \otimes K) &\to (M \otimes N) \otimes K \ , \\ m \otimes (n \otimes k) &\mapsto \sum_i (x_i \cdot m \otimes y_i \cdot n) \otimes z_i \cdot k \ . \end{aligned} \tag{2.14}$$

The coherence axioms follow from (H3) and (H4).

Conversely, a finite-dimensional quasi-bialgebra with G-graded comultiplications can be reconstructed from a suitable G-graded monoidal category via a *quasi-fibre functor*. Let $\mathcal{C}$ be a degreewise finite $\mathbb{K}$-linear abelian G-graded monoidal category, and let vec_G denote the category of finite-dimensional G-graded $\mathbb{K}$-vector spaces with untwisted associator. Adapting the setup of [EGNO15, Def. 5.1.1 & 5.12.1] to our context, a normalised quasi-fibre functor

$$(F, \omega) \colon \mathcal{C} \longrightarrow \mathrm{vec}_G \tag{2.15}$$

consists of a G-graded, exact, faithful, $\mathbb{K}$-linear functor F satisfying $F(\mathbb{1}_{\mathcal{C}}) = \mathbb{K}$, together with natural isomorphisms

$$\omega_{X,Y} \colon F(X) \otimes_{\mathbb{K}} F(Y) \xrightarrow{\sim} F(X \otimes_{\mathcal{C}} Y) \ , \qquad X, Y \in \mathcal{C} \ , \tag{2.16}$$

normalised by $\omega_{X,\mathbb{1}_{\mathcal{C}}} = \omega_{\mathbb{1}_{\mathcal{C}},X} = \mathrm{id}_{F(X)}$ for all $X \in \mathcal{C}$. Note that no associativity coherence condition is imposed on ω.

Proposition 2.4. *Let $\mathcal{C} = \bigoplus_{g\in G} \mathcal{C}_g$ be a degreewise finite $\mathbb{K}$-linear abelian G-graded monoidal category, and let*

$$(F, \omega) \colon \mathcal{C} \to \mathrm{vec}_G \tag{2.17}$$

be a normalised quasi-fibre functor. Then there exists a finite-dimensional quasi-bialgebra with G-graded comultiplications $U = \{U_g\}_{g\in G}$ such that $\mathcal{C} \cong \operatorname{Rep} U$ as $\mathbb{K}$-linear abelian G-graded monoidal categories.

Proof. This is a G-graded analogue of the usual Tannakian reconstruction of a quasi-bialgebra; compare [EGNO15, Thm. 5.12.7 & Ex. 5.12.9].

For each $g \in G$, since F_g is exact and faithful, there exists a projective generator $P_g \in \mathcal{C}_g$ such that

$$F_g(X) \cong \mathrm{Hom}_{\mathcal{C}_g}(P_g, X) \tag{2.18}$$

naturally for every $X \in \mathcal{C}_g$. Setting

$$U_g \coloneqq \mathrm{End}_{\mathcal{C}_g}(P_g)^{\mathrm{op}} \cong \mathrm{Nat}(F_g, F_g) \ , \tag{2.19}$$

and denoting by $\operatorname{Rep}_{\mathrm{f.d.}} U_g$ the category of finite dimensional U_g-modules, we obtain an equivalence of $\mathbb{K}$-linear abelian categories

$$\mathcal{C}_g \simeq \operatorname{Rep}_{\mathrm{f.d.}} U_g \ . \tag{2.20}$$

These standard facts may be found, for example, in [EGNO15, Sec. 1.8].

We next construct the G-graded comultiplications, counit, and coassociators on $U = \{U_g\}_{g\in G}$. Transport the monoidal structure of $\mathcal{C}$ along the equivalence $\mathcal{C} \simeq \operatorname{Rep}_{\mathrm{f.d.}} U$, and denote the resulting monoidal product on $\operatorname{Rep}_{\mathrm{f.d.}} U$ by $\odot$, with associator α. Under this identification, F becomes the canonical forgetful functor, and the quasi-monoidal structure is given by natural linear isomorphisms

$$\omega_{M,N} \colon M \otimes_{\mathbb{K}} N \xrightarrow{\sim} M \odot N \ , \qquad M, N \in \operatorname{Rep}_{\mathrm{f.d.}} U \ , \tag{2.21}$$

where we have suppressed the forgetful functor from the notation. The isomorphisms ω further transport the monoidal structure $(\mathrm{Rep}_{\mathrm{f.d.}}U, \odot, \alpha)$ to one on $(\mathrm{Rep}_{\mathrm{f.d.}}U, \otimes_{\mathbb{K}})$. Let $M \in \mathrm{Rep}_{\mathrm{f.d.}}U_a$, $N \in \mathrm{Rep}_{\mathrm{f.d.}}U_b$, $K \in \mathrm{Rep}_{\mathrm{f.d.}}U_c$, and let us denote the module action by a dot. Then, for any $x \in U_{ab}$, $m \in M$, and $n \in N$, the transported U_{ab}-action on $M \otimes_{\mathbb{K}} N$ is defined by

$$x \cdot (m \otimes_{\mathbb{K}} n) \coloneqq \omega_{M,N}^{-1}\Big(x \cdot \big(\omega_{M,N}(m \otimes_{\mathbb{K}} n)\big)\Big) \ , \tag{2.22}$$

and the transported associator

$$\widetilde{\alpha}_{M,N,K} \colon M \otimes_{\mathbb{K}} (N \otimes_{\mathbb{K}} K) \longrightarrow (M \otimes_{\mathbb{K}} N) \otimes_{\mathbb{K}} K \tag{2.23}$$

is given by the composition

$$\widetilde{\alpha}_{M,N,K} \coloneqq (\omega_{M,N}^{-1} \otimes_{\mathbb{K}} \mathrm{id}_K) \circ \omega_{M\odot N,K}^{-1} \circ \alpha_{M,N,K} \circ \omega_{M,N\odot K} \circ (\mathrm{id}_M \otimes_{\mathbb{K}} \omega_{N,K}) \ . \tag{2.24}$$

It is straightforward to check that ω yields an equivalence of monoidal categories

$$(\mathrm{Rep}_{\mathrm{f.d.}}U, \otimes_{\mathbb{K}}, \widetilde{\alpha}) \simeq (\mathrm{Rep}_{\mathrm{f.d.}}U, \odot, \alpha) \ . \tag{2.25}$$

Finally, because the quasi-fibre functor is normalised, the transported unitors are trivial.

For any $a, b \in G$, define

$$\Delta_{a,b} \colon U_{ab} \longrightarrow U_a \otimes_{\mathbb{K}} U_b \ , \qquad \Delta_{a,b}(x) \coloneqq x \cdot (\mathbb{1}_a \otimes_{\mathbb{K}} \mathbb{1}_b) \ , \tag{2.26}$$

where U_a and U_b are regarded as regular left modules. For any $M \in \mathrm{Rep}_{\mathrm{f.d.}}U_a$, $N \in \mathrm{Rep}_{\mathrm{f.d.}}U_b$, and any $m \in M$, $n \in N$, $x \in U_{ab}$, we have the following commutative diagram,

$$\begin{array}{ccc} U_a \otimes_{\mathbb{K}} U_b & \xrightarrow{x\cdot} & U_a \otimes_{\mathbb{K}} U_b \\ \downarrow{\scriptstyle \overline{m}\otimes_{\mathbb{K}}\overline{n}} & & {\scriptstyle \overline{m}\otimes_{\mathbb{K}}\overline{n}}\downarrow \\ M \otimes_{\mathbb{K}} N & \xrightarrow{x\cdot} & M \otimes_{\mathbb{K}} N \end{array} \tag{2.27}$$

where $\overline{m}$ denotes the intertwiner sending $u \in U_a$ to $u \cdot m$, and similarly for $\overline{n}$. Inserting $\mathbb{1}_a \otimes_{\mathbb{K}} \mathbb{1}_b$ at the top-left corner, we obtain

$$x \cdot (m \otimes_{\mathbb{K}} n) = \Delta_{a,b}(x) \cdot (m \otimes_{\mathbb{K}} n) \ . \tag{2.28}$$

It follows directly from the module axioms that $\Delta_{a,b}$ is a unital algebra map.

The tensor unit has underlying vector space $\mathbb{K}$, and its U_1-action is therefore determined by a unital algebra map

$$\varepsilon \colon U_1 \longrightarrow \mathbb{K} \ . \tag{2.29}$$

Since the transported unitors are trivial, we obtain (H1).

For any $a, b, c \in G$, define

$$\Phi_{a,b,c} \coloneqq \widetilde{\alpha}_{U_a,U_b,U_c}\big(\mathbb{1}_a \otimes_{\mathbb{K}} (\mathbb{1}_b \otimes_{\mathbb{K}} \mathbb{1}_c)\big) \in (U_a \otimes_{\mathbb{K}} U_b) \otimes_{\mathbb{K}} U_c \ . \tag{2.30}$$

By naturality of $\widetilde{\alpha}$, for any $M \in \mathrm{Rep}_{\mathrm{f.d.}}U_a$, $N \in \mathrm{Rep}_{\mathrm{f.d.}}U_b$, $K \in \mathrm{Rep}_{\mathrm{f.d.}}U_c$, and any $m \in M$, $n \in N$, $k \in K$, we have the following commutative diagram.

$$\begin{array}{ccc} U_a \otimes_{\mathbb{K}} (U_b \otimes_{\mathbb{K}} U_c) & \xrightarrow{\widetilde{\alpha}_{U_a,U_b,U_c}} & (U_a \otimes_{\mathbb{K}} U_b) \otimes_{\mathbb{K}} U_c \\ {\scriptstyle \overline{m}\otimes_{\mathbb{K}}(\overline{n}\otimes_{\mathbb{K}}\overline{k})}\downarrow & & \downarrow{\scriptstyle (\overline{m}\otimes_{\mathbb{K}}\overline{n})\otimes_{\mathbb{K}}\overline{k}} \\ M \otimes_{\mathbb{K}} (N \otimes_{\mathbb{K}} K) & \xrightarrow{\widetilde{\alpha}_{M,N,K}} & (M \otimes_{\mathbb{K}} N) \otimes_{\mathbb{K}} K \end{array} \tag{2.31}$$

Inserting $\mathbb{1}_a \otimes_{\mathbb{K}} (\mathbb{1}_b \otimes_{\mathbb{K}} \mathbb{1}_c)$ at the top-left corner, we obtain

$$\widetilde{\alpha}_{M,N,K}\big(m \otimes_{\mathbb{K}} (n \otimes_{\mathbb{K}} k)\big) = \Phi_{a,b,c} \cdot \big((m \otimes_{\mathbb{K}} n) \otimes_{\mathbb{K}} k\big) \ . \tag{2.32}$$

Applying the same argument to $\widetilde{\alpha}^{-1}$ shows that $\Phi_{a,b,c}$ is invertible. Since $\widetilde{\alpha}_{U_a,U_b,U_c}$ is a U_{abc}-intertwiner, for any $x \in U_{abc}$ the following diagram commutes.

$$\begin{array}{ccc} U_a \otimes_{\mathbb{K}} (U_b \otimes_{\mathbb{K}} U_c) & \xrightarrow{\widetilde{\alpha}_{U_a,U_b,U_c}} & (U_a \otimes_{\mathbb{K}} U_b) \otimes_{\mathbb{K}} U_c \\ \downarrow{\scriptstyle x\cdot} & & {\scriptstyle x\cdot}\downarrow \\ U_a \otimes_{\mathbb{K}} (U_b \otimes_{\mathbb{K}} U_c) & \xrightarrow{\widetilde{\alpha}_{U_a,U_b,U_c}} & (U_a \otimes_{\mathbb{K}} U_b) \otimes_{\mathbb{K}} U_c \end{array} \tag{2.33}$$

Inserting $\mathbb{1}_a \otimes_{\mathbb{K}} (\mathbb{1}_b \otimes_{\mathbb{K}} \mathbb{1}_c)$ at the top-left corner and using (2.28), we obtain (H2).

Finally, the coherence conditions for $\widetilde{\alpha}$ implies (H3) and (H4). Hence $U = \{U_g\}_{g\in G}$ is a quasi-bialgebra with G-graded comultiplications.

□

Remark 2.5. The reconstruction above also works in presentable setting. Let $\mathcal{C}$ be a degreewise cocomplete $\mathbb{K}$-linear abelian G-graded monoidal category, and let

$$(F, \omega)\colon \mathcal{C} \longrightarrow \mathrm{Vec}_G \tag{2.34}$$

be a normalised quasi-fibre functor to the category of possibly infinite-dimensional G-graded vector spaces. Assume moreover that, for each $g \in G$, the restriction $F_g \coloneqq F_{|\mathcal{C}_g}$ is cocontinuous and admits a left adjoint

$$L_g\colon \mathrm{Vec} \longrightarrow \mathcal{C}_g \ . \tag{2.35}$$

Setting $P_g \coloneqq L_g(\mathbb{K})$, adjunction gives

$$F_g(X) \cong \mathrm{Hom}_{\mathcal{C}_g}(P_g, X) \tag{2.36}$$

naturally in $X \in \mathcal{C}_g$. Exactness, faithfulness, and cocontinuity of F_g imply that P_g is respectively projective, a generator, and compact. Hence, with $U_g \coloneqq \mathrm{End}_{\mathcal{C}_g}(P_g)^{\mathrm{op}}$, we have an equivalence of $\mathbb{K}$-linear abelian categories $\mathcal{C}_g \simeq \mathrm{Mod}U_g$ following [Mi65, Ch. IV, Thm. 4.1].

Denote by $\mathrm{Rep}_{\mathrm{f.d.}}U$ the subcategory of finite-dimensional left U-modules. If we want to endow $\mathrm{Rep}_{\mathrm{f.d.}}U$ with a rigid structure, we need to define module structures on dual vector spaces, which motivates the following definition.

Definition 2.6. A quasi-bialgebra with G-graded comultiplications $U = \{U_g\}_{g\in G}$ is called a *quasi-Hopf G-coalgebra* if it admits

- a family of anti-multiplicative linear isomorphisms called *antipodes* $S = \{S_a\colon U_a \to U_{a^{-1}}\}_{a\in G}$;
- a family of *evaluation elements* $\{\boldsymbol{\alpha}_a \in U_a\}_{a\in G}$;
- a family of *coevaluation elements* $\{\boldsymbol{\beta}_a \in U_a\}_{a\in G}$,

such that for any $a \in G$, $x \in U_1$,

$$\sum_{(x)} S_{a^{-1}}(x_{(1,a^{-1})})\, \boldsymbol{\alpha}_a\, x_{(2,a)} = \varepsilon(x)\boldsymbol{\alpha}_a \ , \qquad \sum_{(x)} x_{(1,a)}\, \boldsymbol{\beta}_a\, S_{a^{-1}}(x_{(2,a^{-1})}) = \varepsilon(x)\boldsymbol{\beta}_a \ , \tag{H5}$$

$$\sum_i p_i\, \boldsymbol{\beta}_a\, S_{a^{-1}}(q_i)\, \boldsymbol{\alpha}_a\, r_i = \mathbb{1}_a \ , \qquad \sum_j S_a(x_j)\, \boldsymbol{\alpha}_{a^{-1}}\, y_j\, \boldsymbol{\beta}_{a^{-1}}\, S_a(z_j) = \mathbb{1}_{a^{-1}} \ , \tag{H6}$$

where we use Sweedler notation $\Delta_{g,k}(u) = \sum_{(u)} u_{(1,g)} \otimes u_{(2,k)}$, and we write $\Phi^{-1}_{a,a^{-1},a} = \sum_i p_i \otimes q_i \otimes r_i$ and $\Phi_{a,a^{-1},a} = \sum_j x_j \otimes y_j \otimes z_j$.

Remark 2.7. If $\{U_g\}_{g\in G}$ is a quasi-Hopf G-coalgebra for a finite group G, then $\bigoplus_{g\in G} U_g$ is a quasi-Hopf algebra. Writing $x_g \in U_g$ for homogeneous element in degree g, we have

- the multiplication and the unit given by

$$\mu(\bigoplus_{a\in G} x_a \otimes \bigoplus_{b\in G} y_b) = \bigoplus_{a\in G} x_a y_a \ , \qquad \eta(1) = \bigoplus_{g\in G} \eta_g(1) \ ; \tag{2.38}$$

- the comultiplication and the counit given by

$$\Delta(\bigoplus_{g\in G} x_g) = \bigoplus_{g\in G} \bigoplus_{ab=g} \Delta_{a,b}(x_g) \ , \qquad \varepsilon(\bigoplus_{g\in G} x_g) = \varepsilon(x_1) \ ; \tag{2.39}$$

- the antipode given by

$$S(\bigoplus_{g\in G} x_g) = \bigoplus_{g\in G} S_g(x_g) \ ; \tag{2.40}$$

- the coassociator, and the evaluation and coevaluation element given by

$$\Phi = \bigoplus_{a,b,c\in G} \Phi_{a,b,c} \ , \qquad \boldsymbol{\alpha} = \bigoplus_{g\in G} \boldsymbol{\alpha}_g \ , \qquad \boldsymbol{\beta} = \bigoplus_{g\in G} \boldsymbol{\beta}_g \ . \tag{2.41}$$

Let U be a quasi-Hopf G-coalgebra. For any $M \in \mathrm{Rep}_{\mathrm{f.d.}} U_a$, its dual vector space M^* can be endowed with a $U_{a^{-1}}$-module structure as follows: for any $x \in U_{a^{-1}}$, $f \in M^*$, and $m \in M$, we define

$$(x \cdot f)(m) \coloneqq f(S_{a^{-1}}(x) \cdot m) \ . \tag{2.42}$$

We can further define an evaluation map by

$$\mathrm{ev}_M \colon M^* \otimes M \to \mathbb{K} \ , \qquad f \otimes m \mapsto f(\boldsymbol{\alpha}_a \cdot m) \ , \tag{2.43}$$

and a coevaluation map by

$$\mathrm{coev}_M \colon \mathbb{K} \to M \otimes M^* \ , \qquad 1 \mapsto \boldsymbol{\beta}_a \cdot m_i \otimes m^i \ , \tag{2.44}$$

where $\{m_i\}$ is a basis of M with dual basis $\{m^i\}$, and we use the Einstein summation convention. By (H5), the evaluation and coevaluation maps intertwine the U_1-action. Moreover, we have

$$(\mathrm{id}_M \otimes \mathrm{ev}_M) \circ \alpha^{-1}_{M,M^*,M} \circ (\mathrm{coev}_M \otimes \mathrm{id}_M) = \mathrm{id}_M \ , \tag{2.45a}$$

$$(\mathrm{ev}_M \otimes \mathrm{id}_{M^*}) \circ \alpha_{M^*,M,M^*} \circ (\mathrm{id}_{M^*} \otimes \mathrm{coev}_M) = \mathrm{id}_{M^*} \ , \tag{2.45b}$$

following (H6). Consequently, the category $\mathrm{Rep}_{\mathrm{f.d.}} U$ is rigid.

2.3. **H-Crossed quasi-Hopf G-coalgebras.** Let H be a group acting on G by group automorphisms. We write the action of a group element $h \in H$ on $g \in G$ as $h \triangleright g$.

Definition 2.8. An *H-crossed quasi-Hopf G-coalgebra* is a quasi-Hopf G-coalgebra $U = \{U_g\}_{g\in G}$ together with

- a family of algebra isomorphisms $\{\gamma_{h,a} \colon U_a \to U_{h\triangleright a}\}_{(h,a)\in H\times G}$;
- a family of invertible elements $\{\Omega^h_{a,b} \in U_a \otimes U_b\}_{(h,a,b)\in H\times G\times G}$,

such that the following conditions hold for any $h, h' \in H$, $a, b, c \in G$, and any $x \in U_{ab}$:

$$\gamma_{1,a} = \mathrm{id}_{U_a} \ ; \tag{C1}$$

$$\gamma_{h'h,a} = \gamma_{h',h\triangleright a} \circ \gamma_{h,a} \ ; \tag{C2}$$

$$\varepsilon = \varepsilon \circ \gamma_{h,1} \ ; \tag{C3}$$

$$\Delta_{a,b}(x) \cdot \Omega^h_{a,b} = \Omega^h_{a,b} \cdot (\gamma^{-1}_{h,a} \otimes \gamma^{-1}_{h,b}) \circ \Delta_{h\triangleright a, h\triangleright b} \circ \gamma_{h,ab}(x) \ ; \tag{C4}$$

$$\Omega^{h'h}_{a,b} = \Omega^h_{a,b} \cdot (\gamma^{-1}_{h,a} \otimes \gamma^{-1}_{h,b}) \Omega^{h'}_{h\triangleright a, h\triangleright b} \ ; \tag{C5}$$

$$(\varepsilon \otimes \mathrm{id}_{U_a}) \Omega^h_{1,a} = \mathbb{1}_a = (\mathrm{id}_{U_a} \otimes \varepsilon) \Omega^h_{a,1} \ ; \tag{C6}$$

$$\Phi_{a,b,c}\cdot(\mathrm{id}_{U_a}\otimes\Delta_{b,c})\Omega^h_{a,bc}\cdot(\mathbb{1}_a\otimes\Omega^h_{b,c}) = (\Delta_{a,b}\otimes\mathrm{id}_{U_c})\Omega^h_{ab,c}\cdot(\Omega^h_{a,b}\otimes\mathbb{1}_c)\cdot(\gamma^{-1}_{h,a}\otimes\gamma^{-1}_{h,b}\otimes\gamma^{-1}_{h,c})\Phi_{h\triangleright a,h\triangleright b,h\triangleright c}\ . \tag{C7}$$

Let $U=\{U_g\}_{g\in G}$ be an H-crossed quasi-Hopf G-coalgebra. As we explain in the following, the data γ and Ω induce an H-action on $\mathrm{Rep}\,U$, i.e. a strict monoidal functor

$$\varphi\colon \underline{H}\longrightarrow \mathrm{Aut}_\otimes(\mathrm{Rep}\,U)\ ,\qquad h\longmapsto {}^h(-)\ . \tag{2.46}$$

For $h\in H$ and $M\in\mathrm{Rep}\,U_g$, define ${}^hM$ to have the same underlying vector space as M, with $U_{h\triangleright g}$-action

$$\rho_{{}^hM}\coloneqq\rho_M\circ(\gamma^{-1}_{h,g}\otimes\mathrm{id}_M)\ , \tag{2.47}$$

where $\rho_M\colon U_g\otimes M\to M$ is the action map for M. For a U_g-intertwiner $f\colon M\to N$, define

$${}^hf\colon {}^hM\longrightarrow {}^hN \tag{2.48}$$

to be the same underlying linear map as f. Since $\gamma_{h,g}$ is an algebra isomorphism, ${}^hM$ is a $U_{h\triangleright g}$-module, and the U_g-linearity of f implies that ${}^hf$ is $U_{h\triangleright g}$-linear. Hence each ${}^h(-)$ is an autoequivalence of $\mathrm{Rep}\,U$, with inverse ${}^{h^{-1}}(-)$. Conditions (C1) and (C2) imply

$${}^{1_H}(-)=\mathrm{id}_{\mathrm{Rep}\,U}\ ,\qquad {}^{h'}({}^h(-))={}^{h'h}(-)\ , \tag{2.49}$$

hence the assignment $h\mapsto {}^h(-)$ defines a strict moniodal functor $\underline{H}\to\mathrm{Aut}(\mathrm{Rep}\,U)$. We next show that each ${}^h(-)$ is monoidal. For any $L\in\mathrm{Rep}\,U_a$ and $K\in\mathrm{Rep}\,U_b$, define the structure morphism of ${}^h(-)$ to be

$$h_{L,K}\colon {}^hL\otimes{}^hK\longrightarrow {}^h(L\otimes K)\ ,\qquad l\otimes k\longmapsto \Omega^h_{a,b}\cdot(l\otimes k)\ . \tag{2.50}$$

Here by $\Omega^h_{a,b}\cdot(l\otimes k)$ we mean $\rho_{L\otimes K}\big(\Omega^h_{a,b}\otimes(l\otimes k)\big)$. Condition (C4) implies that $h_{L,K}$ is a $U_{h\triangleright(ab)}$-intertwiner, and (C3) shows that ${}^h(-)$ preserves the tensor unit. Finally, conditions (C5)–(C7) yield the coherence conditions (2.4a)–(2.4c), respectively.

Conversely, every monoidal structure of ${}^h(-)$ arises in this way:

Proposition 2.9. *Let H be a group acting on G via group automorphisms, and let $U=\{U_g\}_{g\in G}$ be a quasi-Hopf G-coalgebra. Suppose that there exists a family of algebra isomorphisms*

$$\{\gamma_{(h,a)}\colon U_a\to U_{h\triangleright a}\}_{(h,a)\in H\times G} \tag{2.51}$$

satisfying (C1)–(C3). Via (2.47) and (2.48), this determines a strict monoidal functor $\varphi\colon\underline{H}\to\mathrm{Aut}(\mathrm{Rep}\,U)$. Suppose, moreover, that each ${}^h(-)=\varphi(h)$ is strictly unital monoidal so that φ lifts to a strict monoidal functor as in (2.46). Then there exists a family of invertible elements

$$\{\Omega^h_{a,b}\in U_a\otimes U_b\}_{(h,a,b)\in H\times G\times G} \tag{2.52}$$

satisfying (C4)–(C7) such that the monoidal structure of ${}^h(-)$ is given as in (2.50).

Proof. For any $h\in H$, $a,b\in G$, set

$$\Omega^h_{a,b}=h_{U_a,U_b}(\mathbb{1}_a\otimes\mathbb{1}_b)\ , \tag{2.53}$$

where U_a, U_b are regular representations of U_a and H_b respectively, and $h_{U_a,U_b}\colon {}^hU_a\otimes{}^hU_b\to{}^h(U_a\otimes U_b)$ is the structure morphism for ${}^h(-)$. By naturality of $h_{-,-}$, for any $M\in\mathrm{Rep}\,U_a$ and $N\in\mathrm{Rep}\,U_b$ we have the following commutative diagram:

$$\begin{array}{ccc} {}^hU_a\otimes{}^hU_b & \xrightarrow{h_{U_a,U_b}} & {}^h(U_a\otimes U_b)\\ \downarrow{\scriptstyle {}^h\overline{m}\otimes{}^h\overline{n}} & & {}^h(\overline{m}\otimes\overline{n})\downarrow\\ {}^hM\otimes{}^hN & \xrightarrow{h_{M,N}} & {}^h(M\otimes N)\end{array} \tag{2.54}$$

where $\overline{m}: U_a \to M$ is the U_a-intertwiner sending $x \in U_a$ to $x \cdot m$, and similarly for $\overline{n}$. Inserting $\mathbb{1}_a \otimes \mathbb{1}_b$ at the top-left corner, we then get $h_{M,N}(m \otimes n) = \Omega^h_{a,b} \cdot (m \otimes n)$ as in (2.50).

Since h_{U_a,U_b} is a U_{ab}-intertwiner, the following diagram commutes for any $x \in U_{ab}$:

$$\begin{array}{ccc} {}^hU_a \otimes {}^hU_b & \xrightarrow{h_{U_a,U_b}} & {}^h(U_a \otimes U_b) \\ \downarrow{\scriptstyle \gamma_{h,ab}(x)\cdot} & & {\scriptstyle \gamma_{h,ab}(x)\cdot}\downarrow \\ {}^hU_a \otimes {}^hU_b & \xrightarrow{h_{U_a,U_b}} & {}^h(U_a \otimes U_b) \end{array} \tag{2.55}$$

Inserting $\mathbb{1}_a \otimes \mathbb{1}_b$ at the top-left corner, we obtain (C4). Finally, (C5)–(C7) follow from (2.4a)–(2.4c), respectively.

□

2.4. **χ-Crossed ribbon quasi-Hopf algebra.** Let $\chi\colon G \to H$ be a crossed module, and let $U = \{U_g\}_{g\in G}$ be an H-crossed quasi-Hopf G-coalgebra.

Definition 2.10. A family of *χ-crossed R-matrices* of U is a family of invertible elements

$$R = \{R_{a,b} = \sum_i (R'_i)_a \otimes (R''_i)_b \in U_a \otimes U_b\}_{(a,b)\in G^{\times 2}} \ , \tag{2.56}$$

such that for any $a, b, c \in G$, $h \in H$, $x \in U_{ab}$, the following conditions hold:

$$R_{a,b}\Delta_{a,b}(x) = \Big(\tau \circ (\gamma^{-1}_{\chi_a,b} \otimes \mathrm{id}_{U_a}) \circ \Delta_{aba^{-1},a}(x)\Big) R_{a,b} \ , \tag{R1}$$

$$\begin{aligned}(\Delta_{a,b} \otimes \mathrm{id}_{U_c})R_{ab,c} = \Big((\gamma^{-1}_{\chi_{ab},c} \otimes \mathrm{id}_{U_a} \otimes \mathrm{id}_{U_b})\Phi^{-1}_{abcb^{-1}a^{-1},a,b}\Big)_{231}\Big((\mathrm{id}_{U_a} \otimes \gamma^{-1}_{\chi_b,c})R_{a,bcb^{-1}}\Big)_{1b3} \\ \cdot\Big((\mathrm{id}_{U_a} \otimes \gamma^{-1}_{\chi_b,c} \otimes \mathrm{id}_{U_b})\Phi_{a,bcb^{-1},b}\Big)_{132}(R_{b,c})_{a23}\Phi^{-1}_{a,b,c} \ ,\end{aligned} \tag{R2}$$

$$\begin{aligned}(\mathrm{id}_{U_a} \otimes \Delta_{b,c})R_{a,bc} = (\Omega^{\chi_a}_{b,c})_{a23}\Big((\gamma^{-1}_{\chi_a,b} \otimes \gamma^{-1}_{\chi_a,c} \otimes \mathrm{id}_{U_a})\Phi_{aba^{-1},aca^{-1},a}\Big)_{312} \\ \cdot(R_{a,c})_{1b3}\Big((\gamma^{-1}_{\chi_a,b} \otimes \mathrm{id}_{U_a} \otimes \mathrm{id}_{U_c})\Phi^{-1}_{aba^{-1},a,c}\Big)_{213}(R_{a,b})_{12c}\Phi_{a,b,c} \ ,\end{aligned} \tag{R3}$$

$$\Big(\tau \circ (\gamma^{-1}_{\chi_a,b} \otimes \mathrm{id}_{U_a})\Omega^h_{aba^{-1},a}\Big)(\gamma^{-1}_{h,a} \otimes \gamma^{-1}_{h,b})R_{h\triangleright a,h\triangleright b} = R_{a,b}\Omega^h_{a,b} \ , \tag{R4}$$

where $\tau(x \otimes y) = y \otimes x$ is the usual transposition, for any $X = \sum_i x_i \otimes y_i \otimes z_i$, we write $X_{231} = \sum_i y_i \otimes z_i \otimes x_i$, and for any $T = \sum_j p_j \otimes q_j$, we write $(T)_{1a3} = \sum_j p_j \otimes \mathbb{1}_a \otimes q_j$, and so on. An H-crossed quasi-Hopf G-coalgebra admitting a family of χ-crossed R-matrices is called a *χ-crossed quasitriangular quasi-Hopf algebra*.

The representation category of a χ-crossed quasitriangular quasi-Hopf algebra $U = \{U_g\}_{g\in G}$ is χ-crossed braided: for any $M \in \mathrm{Rep}\, U_a$ and $N \in \mathrm{Rep}\, U_b$ we define a χ-crossed braiding by

$$C_{M,N}\colon M \otimes N \to {}^{\chi_a}N \otimes M \ , \qquad m \otimes n \mapsto \tau\big(R_{a,b} \cdot (m \otimes n)\big) \ . \tag{2.58}$$

$C_{M,N}$ is indeed a U_{ab}-intertwiner according to (R1). The condition (B1) follows by (2.48). Conditions (B2)–(B4) follow from (R2)–(R4), respectively. Conversely, all χ-crossed braiding comes from this way in the following sense:

Proposition 2.11. *Let $U = \{U_g\}_{g\in G}$ be an H-crossed quasi-Hopf G-coalgebra. Suppose that $\mathrm{Rep}\, U$ is equipped with a χ-crossed braiding with respect to the H-action induced by γ and Ω as described above Proposition 2.9. Then there exists a unique family of χ-crossed R-matrices*

$$\{R_{a,b} \in U_a \otimes U_b\}_{(a,b)\in G\times G} \tag{2.59}$$

such that the χ-crossed braiding on $\mathrm{Rep}\, U$ is given by (2.58).

Proof. For any $a, b \in G$, define

$$R_{a,b} = \tau\Big(C_{U_a,U_b}(\mathbb{1}_a \otimes \mathbb{1}_b)\Big) \ . \tag{2.60}$$

The proof follows the idea of Proposition 15.2.2 in [Ka95] for braided quasi-bialgebras, and proceeds similarly to the proofs of Proposition 2.4 and 2.9. □

For any $h \in H$ and $a \in G$, set $b = h \triangleright a$ and write

$$\Phi_{b^{-1},b,b^{-1}} = \sum_j \Phi_j^{(1)} \otimes \Phi_j^{(2)} \otimes \Phi_j^{(3)} \ , \qquad \Omega^h_{a^{-1},a} = \sum_i \Omega_i^{(1)} \otimes \Omega_i^{(2)} \ . \tag{2.61}$$

For $M \in \mathrm{Rep}_{\mathrm{f.d.}} U_a$, recall the canonical isomorphism $\nu_{h,M} \colon {}^h(M^*) \to ({}^hM)^*$ as in Remark 2.2. We define

$$\mathfrak{d}_{h,a} \coloneqq \sum_{i,j} S_{a^{-1}}\left(\Omega_i^{(1)} \gamma^{-1}_{h,a^{-1}}\big(\Phi_j^{(1)}\big)\right) \boldsymbol{\alpha}_a \Omega_i^{(2)} \gamma^{-1}_{h,a}\left(\Phi_j^{(2)} \boldsymbol{\beta}_b S_{b^{-1}}\big(\Phi_j^{(3)}\big)\right) \in U_a \ . \tag{2.62}$$

For every $f \in {}^h(M^*)$, and $m \in {}^hM$, a straightforward but lengthy computation shows that

$$\big(\nu_{h,M}(f)\big)(m) = f\big(\mathfrak{d}_{h,a} \cdot m\big) \ , \tag{2.63}$$

where on the right-hand side, we view f and m as elements of M^* and M, respectively.

Definition 2.12. A χ-crossed quasitriangular quasi-Hopf algebra U is called *χ-crossed ribbon* if it admits a family of *χ-crossed ribbon elements*, i.e. a family of invertible elements

$$v = \{v_g \in U_g\}_{g \in G} \tag{2.64}$$

such that for any $a, b \in G$, $h \in H$, $x \in U_a$, we have

$$\gamma^{-1}_{\chi_a,a}(x) = v_a^{-1} x v_a \ , \tag{RE1}$$

$$\Delta_{a,b}(v_{ab}^{-1}) = \Omega^{\chi_{ab}}_{a,b}\Big(\tau(\gamma^{-1}_{\chi_{ab},b} \otimes \gamma^{-1}_{\chi_a,a}) R_{aba^{-1},a}\Big)\Big((\gamma^{-1}_{\chi_a,a} \otimes \gamma^{-1}_{\chi_b,b}) R_{a,b}\Big)\Big(v_a^{-1} \otimes v_b^{-1}\Big) \ , \tag{RE2}$$

$$v_{h \triangleright a} = \gamma_{h,a}(v_a) \ , \tag{RE3}$$

$$S_{a^{-1}}(v_{a^{-1}}) \mathfrak{d}_{\chi_a,a} = v_a \ . \tag{RE4}$$

χ-crossed ribbon elements $\{v_g\}_{g \in G}$ give rise to a χ-crossed ribbon structure on $\mathrm{Rep}_{f.d} U$. Explicitly, for any $a \in G$ and for any $M \in \mathrm{Rep}_{f.d} U_a$, the χ-crossed balancing isomorphism is given by

$$\vartheta_M \colon M \to {}^{\chi_a}M \ , \qquad m \mapsto v_a^{-1} \cdot m \ . \tag{2.66}$$

By (RE1) this is indeed a U_a-intertwiner. Condition (BL1) is satisfied folowing (2.48). Conditions (BL2) and (BL3) correspond to (RE2) and (RE3), respectively. It remains to check the final condition (RB1). For $f \in {}^{\chi_a}(M^*)$ and $m \in M$, using (2.63), we have

$$\begin{aligned} \left((\vartheta_M)^* \circ \nu_{\chi_a,M}\right)(f)(m) &= \nu_{\chi_a,M}(f)\left(v_a^{-1} \cdot m\right) \\ &= f(\mathfrak{d}_{\chi_a,a} v_a^{-1} \cdot m) \ . \end{aligned} \tag{2.67}$$

On the other hand,

$$\begin{aligned} \vartheta_{\chi_a(M^*)}(f)(m) &= \big(\gamma^{-1}_{\chi_a,a^{-1}}(v_{a^{-1}}^{-1}) \cdot f\big)(m) \\ &= f\big(S_{a^{-1}} \circ \gamma^{-1}_{\chi_a,a^{-1}}(v_{a^{-1}}^{-1})\big) \\ &= f\big(S_{a^{-1}}(v_{a^{-1}}^{-1})\big) \ , \end{aligned} \tag{2.68}$$

where in the last step we used (RE3). Following (RE4), we have (2.67) = (2.68), which is exactly (RB1).

In the same spirit as Propositions 2.4, 2.9, and 2.11, we present the following result:

Proposition 2.13. *For any χ-crossed quasitriangular quasi-Hopf algebra $U = \{U_g\}_{g\in G}$ such that each U_g is finite-dimensional and $\mathrm{Rep}_{\mathrm{f.d.}}U$ is equipped with a χ-crossed ribbon structure, there exists a unique family of χ-crossed ribbon elements such that the χ-crossed ribbon structure on $\mathrm{Rep}_{\mathrm{f.d.}}U$ is given by (2.66).*

Example 2.14. Let $\chi\colon G \to H$ be a crossed module. In [GRW26, Sec. 4], we gave a cohomological classification of χ-crossed braided structures on the category of finite-dimensional G-graded vector spaces. Consider a class in $H^3_{\mathrm{cm}}(\chi;\mathbb{K}^\times)$ (cf. [GRW26, Prop. 4.16]) with a normalised representative $(\omega,\Omega,1;R)$, where

$$\omega\colon G^3 \to \mathbb{K}^\times\ , \qquad \Omega\colon H\times G^2 \to \mathbb{K}^\times\ , \qquad R\colon G^2 \to \mathbb{K}^\times \tag{2.69}$$

are functions satisfying the cocycle and crossed braiding conditions of [GRW26, Sec. 3.2 & 4.4]. We define a χ-crossed ribbon quasi-Hopf algebra $U = \{U_g\}_{g\in G}$ by the following data, for all $a,b,c \in G$ and $h \in H$:

- U_a is a one-dimensional algebra with the unit denoted by $\mathbb{1}_a$;
- the coproducts and counit are
$$\Delta_{a,b}(\mathbb{1}_{ab}) = \mathbb{1}_a \otimes \mathbb{1}_b\ , \qquad \varepsilon(\mathbb{1}_{1_G}) = 1\ ; \tag{2.70}$$
- the coassociators are
$$\Phi_{a,b,c} = \omega(a,b,c)\,\mathbb{1}_a \otimes \mathbb{1}_b \otimes \mathbb{1}_c\ ; \tag{2.71}$$
- the antipodes, evaluation and coevaluation elements are
$$S_a(\mathbb{1}_a) = \mathbb{1}_{a^{-1}}\ , \qquad \boldsymbol{\alpha}_a = \omega(a,a^{-1},a)\mathbb{1}_a\ , \qquad \boldsymbol{\beta}_a = \mathbb{1}_a\ ; \tag{2.72}$$
- the crossed action is given by
$$\gamma_{h,a}(\mathbb{1}_a) = \mathbb{1}_{h\triangleright a}\ , \qquad \Omega^h_{a,b} = \Omega(h,a,b)\,\mathbb{1}_a \otimes \mathbb{1}_b\ ; \tag{2.73}$$
- the χ-crossed R-matrices and ribbon elements are
$$R_{a,b} = R(a,b)\,\mathbb{1}_a \otimes \mathbb{1}_b\ , \qquad v_a = R(a,a)^{-1}\mathbb{1}_a\ . \tag{2.74}$$

All axioms of a χ-crossed ribbon quasi-Hopf algebra are satisfied because it is reconstructed from the χ-crossed ribbon category $\mathrm{vec}_G^{(\omega,\Omega,1;R)}$.

2.5. **χ-Crossed ribbon Hopf algebra.** Because the quasi-Hopf structure involves substantial coherence data, we summarise the definition of a χ-crossed ribbon Hopf algebra below for clarity.

Definition 2.15. Let G and H be two groups and let H act on G by group automorphisms, where the action of a group element h is denoted by $h\triangleright$. An *H-crossed Hopf G-coalgebra* is a family of associative unital algebras $U = \{U_g\}_{g\in G}$ over a ground field $\mathbb{K}$ with

- a family of comultiplications $\Delta = \{\Delta_{a,b}\colon U_{ab} \to U_a \otimes U_b\}_{a,b\in G}$;
- a counit $\varepsilon = \{\varepsilon\colon U_1 \to \mathbb{K}\}$;
- a family of antipodes $S = \{S_a\colon U_a \to U_{a^{-1}}\}_{a\in G}$;
- a family of algebra isomorphisms $\{\gamma_{(h,a)}\colon U_a \to U_{h\triangleright a}\}_{(h,a)\in H\times G}$;
- a family of invertible elements $\{\Omega^h_{a,b} \in U_a \otimes U_b\}_{(h,a,b)\in H\times G\times G}$,

such that for any $a,b,c \in G$, $h,h' \in H$, $u \in U_1$, $x \in U_{ab}$, the following conditions hold:

$$(\varepsilon \otimes \mathrm{id}_{U_a}) \circ \Delta_{1,a} = \mathrm{id}_{U_a} = (\mathrm{id}_{U_a} \otimes \varepsilon) \circ \Delta_{a,1}\ ; \tag{H′1}$$

$$(\Delta_{a,b} \otimes \mathrm{id}_{U_c}) \circ \Delta_{ab,c} = (\mathrm{id}_{U_a} \otimes \Delta_{b,c}) \circ \Delta_{a,bc}\ ; \tag{H′2}$$

$$\sum_{(u)} S_{a^{-1}}(u_{(1,a^{-1})})u_{(2,a)} = \varepsilon(u), \qquad \sum_{(u)} u_{(1,a)}S_{a^{-1}}(u_{(2,a^{-1})}) = \varepsilon(u)\ ; \tag{H′5}$$

$$\gamma_{1,a} = \mathrm{id}_{U_a}; \tag{C1}$$

$$\gamma_{h'h,a} = \gamma_{h',h\triangleright a} \circ \gamma_{h,a} \ ; \tag{C2}$$

$$\varepsilon = \varepsilon \circ \gamma_{(h,1)} \ ; \tag{C3}$$

$$\Delta_{a,b}(x) \cdot \Omega^h_{a,b} = \Omega^h_{a,b} \cdot (\gamma^{-1}_{h,a} \otimes \gamma^{-1}_{h,b}) \circ \Delta_{h\triangleright a, h\triangleright b} \circ \gamma_{h,ab}(x) \ ; \tag{C4}$$

$$\Omega^{h'h}_{a,b} = \Omega^h_{a,b} \cdot (\gamma^{-1}_{h,a} \otimes \gamma^{-1}_{h,b}) \Omega^{h'}_{h\triangleright a, h\triangleright b} \ ; \tag{C5}$$

$$(\varepsilon \otimes \mathrm{id}_{U_a}) \Omega^h_{1,a} = \mathbb{1}_a = (\mathrm{id}_{U_a} \otimes \varepsilon) \Omega^h_{a,1} \ ; \tag{C6}$$

$$(\mathrm{id}_{U_a} \otimes \Delta_{b,c}) \Omega^h_{a,bc} \cdot (\mathbb{1}_a \otimes \Omega^h_{b,c}) = (\Delta_{a,b} \otimes \mathrm{id}_{U_c}) \Omega^h_{ab,c} \cdot (\Omega^h_{a,b} \otimes \mathbb{1}_c) \ . \tag{C'7}$$

Moreover, if $\chi\colon G \to H$ is a crossed module with H acting on G as above, then U is called a *χ-crossed ribbon Hopf algebra* if there exist

- a family of χ-crossed R-matrices $R = \{R_{a,b} = \sum_i (R'_i)_a \otimes (R''_i)_b \in U_a \otimes U_b\}_{(a,b)\in G^{\times 2}}$;
- a family of χ-crossed ribbon elements $v = \{v_g \in U_g\}_{g\in G}$,

such that for any $a, b, c \in G$, $h \in H$, and for any $x \in U_{ab}$, $y \in U_a$, the following conditions hold:

$$R_{a,b} \Delta_{a,b}(x) = \Big(\tau \circ (\gamma^{-1}_{\chi_a,b} \otimes \mathrm{id}_{U_a}) \circ \Delta_{aba^{-1},a}(x)\Big) R_{a,b} \ ; \tag{R1}$$

$$(\Delta_{a,b} \otimes \mathrm{id}_{U_c}) R_{ab,c} = \Big((\mathrm{id}_{U_a} \otimes \gamma^{-1}_{\chi_b,c}) R_{a,bcb^{-1}}\Big)_{1b3} (R_{b,c})_{a23} \ ; \tag{R'2}$$

$$(\mathrm{id}_{U_a} \otimes \Delta_{b,c}) R_{a,bc} = (\Omega^{\chi_a}_{b,c})_{a23} (R_{a,c})_{1b3} (R_{a,b})_{12c} \ ; \tag{R'3}$$

$$\Big(\tau \circ (\gamma^{-1}_{\chi_a,b} \otimes \mathrm{id}_{U_a}) \Omega^h_{aba^{-1},a}\Big) (\gamma^{-1}_{h,a} \otimes \gamma^{-1}_{h,b}) R_{h\triangleright a, h\triangleright b} = R_{a,b} \Omega^h_{a,b} \ ; \tag{R4}$$

$$\gamma^{-1}_{\chi_a,a}(y) = v_a^{-1} y v_a \ ; \tag{RE1}$$

$$\Delta_{a,b}(v_{ab}^{-1}) = \Omega^{\chi_{ab}}_{a,b} \Big(\tau(\gamma^{-1}_{\chi_{ab},b} \otimes \gamma^{-1}_{\chi_a,a}) R_{aba^{-1},a}\Big) \Big((\gamma^{-1}_{\chi_a,a} \otimes \gamma^{-1}_{\chi_b,b}) R_{a,b}\Big) \Big(v_a^{-1} \otimes v_b^{-1}\Big) \ ; \tag{RE2}$$

$$v_{h\triangleright a} = \gamma_{h,a}(v_a) \ ; \tag{RE3}$$

$$\sum_i S_{a^{-1}}(\Omega^{(1)}_i v_{a^{-1}}) \Omega^{(2)}_i = v_a \ , \tag{RE'4}$$

where $\Omega^{\chi_a}_{a^{-1},a} = \sum_i \Omega^{(1)}_i \otimes \Omega^{(2)}_i$, $\tau(x \otimes y) = y \otimes x$ is the usual transposition, and for any $T = \sum_j p_j \otimes q_j$, we write $(T)_{1a3} = \sum_j p_j \otimes \mathbb{1}_a \otimes q_j$ and so on.

Note that the conditions with "′" in tags differ from their corresponding ones in the quasi-version.

3. Graded coends

Let H and G be groups, with H acting on G by group automorphisms. For a braided monoidal category $\mathcal{B}$ equipped with an H-action by braided autoequivalences, we introduce the notion of an *H-crossed Hopf G-algebra* in $\mathcal{B}$ (Definition 3.1). Given a crossed module $\chi\colon G \to H$ and a χ-crossed ribbon category $\mathcal{C}$, we show that the graded coends as in (3.15) form an H-crossed Hopf G-algebra in the neutral component of $\mathcal{C}$ (Theorem 3.6).

3.1. **Categorical H-crossed Hopf G-algebras.** We first recall the notion of a *Hopf group-algebra* in a braided monoidal category [Vi00a, Sec. 3.3.1]. Let $\mathcal{B}$ be a braided monoidal category and G be a group. We write $C_{U,V}\colon U \otimes V \to V \otimes U$ for the braiding of $\mathcal{B}$.

A *Hopf G-algebra object* in $\mathcal{B}$ is a family of objects

$$A \coloneqq \{A_g\}_{g\in G} \tag{3.1}$$

together with

- a family of *multiplications* $\mu = \{\mu_{a,b}\colon A_a \otimes A_b \to A_{ab}\}_{a,b\in G}$;
- a *unit* $\eta\colon \mathbb{1} \to A_{1_G}$;

- a family of *comultiplications* $\Delta = \{\Delta_a \colon A_a \to A_a \otimes A_a\}_{a\in G}$;
- a family of *counits* $\varepsilon = \{\varepsilon_a \colon A_a \to \mathbb{1}\}_{a\in G}$;
- a family of morphisms called *antipodes* $S = \{S_a \colon A_a \to A_{a^{-1}}\}_{a\in G}$,

such that, for all $a, b, c \in G$, these data satisfy:

$$\mu_{ab,c} \circ (\mu_{a,b} \otimes \mathrm{id}_{A_c}) = \mu_{a,bc} \circ (\mathrm{id}_{A_a} \otimes \mu_{b,c}) \ , \tag{CH1}$$

$$\mu_{1_G,a} \circ (\eta \otimes \mathrm{id}_{A_a}) = \mathrm{id}_{A_a} = \mu_{a,1_G} \circ (\mathrm{id}_{A_a} \otimes \eta) \ , \tag{CH2}$$

$$(\Delta_a \otimes \mathrm{id}_{A_a}) \circ \Delta_a = (\mathrm{id}_{A_a} \otimes \Delta_a) \circ \Delta_a \ , \tag{CH3}$$

$$(\varepsilon_a \otimes \mathrm{id}_{A_a}) \circ \Delta_a = \mathrm{id}_{A_a} = (\mathrm{id}_{A_a} \otimes \varepsilon_a) \circ \Delta_a \ , \tag{CH4}$$

$$\Delta_{ab} \circ \mu_{a,b} = (\mu_{a,b} \otimes \mu_{a,b}) \circ (\mathrm{id}_{A_a} \otimes C_{A_a,A_b} \otimes \mathrm{id}_{A_b}) \circ (\Delta_a \otimes \Delta_b) \ , \tag{CH5}$$

$$\varepsilon_{ab} \circ \mu_{a,b} = \varepsilon_a \otimes \varepsilon_b \ , \tag{CH6}$$

$$\Delta_{1_G} \circ \eta = \eta \otimes \eta \ , \tag{CH7}$$

$$\varepsilon_{1_G} \circ \eta = \mathrm{id}_{\mathbb{1}} \ , \tag{CH8}$$

$$\mu_{a^{-1},a} \circ (S_a \otimes \mathrm{id}_{A_a}) \circ \Delta_a = \eta \circ \varepsilon_a = \mu_{a,a^{-1}} \circ (\mathrm{id}_{A_a} \otimes S_a) \circ \Delta_a \ , \tag{CH9}$$

where we have suppressed associators and unitors of $\mathcal{B}$.

Let H be a group acting on G by group automorphisms. We denote the action of $h \in H$ on $g \in G$ by $h \triangleright g$. Let $\underline{H}$ denote the discrete monoidal category associated to H. Suppose that we are given a monoidal functor

$$\varphi : \underline{H} \longrightarrow \mathrm{Aut}_{\mathrm{br}}(\mathcal{B}) \ , \qquad h \longmapsto {}^h(-) \ , \tag{3.2}$$

where $\mathrm{Aut}_{\mathrm{br}}(\mathcal{B})$ is the 2-group whose objects are braided autoequivalences of $\mathcal{B}$ and whose morphisms are monoidal natural isomorphisms. $\mathrm{Aut}_{\mathrm{br}}(\mathcal{B})$ is strictly monoidal with the tensor product on objects given by composition of functors, and on morphisms given by horizontal composition of natural transformations. For all $h \in H$, and all $X, Y \in \mathcal{B}$ we write

$$h_{X,Y} \colon {}^hX \otimes {}^hY \xrightarrow{\cong} {}^h(X \otimes Y) \ , \qquad h_{\mathbb{1}} \colon \mathbb{1} \xrightarrow{\cong} {}^h\mathbb{1} \tag{3.3}$$

for the monoidal structure isomorphisms of ${}^h(-)$. For all $h_1, h_2 \in H$, we write the monoidal structure of the functor φ as

$$T^{h_1,h_2} \colon {}^{h_1}({}^{h_2}(-)) \xRightarrow{\sim} {}^{h_1h_2}(-) \ , \qquad T_0 \colon \mathrm{id}_{\mathcal{B}} \xRightarrow{\cong} {}^{1_H}(-) \ . \tag{3.4}$$

Definition 3.1. An *H-crossed Hopf G-algebra* in $\mathcal{B}$ is a Hopf G-algebra $A = \{A_g\}_{g\in G}$ in $\mathcal{B}$, together with a family of isomorphisms

$$\{\rho_{h,g} \colon {}^hA_g \xrightarrow{\sim} A_{h\triangleright g}\}_{h\in H,\ g\in G} \tag{3.5}$$

such that, for all $a, b \in G$, $h, h_1, h_2 \in H$, the following conditions hold:

$$\rho_{1_H,a} \circ (T_0)_{A_a} = \mathrm{id}_{A_a} \ , \tag{CCH1}$$

$$\rho_{h_1,h_2\triangleright a} \circ {}^{h_1}(\rho_{h_2,a}) = \rho_{h_1h_2,a} \circ T^{h_1,h_2}_{A_a} \ , \tag{CCH2}$$

$$\rho_{h,ab} \circ {}^h\mu_{a,b} \circ h_{A_a,A_b} = \mu_{h\triangleright a,h\triangleright b} \circ (\rho_{h,a} \otimes \rho_{h,b}) \ , \tag{CCH3}$$

$$\rho_{h,1_G} \circ {}^h\eta \circ h_{\mathbb{1}} = \eta \ , \tag{CCH4}$$

$$\Delta_{h\triangleright a} \circ \rho_{h,a} = (\rho_{h,a} \otimes \rho_{h,a}) \circ h^{-1}_{A_a,A_a} \circ {}^h\Delta_a \ , \tag{CCH5}$$

$$\varepsilon_{h\triangleright a} \circ \rho_{h,a} = h^{-1}_{\mathbb{1}} \circ {}^h\varepsilon_a \ . \tag{CCH6}$$

Lemma 3.2. *Let $\big(\{A_g\}_{g\in G}, \{\rho_{h,g}\}_{h\in H,\ g\in G}\big)$ be an H-crossed Hopf G-algebra in $\mathcal{B}$ as above. Then it holds that, for all $h \in H$, $a \in G$,*

$$S_{h\triangleright a} \circ \rho_{h,a} = \rho_{h,a^{-1}} \circ {}^hS_a \ . \tag{CCH7}$$

Proof. For any $h \in H$ and $a, b, c \in G$, consider the graded convolution

$$\mathrm{Hom}_{\mathcal{B}}({}^h A_a, A_b) \times \mathrm{Hom}_{\mathcal{B}}({}^h A_a, A_c) \longrightarrow \mathrm{Hom}_{\mathcal{B}}({}^h A_a, A_{bc}) \tag{3.6}$$

defined by

$$f * g \coloneqq \mu_{b,c} \circ (f \otimes g) \circ h^{-1}_{A_a, A_a} \circ {}^h\Delta_a \ . \tag{3.7}$$

Its unit is the morphism

$$\eta \circ h_{\mathbb{1}}^{-1} \circ {}^h\varepsilon_a \colon {}^h A_a \longrightarrow A_{1_G} \ . \tag{3.8}$$

We now show that both $S_{h \triangleright a} \circ \rho_{h,a}$ and $\rho_{h,a^{-1}} \circ {}^h S_a$ are convolution inverses of

$$\rho_{h,a} \colon {}^h A_a \longrightarrow A_{h \triangleright a} \ . \tag{3.9}$$

For $S_{h \triangleright a} \circ \rho_{h,a}$,

$$\begin{aligned} &\mu_{(h \triangleright a)^{-1},\, h \triangleright a} \circ \big((S_{h \triangleright a} \circ \rho_{h,a}) \otimes \rho_{h,a} \big) \circ h^{-1}_{A_a, A_a} \circ {}^h\Delta_a \\ &= \mu_{(h \triangleright a)^{-1},\, h \triangleright a} \circ (S_{h \triangleright a} \otimes \mathrm{id}) \circ \Delta_{h \triangleright a} \circ \rho_{h,a} = \eta \circ \varepsilon_{h \triangleright a} \circ \rho_{h,a} = \eta \circ h_{\mathbb{1}}^{-1} \circ {}^h\varepsilon_a \ . \end{aligned} \tag{3.10}$$

where the first equality uses (CCH5), the second uses (CH9), and the third uses (CCH6). The other inverse identity can be computed analogously. On the other hand, for $\rho_{h,a^{-1}} \circ {}^h S_a$,

$$\begin{aligned} &\mu_{h \triangleright (a^{-1}),\, h \triangleright a} \circ \big((\rho_{h,a^{-1}} \circ {}^h S_a) \otimes \rho_{h,a} \big) \circ h^{-1}_{A_a, A_a} \circ {}^h\Delta_a \\ &= \mu_{h \triangleright (a^{-1}),\, h \triangleright a} \circ (\rho_{h,a^{-1}} \otimes \rho_{h,a}) \circ ({}^h S_a \otimes \mathrm{id}_{{}^h A_a}) \circ h^{-1}_{A_a, A_a} \circ {}^h\Delta_a \\ &= \mu_{h \triangleright (a^{-1}),\, h \triangleright a} \circ (\rho_{h,a^{-1}} \otimes \rho_{h,a}) \circ h^{-1}_{A_{a^{-1}}, A_a} \circ {}^h(S_a \otimes \mathrm{id}_{A_a}) \circ {}^h\Delta_a \\ &= \rho_{h,1_G} \circ {}^h\mu_{a^{-1},a} \circ {}^h(S_a \otimes \mathrm{id}_{A_a}) \circ {}^h\Delta_a \\ &= \rho_{h,1_G} \circ {}^h\big(\mu_{a^{-1},a} \circ (S_a \otimes \mathrm{id}_{A_a}) \circ \Delta_a\big) \\ &= \rho_{h,1_G} \circ {}^h(\eta \circ \varepsilon_a) = \rho_{h,1_G} \circ {}^h\eta \circ {}^h\varepsilon_a = \eta \circ h_{\mathbb{1}}^{-1} \circ {}^h\varepsilon_a \ , \end{aligned} \tag{3.11}$$

where the second equality follows from naturality of $h_{-,-}$, the third from (CCH3), the fifth from (CH9), and the last from (CCH4). Again, similar computation shows the other inverse identity.

□

3.2. **Graded coends in a rigid G-graded monoidal category.** We recall *graded coends* in a rigid G-graded monoidal category [Vi00a, Sec. 3.2]. Let $\mathcal{C}$ be a rigid G-graded monoidal category. For each $g \in G$, consider the functor

$$F_g \colon \mathcal{C}_g^{\mathrm{op}} \times \mathcal{C}_g \to \mathcal{C}_{1_G} \ , \qquad (X, Y) \mapsto X^* \otimes Y \ , \qquad (f, h) \mapsto f^* \otimes h \ . \tag{3.12}$$

Let $T \in \mathcal{C}_{1_G}$ be an object. A *dinatural transformation* $q \colon F_g \dddot{\to} T$ is a family of morphisms

$$\{q_X \colon X^* \otimes X \longrightarrow T\}_{X \in \mathcal{C}_g} \tag{3.13}$$

such that, for every morphism $f \colon X \to Y$ in $\mathcal{C}_g$, the following diagram commutes:

$$\begin{array}{ccc} Y^* \otimes X & \xrightarrow{f^* \otimes \mathrm{id}_X} & X^* \otimes X \\ \downarrow{\scriptstyle \mathrm{id}_{Y^*} \otimes f} & & {\scriptstyle q_X}\downarrow \\ Y^* \otimes Y & \xrightarrow{\quad q_Y \quad} & T \end{array} \tag{3.14}$$

A *coend* of F_g is an object denoted by

$$\mathcal{L}_g \coloneqq \int^{X \in \mathcal{C}_g} X^* \otimes X \in \mathcal{C}_{1_G} \ , \tag{3.15}$$

together with a dinatural transformation

$$\{i_X^g \colon X^* \otimes X \longrightarrow \mathcal{L}_g\}_{X \in \mathcal{C}_g} \tag{3.16}$$

such that the following universal property holds. For every object $T \in \mathcal{C}_{1_G}$ and every dinatural transformation $q\colon F_g \overset{\cdot\cdot}{\to} T$, there exists a unique morphism

$$l_T\colon \mathcal{L}_g \longrightarrow T \tag{3.17}$$

such that, for all $X \in \mathcal{C}_g$, one has

$$q_X = l_T \circ i_X^g \ . \tag{3.18}$$

Remark 3.3. Following the universal property, the coend, when it exists, is unique up to unique isomorphism.

Suppose that F_g admits a coend for every $g \in G$.

Lemma 3.4 ([Vi00a, Lem. 3.2])**.** *Let $a, b \in G$ and $Z \in \mathcal{C}_{1_G}$. Suppose that for every $X \in \mathcal{C}_a$ and $Y \in \mathcal{C}_b$, we are given a morphism*

$$\xi_{X,Y}\colon (X^* \otimes X) \otimes (Y^* \otimes Y) \longrightarrow Z \tag{3.19}$$

in $\mathcal{C}_{1_G}$, such that for every $f\colon X \to X'$ in $\mathcal{C}_a$ and $h\colon Y \to Y'$ in $\mathcal{C}_b$, the following diagram commutes:

$$\begin{array}{ccc} (X'^* \otimes X) \otimes (Y'^* \otimes Y) & \xrightarrow{(f^* \otimes \mathrm{id}_X) \otimes (h^* \otimes \mathrm{id}_Y)} & (X^* \otimes X) \otimes (Y^* \otimes Y) \\ \downarrow{\scriptstyle (\mathrm{id}_{X'^*} \otimes f) \otimes (\mathrm{id}_{Y'^*} \otimes h)} & & {\scriptstyle \xi_{X,Y}} \downarrow \\ (X'^* \otimes X') \otimes (Y'^* \otimes Y') & \xrightarrow{\qquad\qquad \xi_{X',Y'} \qquad\qquad} & Z \end{array} \tag{3.20}$$

Then there exists a unique morphism

$$l\colon \mathcal{L}_a \otimes \mathcal{L}_b \longrightarrow Z \tag{3.21}$$

satisfying

$$\xi_{X,Y} = l \circ (i_X^a \otimes i_Y^b) \tag{3.22}$$

for all $X \in \mathcal{C}_a$ and $Y \in \mathcal{C}_b$.

3.3. **Graded coends as H-crossed Hopf G-algebras.** Let $\chi\colon G \to H$ be a crossed module, and $\mathcal{C}$ be a χ-crossed ribbon category. We use the notation introduced in (2.3) for the monoidal structure isomorphism of each ${}^h(-)$. Exceptionally, in this subsection we *do not* assume the H-action on $\mathcal{C}$ to be strict. For any $h_1, h_2 \in H$, we write

$$T^{h_1,h_2}\colon {}^{h_1}({}^{h_2}(-)) \xRightarrow{\sim} {}^{h_1h_2}(-) \ , \qquad T_0\colon \mathrm{id}_{\mathcal{C}} \xRightarrow{\cong} {}^{1_H}(-) \tag{3.23}$$

for its monoidal structure. However, without loss of generality, we may assume ${}^{1_H}(-) = \mathrm{id}$ and $T_0 = \mathrm{id}$ [GRW26, Rem. 2.5]. We show that the graded coends as in (3.15) have the structure of an H-crossed Hopf G-algebra in the neutral component $\mathcal{C}_{1_G}$.

For any $g \in G$ and $X \in \mathcal{C}_g$, $Y \in \mathcal{C}$, we denote by

$$C_{X,Y}\colon X \otimes Y \longrightarrow {}^{\chi_g}Y \otimes X \tag{3.24}$$

the χ-crossed braiding of $\mathcal{C}$, and by

$$\vartheta_X\colon X \longrightarrow {}^{\chi_g}X \tag{3.25}$$

the χ-crossed ribbon twist. The neutral component of a χ-crossed ribbon category is ribbon, and we denote its braiding and ribbon twist also by C and ϑ, respectively. Condition (B4) implies that the H-action on $\mathcal{C}$ restricts to an action on $\mathcal{C}_{1_G}$ by braided autoequivalences. By our unitality assumption, both $h_{\mathbb{1}}$ and T_0 in (3.3) and (3.4) are identities.

Suppose that the functors F_g given in (3.12) admit coends for all $g \in G$. Following [Vi00a, Sec. 3.3.2], we first recall that the graded coends

$$\mathcal{L} = \{\mathcal{L}_g\}_{g \in G} \tag{3.26}$$

defined in (3.15) have the structure of a Hopf G-algebra in $\mathcal{C}_{1_G}$. We will only sketch the construction here and refer the reader to [Vi00a, Thm. 3.5] for a proof. In the following, we suppress unitors, associators, and the canonical identifications

$$(X \otimes Y)^* \cong Y^* \otimes X^* \tag{3.27}$$

for readability.

- For any $a, b \in G$, $X \in \mathcal{C}_a$, and $Y \in \mathcal{C}_b$, consider the morphism
$$X^* \otimes X \otimes Y^* \otimes Y \xrightarrow{C_{X^*\otimes X, Y^*} \otimes \mathrm{id}_Y} Y^* \otimes X^* \otimes X \otimes Y \xrightarrow{i^{ab}_{X\otimes Y}} \mathcal{L}_{ab} \ . \tag{3.28}$$
This family satisfies the condition of Lemma 3.4. Hence it induces a unique morphism
$$\mu_{a,b} \colon \mathcal{L}_a \otimes \mathcal{L}_b \longrightarrow \mathcal{L}_{ab} \tag{3.29}$$
such that, for all $X \in \mathcal{C}_a$ and $Y \in \mathcal{C}_b$,
$$\mu_{a,b} \circ (i^a_X \otimes i^b_Y) = i^{ab}_{X\otimes Y} \circ \left(C_{X^*\otimes X, Y^*} \otimes \mathrm{id}_Y\right) \ . \tag{3.30}$$
These morphisms define the multiplications.
- The unit $\eta : \mathbb{1} \longrightarrow \mathcal{L}_{1_G}$ is defined by the composite
$$\mathbb{1} \xrightarrow{\cong} \mathbb{1}^* \otimes \mathbb{1} \xrightarrow{i^{1_G}_{\mathbb{1}}} \mathcal{L}_{1_G} \ . \tag{3.31}$$
- For any $a \in G$ and $X \in \mathcal{C}_a$, consider the morphism
$$X^* \otimes X \xrightarrow{\mathrm{id}_{X^*} \otimes \mathrm{coev}_X \otimes \mathrm{id}_X} X^* \otimes X \otimes X^* \otimes X \xrightarrow{i^a_X \otimes i^a_X} \mathcal{L}_a \otimes \mathcal{L}_a \ . \tag{3.32}$$
This family is dinatural in X, hence induces a unique morphism
$$\Delta_a \colon \mathcal{L}_a \longrightarrow \mathcal{L}_a \otimes \mathcal{L}_a \tag{3.33}$$
such that, for all $X \in \mathcal{C}_a$,
$$\Delta_a \circ i^a_X = (i^a_X \otimes i^a_X) \circ (\mathrm{id}_{X^*} \otimes \mathrm{coev}_X \otimes \mathrm{id}_X) \ . \tag{3.34}$$
These morphisms define the comultiplications.
- For any $a \in G$ and $X \in \mathcal{C}_a$, the evaluation morphism $\mathrm{ev}_X \colon X^* \otimes X \longrightarrow \mathbb{1}$ is dinatural in X, hence induces a unique morphism
$$\varepsilon_a \colon \mathcal{L}_a \longrightarrow \mathbb{1} \tag{3.35}$$
such that, for all $X \in \mathcal{C}_a$,
$$\varepsilon_a \circ i^a_X = \mathrm{ev}_X \ . \tag{3.36}$$
These morphisms define the counits.
- For any $a \in G$ and $X \in \mathcal{C}_a$, the composite
$$\begin{aligned} X^* \otimes X &\xrightarrow{\mathrm{id}_{X^*} \otimes \vartheta_X} X^* \otimes {}^{\chi_a}X \xrightarrow{C_{X^*, {}^{\chi_a}X}} {}^{\chi_{a^{-1}}}\left({}^{\chi_a}X\right) \otimes X^* \\ &\xrightarrow{T_X^{\chi_{a^{-1}}, \chi_a} \otimes \mathrm{id}_{X^*}} X \otimes X^* \xrightarrow{\cong} X^{**} \otimes X^* \xrightarrow{i^{a^{-1}}_{X^*}} \mathcal{L}_{a^{-1}} \end{aligned} \tag{3.37}$$
is dinatural in X, hence induces a unique morphism
$$S_a \colon \mathcal{L}_a \longrightarrow \mathcal{L}_{a^{-1}} \tag{3.38}$$
such that, for all $X \in \mathcal{C}_a$,
$$S_a \circ i^a_X = i^{a^{-1}}_{X^*} \circ (\delta_X \otimes \mathrm{id}_{X^*}) \circ \left(T_X^{\chi_{a^{-1}}, \chi_a} \otimes \mathrm{id}_{X^*}\right) \circ C_{X^*, {}^{\chi_a}X} \circ (\mathrm{id}_{X^*} \otimes \vartheta_X) \ , \tag{3.39}$$
where $\delta_X \colon X \to X^{**}$ is the pivotal structure induced by the crossed ribbon structure on $\mathcal{C}$ as in [GRW26, Cor. 4.15]. These morphisms define the antipodes.

It remains to define the H-crossed structure on $\{\mathcal{L}_g\}_{g\in g}$. To this end, we first need the following technical lemma.

Lemma 3.5. *Let $\mathcal{D},\mathcal{D}',\mathcal{B},\mathcal{B}'$ be categories, and*

$$E\colon \mathcal{D}\longrightarrow\mathcal{D}'\ ,\qquad K\colon \mathcal{B}\longrightarrow\mathcal{B}' \tag{3.40}$$

be equivalences. Let

$$F\colon \mathcal{D}^{\mathrm{op}}\times\mathcal{D}\longrightarrow\mathcal{B}\ ,\qquad F'\colon (\mathcal{D}')^{\mathrm{op}}\times\mathcal{D}'\longrightarrow\mathcal{B}' \tag{3.41}$$

be functors, and suppose that there is a natural isomorphism

$$\lambda_{X,Y}\colon K(F(X,Y))\xrightarrow{\cong}F'(E(X),E(Y))\ . \tag{3.42}$$

Assume that the coends

$$L=\int^{X\in\mathcal{D}}F(X,X)\ ,\qquad L'=\int^{Y\in\mathcal{D}'}F'(Y,Y) \tag{3.43}$$

exist, with structure morphisms

$$i_X\colon F(X,X)\longrightarrow L\ ,\qquad j_Y\colon F'(Y,Y)\longrightarrow L'\ . \tag{3.44}$$

Then there is a canonical isomorphism

$$\rho : K(L)\xrightarrow{\cong}L' \tag{3.45}$$

characterised by

$$\rho\circ K(i_X)=j_{E(X)}\circ\lambda_{X,X} \tag{3.46}$$

for all $X\in\mathcal{D}$.

Proof. Since K is an equivalence, $K(L)$ is a coend of $K\circ F$. Hence, for every $Z\in\mathcal{B}'$, we have natural bijections

$$\begin{aligned}&\mathrm{Hom}_{\mathcal{B}'}(K(L),Z)\cong\mathrm{DiNat}\big(K\circ F,Z\big)\\ &\cong\mathrm{DiNat}\big(F'\circ(E^{\mathrm{op}}\times E),Z\big)\cong\mathrm{DiNat}(F',Z)\cong\mathrm{Hom}_{\mathcal{B}'}(L',Z)\ .\end{aligned} \tag{3.47}$$

Here the second bijection is induced by λ, while the third is induced by the equivalence E. This shows that L' is also a coend of $K\circ F$, hence we conclude the proof. □

Fix $h\in H$ and $g\in G$. We apply Lemma 3.5 with

$$\mathcal{D}=\mathcal{C}_g\ ,\qquad \mathcal{D}'=\mathcal{C}_{h\triangleright g}\ ,\qquad \mathcal{B}=\mathcal{B}'=\mathcal{C}_{1_G}\ , \tag{3.48}$$

with the equivalences given by

$$E={}^h(-)\colon\mathcal{C}_g\longrightarrow\mathcal{C}_{h\triangleright g}\ ,\qquad K={}^h(-)\colon\mathcal{C}_{1_G}\longrightarrow\mathcal{C}_{1_G}\ . \tag{3.49}$$

We further take

$$F=F_g\ ,\qquad F'=F_{h\triangleright g}\ . \tag{3.50}$$

Then, for $X,Y\in\mathcal{C}_g$, one has

$$K\big(F(X,Y)\big)={}^h(X^*\otimes Y)\ , \tag{3.51}$$

while

$$F'\big(E(X),E(Y)\big)=F_{h\triangleright g}({}^hX,{}^hY)=({}^hX)^*\otimes{}^hY\ . \tag{3.52}$$

The monoidal structure of ${}^h(-)$, together with the canonical isomorphism

$$\nu_{h,X}:{}^h(X^*)\xrightarrow{\cong}({}^hX)^* \tag{3.53}$$

from Remark 2.2, gives a natural isomorphism

$$\lambda^h_{X,Y}\coloneqq(\nu_{h,X}\otimes\mathrm{id}_{{}^hY})\circ h^{-1}_{X^*,Y}\colon{}^h(X^*\otimes Y)\xrightarrow{\cong}({}^hX)^*\otimes{}^hY\ . \tag{3.54}$$

Thus the assumptions of Lemma 3.5 are satisfied, with

$$L = \mathcal{L}_g \ , \qquad L' = \mathcal{L}_{h\triangleright g} \ . \tag{3.55}$$

We therefore obtain a canonical isomorphism

$$\rho_{h,g} \colon {}^h\mathcal{L}_g \xrightarrow{\cong} \mathcal{L}_{h\triangleright g} \tag{3.56}$$

characterised by the condition that, for every $X \in \mathcal{C}_g$,

$$\rho_{h,g} \circ {}^h i^g_X = i^{h\triangleright g}_{{}^h X} \circ \lambda^h_{X,X} \ . \tag{3.57}$$

Theorem 3.6. *The Hopf G-algebra*

$$\mathcal{L} = \{\mathcal{L}_g\}_{g\in G} \tag{3.58}$$

together with the isomorphisms

$$\rho_{h,g} \colon {}^h\mathcal{L}_g \xrightarrow{\cong} \mathcal{L}_{h\triangleright g} \tag{3.59}$$

defined by (3.57) is an H-crossed Hopf G-algebra in $\mathcal{C}_{1_G}$.

The proof is rather technical. We first establish two lemmas.

Lemma 3.7. *For every $h, h_1, h_2 \in H$ and $X, Y \in \mathcal{C}$, the canonical isomorphisms $\nu_{h,X}$ satisfy the following coherence properties:*

$$\begin{array}{ccc} {}^h(X^*) \otimes {}^h X & \xrightarrow{\nu_{h,X}\otimes \mathrm{id}_{{}^h X}} & ({}^h X)^* \otimes {}^h X \\ \downarrow{\scriptstyle h_{X^*,X}} & & \downarrow{\scriptstyle \mathrm{ev}_{{}^h X}} \\ {}^h(X^* \otimes X) & \xrightarrow{{}^h\mathrm{ev}_X} & \mathbb{1} \end{array} \tag{3.60a}$$

$$\begin{array}{ccc} \mathbb{1} & \xrightarrow{{}^h\mathrm{coev}_X} & {}^h(X \otimes X^*) \\ \downarrow{\scriptstyle \mathrm{coev}_{{}^h X}} & & \downarrow{\scriptstyle h^{-1}_{X,X^*}} \\ {}^h X \otimes ({}^h X)^* & \xrightarrow{\mathrm{id}\otimes \nu^{-1}_{h,X}} & {}^h X \otimes {}^h(X^*) \end{array} \tag{3.60b}$$

$$\begin{array}{ccccc} {}^{h_1}\big({}^{h_2}(X^*)\big) & \xrightarrow{{}^{h_1}(\nu_{h_2,X})} & {}^{h_1}\big(({}^{h_2}X)^*\big) & \xrightarrow{\nu_{h_1,{}^{h_2}X}} & \big({}^{h_1}({}^{h_2}X)\big)^* \\ \downarrow{\scriptstyle T^{h_1,h_2}_{X^*}} & & & & \downarrow{\scriptstyle ((T^{h_1,h_2}_X)^{-1})^*} \\ {}^{h_1h_2}(X^*) & & \xrightarrow{\qquad\qquad \nu_{h_1h_2,X} \qquad\qquad} & & ({}^{h_1h_2}X)^* \end{array} \tag{3.60c}$$

$$\begin{array}{ccc} {}^h(Y^*) \otimes {}^h(X^*) & \xrightarrow{h_{Y^*,X^*}} & {}^h\big((X\otimes Y)^*\big) \\ \downarrow{\scriptstyle \nu_{h,Y}\otimes\nu_{h,X}} & & \downarrow{\scriptstyle \nu_{h,X\otimes Y}} \\ ({}^h Y)^* \otimes ({}^h X)^* & \xrightarrow{(h^{-1}_{X,Y})^*} & \big({}^h(X\otimes Y)\big)^* \end{array} \tag{3.60d}$$

Proof. The compatibility (3.60a) is the defining property $\nu_{h,X}$, and (3.60b) is the corresponding compatibility with coevaluation.

Note that both ${}^{h_1}({}^{h_2}(X^*))$ and $({}^{h_1h_2}X)^*$ are duals of ${}^{h_1}({}^{h_2}X)$. The two paths in (3.60c) give isomorphisms between these two duals which are compatible with evaluation, and hence they agree by uniqueness of the canonical isomorphism between duals. Similarly, (3.60d) follows from the uniqueness of the canonical identification of the dual of ${}^h(X\otimes Y)$. □

Lemma 3.8. *For every $h, h_1, h_2 \in H$ and $X, Y \in \mathcal{C}$, the natural isomorphisms*

$$\lambda^h_{X,Y} \colon {}^h(X^* \otimes Y) \xrightarrow{\cong} ({}^hX)^* \otimes {}^hY \tag{3.61}$$

defined in (3.54) satisfy the following coherence properties:

$$\begin{array}{ccc} {}^h(X^* \otimes X) & \xrightarrow{\lambda^h_{X,X}} & ({}^hX)^* \otimes {}^hX \\ \downarrow{\scriptstyle {}^h\mathrm{ev}_X} & & \downarrow{\scriptstyle \mathrm{ev}_{{}^hX}} \\ \mathbb{1} & = & \mathbb{1} \end{array} \tag{3.62a}$$

$$\begin{array}{ccccc} {}^h(X^* \otimes X) & & \xrightarrow{\lambda^h_{X,X}} & & ({}^hX)^* \otimes {}^hX \\ \downarrow{\scriptstyle {}^h(\mathrm{id}_{X^*} \otimes \mathrm{coev}_X \otimes \mathrm{id}_X)} & & & & \downarrow{\scriptstyle \mathrm{id}_{({}^hX)^*} \otimes \mathrm{coev}_{{}^hX} \otimes \mathrm{id}_{{}^hX}} \\ {}^h\big(X^* \otimes X \otimes X^* \otimes X\big) & \xrightarrow[h^{-1}_{X^* \otimes X, X^* \otimes X}]{} & {}^h(X^* \otimes X) \otimes {}^h(X^* \otimes X) & \xrightarrow[\lambda^h_{X,X} \otimes \lambda^h_{X,X}]{} & ({}^hX)^* \otimes {}^hX \otimes ({}^hX)^* \otimes {}^hX \end{array} \tag{3.62b}$$

$$\begin{array}{ccccc} {}^{h_1}\big({}^{h_2}(X^* \otimes Y)\big) & \xrightarrow{{}^{h_1}(\lambda^{h_2}_{X,Y})} & {}^{h_1}\big(({}^{h_2}X)^* \otimes {}^{h_2}Y\big) & \xrightarrow{\lambda^{h_1}_{{}^{h_2}X, {}^{h_2}Y}} & \big({}^{h_1}({}^{h_2}X)\big)^* \otimes {}^{h_1}({}^{h_2}Y) \\ \downarrow{\scriptstyle T^{h_1,h_2}_{X^* \otimes Y}} & & & & \downarrow{\scriptstyle ((T^{h_1,h_2}_X)^{-1})^* \otimes T^{h_1,h_2}_Y} \\ {}^{h_1h_2}(X^* \otimes Y) & & \xrightarrow{\lambda^{h_1h_2}_{X,Y}} & & ({}^{h_1h_2}X)^* \otimes {}^{h_1h_2}Y \end{array} \tag{3.62c}$$

$$\lambda^{1_H}_{X,Y} = \mathrm{id}_{X^* \otimes Y} \ , \qquad \lambda^h_{\mathbb{1},\mathbb{1}} = \mathrm{id}_{\mathbb{1}} \ , \tag{3.62d}$$

$$\begin{array}{ccccc} {}^h\big(X^* \otimes X \otimes Y^* \otimes Y\big) & \xrightarrow{h^{-1}_{X^* \otimes X, Y^* \otimes Y}} & {}^h(X^* \otimes X) \otimes {}^h(Y^* \otimes Y) & \xrightarrow{\lambda^h_{X,X} \otimes \lambda^h_{Y,Y}} & ({}^hX)^* \otimes {}^hX \otimes ({}^hY)^* \otimes {}^hY \\ \downarrow{\scriptstyle {}^h(C_{X^* \otimes X, Y^*} \otimes \mathrm{id}_Y)} & & & & \downarrow{\scriptstyle C_{({}^hX)^* \otimes {}^hX, ({}^hY)^*} \otimes \mathrm{id}_{{}^hY}} \\ {}^h\big((X \otimes Y)^* \otimes (X \otimes Y)\big) & \xrightarrow[\lambda^h_{X \otimes Y, X \otimes Y}]{} & \big({}^h(X \otimes Y)\big)^* \otimes {}^h(X \otimes Y) & \xleftarrow[(h^{-1}_{X,Y})^* \otimes h_{X,Y}]{} & ({}^hY)^* \otimes ({}^hX)^* \otimes {}^hX \otimes {}^hY \end{array} \tag{3.62e}$$

Proof. By substituting $\lambda^h_{X,X} = (\nu_{h,X} \otimes \mathrm{id}_{{}^hX}) \circ h^{-1}_{X^*,X}$, the commutativity of (3.62a) follows directly from (3.60a), while (3.62b) follows from (3.60b) together with (2.4c). Similarly, by expanding $\lambda^h_{X,Y}$, the commutativity of (3.62c) follows from (3.60c) and the monoidality of T^{h_1,h_2}. The identities in (3.62d) follow immediately from our strict unitality assumption. Finally, after expanding the three λ^h, the commutativity of (3.62e) follows from (3.60d), (B1), (B4), and (2.4c). □

Proof of Theorem 3.6. The Hopf G-algebra structure on $\mathcal{L}$ follows from [Vi00a, Thm. 3.5]. It remains to verify (CCH1)–(CCH6). We repeatedly use the fact that a morphism out of a coend is uniquely determined by its composites with the universal dinatural morphisms.

For (CCH1), we need to show

$$\rho_{1_H,g} = \mathrm{id}_{\mathcal{L}_g} \ . \tag{3.63}$$

By (3.57) and (3.62d), for every $X \in \mathcal{C}_g$,

$$\rho_{1_H,g} \circ i^g_X = i^g_X \circ \lambda^{1_H}_{X,X} = i^g_X \ , \tag{3.64}$$

which implies (CCH1).

For (CCH2), we need to show

$$\rho_{h_1, h_2 \triangleright g} \circ {}^{h_1}(\rho_{h_2,g}) = \rho_{h_1h_2,g} \circ T^{h_1,h_2}_{\mathcal{L}_g} \ . \tag{3.65}$$

Let $X \in \mathcal{C}_g$ and precompose both sides with ${}^{h_1}({}^{h_2}i^g_X)$. By (3.57), the left-hand side becomes

$$i^{(h_1h_2)\triangleright g}_{{}^{h_1}({}^{h_2}X)} \circ \lambda^{h_1}_{{}^{h_2}X,{}^{h_2}X} \circ {}^{h_1}\big(\lambda^{h_2}_{X,X}\big) \ . \tag{3.66}$$

On the other hand,

$$\begin{aligned} &\rho_{h_1h_2,g} \circ T^{h_1,h_2}_{\mathcal{L}_g} \circ {}^{h_1}({}^{h_2}i^g_X) \\ &= i^{(h_1h_2)\triangleright g}_{{}^{h_1h_2}X} \circ \lambda^{h_1h_2}_{X,X} \circ T^{h_1,h_2}_{X^*\otimes X} \\ &= i^{(h_1h_2)\triangleright g}_{{}^{h_1h_2}X} \circ \big(((T^{h_1,h_2}_X)^{-1})^* \otimes T^{h_1,h_2}_X\big) \circ \lambda^{h_1}_{{}^{h_2}X,{}^{h_2}X} \circ {}^{h_1}\big(\lambda^{h_2}_{X,X}\big) \\ &= i^{(h_1h_2)\triangleright g}_{{}^{h_1}({}^{h_2}X)} \circ \lambda^{h_1}_{{}^{h_2}X,{}^{h_2}X} \circ {}^{h_1}\big(\lambda^{h_2}_{X,X}\big) \ , \end{aligned} \tag{3.67}$$

which coincides with the left hand side. Here the first equality follows from the naturality of T^{h_1,h_2} and (3.57), the second from (3.62c), and the last from the dinaturality of i with respect to T^{h_1,h_2}_X.

For (CCH3), we need to show

$$\rho_{h,ab} \circ {}^h\mu_{a,b} \circ h_{\mathcal{L}_a,\mathcal{L}_b} = \mu_{h\triangleright a,h\triangleright b} \circ (\rho_{h,a} \otimes \rho_{h,b}) \ . \tag{3.68}$$

Let $X \in \mathcal{C}_a$ and $Y \in \mathcal{C}_b$. By Lemma 3.4, it suffices to precompose both sides with ${}^hi^a_X \otimes {}^hi^b_Y$. We compute

$$\begin{aligned} &\rho_{h,ab} \circ {}^h\mu_{a,b} \circ h_{\mathcal{L}_a,\mathcal{L}_b} \circ ({}^hi^a_X \otimes {}^hi^b_Y) \\ &= \rho_{h,ab} \circ {}^h\mu_{a,b} \circ {}^h(i^a_X \otimes i^b_Y) \circ h_{X^*\otimes X,Y^*\otimes Y} \\ &= \rho_{h,ab} \circ {}^h\big(\mu_{a,b} \circ (i^a_X \otimes i^b_Y)\big) \circ h_{X^*\otimes X,Y^*\otimes Y} \\ &= \rho_{h,ab} \circ {}^hi^{ab}_{X\otimes Y} \circ {}^h\big(C_{X^*\otimes X,Y^*} \otimes \mathrm{id}_Y\big) \circ h_{X^*\otimes X,Y^*\otimes Y} \\ &= i^{h\triangleright(ab)}_{{}^h(X\otimes Y)} \circ \lambda^h_{X\otimes Y,X\otimes Y} \circ {}^h\big(C_{X^*\otimes X,Y^*} \otimes \mathrm{id}_Y\big) \circ h_{X^*\otimes X,Y^*\otimes Y} \\ &= i^{h\triangleright(ab)}_{{}^h(X\otimes Y)} \circ \big((h^{-1}_{X,Y})^* \otimes h_{X,Y}\big) \circ \big(C_{({}^hX)^*\otimes {}^hX,({}^hY)^*} \otimes \mathrm{id}_{{}^hY}\big) \circ (\lambda^h_{X,X} \otimes \lambda^h_{Y,Y}) \\ &= i^{(h\triangleright a)(h\triangleright b)}_{{}^hX\otimes {}^hY} \circ \big(C_{({}^hX)^*\otimes {}^hX,({}^hY)^*} \otimes \mathrm{id}_{{}^hY}\big) \circ (\lambda^h_{X,X} \otimes \lambda^h_{Y,Y}) \\ &= \mu_{h\triangleright a,h\triangleright b} \circ (i^{h\triangleright a}_{{}^hX} \otimes i^{h\triangleright b}_{{}^hY}) \circ (\lambda^h_{X,X} \otimes \lambda^h_{Y,Y}) \\ &= \mu_{h\triangleright a,h\triangleright b} \circ (\rho_{h,a} \otimes \rho_{h,b}) \circ ({}^hi^a_X \otimes {}^hi^b_Y). \end{aligned} \tag{3.69}$$

Here the first equality follows from the naturality of $h_{-,-}$, the third from (3.30), the fourth from (3.57), the fifth from (3.62e), the sixth from the dinaturality of $i^{h\triangleright(ab)}$ with respect to $h_{X,Y}$, the seventh from (3.30), and the last from (3.57).

For (CCH4), we need to show

$$\rho_{h,1_G} \circ {}^h\eta = \eta \ . \tag{3.70}$$

We have

$$\rho_{h,1_G} \circ {}^h\eta = \rho_{h,1_G} \circ {}^hi^{1_G}_{\mathbb{1}} = i^{1_G}_{{}^h\mathbb{1}} \circ \lambda^h_{\mathbb{1},\mathbb{1}} = i^{1_G}_{\mathbb{1}} = \eta \ . \tag{3.71}$$

Here the first equality follows from (3.31), the second from (3.57), the third from (3.62d), and the last again from (3.31).

For (CCH5), we need to show

$$\Delta_{h\triangleright g} \circ \rho_{h,g} = (\rho_{h,g} \otimes \rho_{h,g}) \circ h^{-1}_{\mathcal{L}_g,\mathcal{L}_g} \circ {}^h\Delta_g \ . \tag{3.72}$$

For $X \in \mathcal{C}_g$, we compute

$$\begin{aligned}
&\Delta_{h\triangleright g} \circ \rho_{h,g} \circ {}^h i^g_X \\
&= \Delta_{h\triangleright g} \circ i^{h\triangleright g}_{{}^h X} \circ \lambda^h_{X,X} \\
&= (i^{h\triangleright g}_{{}^h X} \otimes i^{h\triangleright g}_{{}^h X}) \circ \big(\mathrm{id}_{({}^h X)^*} \otimes \mathrm{coev}_{{}^h X} \otimes \mathrm{id}_{{}^h X}\big) \circ \lambda^h_{X,X} \\
&= (i^{h\triangleright g}_{{}^h X} \otimes i^{h\triangleright g}_{{}^h X}) \circ (\lambda^h_{X,X} \otimes \lambda^h_{X,X}) \circ h^{-1}_{X^*\otimes X, X^*\otimes X} \circ {}^h\big(\mathrm{id}_{X^*} \otimes \mathrm{coev}_X \otimes \mathrm{id}_X\big) \\
&= (\rho_{h,g} \otimes \rho_{h,g}) \circ ({}^h i^g_X \otimes {}^h i^g_X) \circ h^{-1}_{X^*\otimes X, X^*\otimes X} \circ {}^h\big(\mathrm{id}_{X^*} \otimes \mathrm{coev}_X \otimes \mathrm{id}_X\big) \\
&= (\rho_{h,g} \otimes \rho_{h,g}) \circ h^{-1}_{\mathcal{L}_g,\mathcal{L}_g} \circ {}^h(i^g_X \otimes i^g_X) \circ {}^h\big(\mathrm{id}_{X^*} \otimes \mathrm{coev}_X \otimes \mathrm{id}_X\big) \\
&= (\rho_{h,g} \otimes \rho_{h,g}) \circ h^{-1}_{\mathcal{L}_g,\mathcal{L}_g} \circ {}^h\Big((i^g_X \otimes i^g_X) \circ (\mathrm{id}_{X^*} \otimes \mathrm{coev}_X \otimes \mathrm{id}_X)\Big) \\
&= (\rho_{h,g} \otimes \rho_{h,g}) \circ h^{-1}_{\mathcal{L}_g,\mathcal{L}_g} \circ {}^h\Delta_g \circ {}^h i^g_X \ .
\end{aligned} \tag{3.73}$$

Here the first equality follows from (3.57), the second from (3.34), the third from (3.62b), the fourth again from (3.57), the fifth from the naturality of $h_{-,-}$, and the last from (3.34).

Finally, for (CCH6), we need to show

$$\varepsilon_{h\triangleright g} \circ \rho_{h,g} = {}^h\varepsilon_g \ . \tag{3.74}$$

For every $X \in \mathcal{C}_g$, we compute

$$\begin{aligned}
&\varepsilon_{h\triangleright g} \circ \rho_{h,g} \circ {}^h i^g_X = \varepsilon_{h\triangleright g} \circ i^{h\triangleright g}_{{}^h X} \circ \lambda^h_{X,X} \\
&= \mathrm{ev}_{{}^h X} \circ \lambda^h_{X,X} = {}^h\mathrm{ev}_X = {}^h\big(\varepsilon_g \circ i^g_X\big) = {}^h\varepsilon_g \circ {}^h i^g_X \ .
\end{aligned} \tag{3.75}$$

Here the first equality follows from (3.57), the second from (3.36), the third from (3.62a), the fourth again from (3.36). □

3.4. **Graded coends in the representation category of a quasi-Hopf G-coalgebra.** Let $U = \{U_g\}_{g\in G}$ be a degreewise finite-dimensional quasi-Hopf G-coalgebra over a ground field $\mathbb{K}$.

For any $g \in G$, denote by

$$(U_g)^* = \mathrm{Hom}_{\mathbb{K}}(U_g, \mathbb{K}) \tag{3.76}$$

the linear dual of U_g. We use the pairing notation

$$\langle \varphi, x\rangle := \varphi(x) \ , \qquad \varphi \in U_g^* \ , \ x \in U_g \ . \tag{3.77}$$

$(U_g)^*$ is a left U_1-module with the following coadjoint action: for $x \in U_1$, $\varphi \in U_g^*$, and $y \in U_g$, set

$$\langle x \cdot \varphi \, , \, y\rangle = \sum_{(x)} \big\langle \varphi \, , \, S_{g^{-1}}\left(x_{(1,g^{-1})}\right) \, y \, x_{(2,g)} \big\rangle \ , \tag{3.78}$$

where we use the Sweedler notation $\Delta_{g^{-1},g}(x) = \sum_{(x)} x_{(1,g^{-1})} \otimes x_{(2,g)}$. We call this U_1-module the *coadjoint representation* of U_g, and denote it by

$$\overline{(U_g)^*} \ . \tag{3.79}$$

Proposition 3.9 ([Vi00a, Lem. 3.6]). *Let $U = \{U_g\}_{g\in G}$ be a degreewise finite-dimensional quasi-Hopf G-coalgebra. Then, for every $g \in G$, the graded coend*

$$\mathcal{L}_g = \int^{M\in \mathrm{Rep}_{\mathrm{f.d.}} U_g} M^* \otimes M \tag{3.80}$$

exists and is given by

$$\mathcal{L}_g \cong \overline{(U_g)^*} \ . \tag{3.81}$$

Under this identification, the universal dinatural transformation

$$i^g_M \colon M^* \otimes M \longrightarrow \overline{(U_g)^*} \ , \qquad M \in \mathrm{Rep}_{\mathrm{f.d.}} U_g \tag{3.82}$$

is given by

$$\big\langle i^g_M(f\otimes m), y\big\rangle = \big\langle f, y\cdot m\big\rangle \ , \qquad f\in M^*, \quad m\in M, \quad y\in U_g \ . \tag{3.83}$$

For every U_1-intertwining dinatural transformation

$$\{q_M\colon M^*\otimes M \longrightarrow T\}_{M\in \mathrm{Rep}_{\mathrm{f.d.}}U_g} \ , \tag{3.84}$$

the unique U_1-module morphism

$$l\colon \overline{(U_g)^*} \longrightarrow T \tag{3.85}$$

such that

$$q_M = l\circ i^g_M \ , \qquad \forall M\in \mathrm{Rep}_{\mathrm{f.d.}}U_g \tag{3.86}$$

is explicitly given by

$$l(\varphi) = q_{U_g}(\varphi\otimes \mathbb{1}_g) \ , \qquad \varphi\in U_g^* \ , \tag{3.87}$$

where U_g is regarded as the regular left U_g-module.

Remark 3.10. The proof follows [Vi00a, Lem. 3.6], where the statement is given for Hopf G-coalgebras rather than quasi-Hopf G-coalgebras. However, the description of the coends is unchanged. The quasi-Hopf structure affects only the induced Hopf G-algebra structure on $\{\mathcal{L}_g\}_{g\in G}$, but not the underlying objects.

4. Twisted-local modules

Following [GRW26, Sec. 5], we recall the construction of a χ-crossed balanced category from twisted-local modules over a doubly central algebra object within an ambient braided category. We then explore a concrete example arising from the representation category of unrolled quantum $\mathfrak{sl}_2$ at an even root of unity (Theorem 4.10).

4.1. **Twisted-local modules.** Let $\mathcal{C}$ be a braided abelian category. Without loss of generality we assume the monoidal structure on $\mathcal{C}$ to be strict.

An *algebra* in $\mathcal{C}$ is an object $A\in\mathcal{C}$ equipped with a multiplication morphism $\mu\colon A\otimes A\to A$ and a unit morphism $\eta\colon \mathbb{1}\to A$ satisfying

$$\mu\circ(\mu\otimes \mathrm{id}_A) = \mu\circ(\mathrm{id}_A\otimes\mu) \ , \qquad \mu\circ(\eta\otimes\mathrm{id}_A) = \mathrm{id}_A = \mu\circ(\mathrm{id}_A\otimes\eta) \ . \tag{4.1}$$

The algebra A is called *commutative* if

$$\mu = \mu\circ C_{A,A} \ , \tag{4.2}$$

where $C_{A,A}$ denotes the braiding of A with itself. For a given algebra $A\in\mathcal{C}$, a *left A-module* is an object $M\in\mathcal{C}$ with a morphism $\rho_M : A\otimes M\to M$ so that,

$$\rho_M\circ(\mathrm{id}_A\otimes\rho_M) = \rho_M\circ(\mu\otimes\mathrm{id}_M) \ , \qquad \mathrm{id}_M = \rho_M\circ(\eta\otimes\mathrm{id}_M) \ . \tag{4.3}$$

Let A be a commutative algebra in $\mathcal{C}$ and let ${}_A\mathcal{C}$ denote the category of left A-modules and A-intertwiners. If the functor $A\otimes - : \mathcal{C}\to\mathcal{C}$ is right exact, then ${}_A\mathcal{C}$ is abelian [EGNO15, Sec. 7.8]. Moreover, the category ${}_A\mathcal{C}$ is monoidal with tensor product given by the relative tensor product $\otimes_A$ [EGNO15, Sec. 8.8]. Explicitly, for any $M,N\in{}_A\mathcal{C}$, this is defined via the cokernel

$$M\otimes_A N = \mathrm{Coker}\big(\rho_M\otimes\mathrm{id}_N - (\mathrm{id}_M\otimes\rho_N)\circ(C_{A,M}\otimes\mathrm{id}_N)\big) \ , \tag{4.4}$$

with the corresponding quotient morphism denoted by $\pi_{M,N}\colon M\otimes N\to M\otimes_A N$.

For later use, we also give the unitors and the associator in ${}_A\mathcal{C}$:

- The left unitor $l_M\colon A\otimes_A M\to M$ is the unique morphism making the following diagram commute:

$$\begin{array}{ccc} A\otimes M & \xrightarrow{\ \pi_{A,M}\ } & A\otimes_A M \\ {\scriptstyle \rho_M}\big\downarrow & \swarrow_{\exists!\, l_M} & \\ M & & \end{array} \tag{4.5}$$

- The right unitor $r_M\colon M\otimes_A A\to M$ is the unique morphism making the following diagram commute:

$$\begin{array}{ccc} M\otimes A & \xrightarrow{\pi_{M,A}} & M\otimes_A A \\ \big\downarrow{\scriptstyle C_{A,M}^{-1}} & & \exists!\, r_M \big\downarrow \\ A\otimes M & \xrightarrow{\rho_M} & M \end{array} \tag{4.6}$$

- the associator $\alpha_{M,N,L}\colon M\otimes_A(N\otimes_A L)\to(M\otimes_A N)\otimes_A L$ is the unique morphism making the following diagram commute:

$$\begin{array}{ccc} M\otimes N\otimes L & \xrightarrow{\pi_{M,N\otimes_A L}\circ(\mathrm{id}_M\otimes\pi_{N,L})} & M\otimes_A(N\otimes_A L) \\ \| & & \exists!\,\alpha_{M,N,L}\big\downarrow \\ M\otimes N\otimes L & \xrightarrow{\pi_{M\otimes_A N,L}\circ(\pi_{M,N}\otimes\mathrm{id}_L)} & (M\otimes_A N)\otimes_A L \end{array} \tag{4.7}$$

Now assume that A admits a half-braiding $\widetilde{\sigma}$ such that the pair $\widetilde{A}=(A,\widetilde{\sigma})$ is a commutative algebra in the Drinfeld centre $Z(\mathcal{C})$. Let $\chi\colon G\to H$ be a crossed module. Let $\varphi\colon G\to \mathrm{Aut}_{\mathcal{C}}(A)$ be a G-action on A (as an object) such that for each $g\in G$, φ_g lifts to an automorphism of $\widetilde{A}$ in $Z(\mathcal{C})$. For any $g\in G$, an A-module M is called *g-local (w.r.t. φ and $\widetilde{A}$)* if

$$\lambda_M=\widetilde{\lambda}_M\circ(\mathrm{id}_M\otimes\varphi_g)\ , \tag{4.8}$$

where $\lambda_M=\rho_M\circ C_{A,M}^{-1}$ and $\widetilde{\lambda}_M=\rho_M\circ\widetilde{\sigma}_M^{-1}$. For each $g\in G$, we denote by ${}_A\mathcal{C}_g$ the full subcategory in ${}_A\mathcal{C}$ consisting of g-local modules. Let $\upsilon\colon H\to\mathrm{Aut}_{\mathcal{C}}(A)$ be an H-action on A by algebra automorphisms, such that each υ_h lifts to an automorphism of $\widetilde{A}$ in $Z(\mathcal{C})$. We further assume:

- the following diagram commutes:

$$\begin{array}{ccc} H\times G & \xrightarrow{\ \triangleright\ } & G \\ \big\downarrow{\scriptstyle \upsilon\times\varphi} & & \varphi\big\downarrow \\ \mathrm{Aut}_{\mathcal{C}}(A)\times\mathrm{Aut}_{\mathcal{C}}(A) & \xrightarrow{\ \lambda\ } & \mathrm{Aut}_{\mathcal{C}}(A) \end{array} \tag{4.9}$$

  where $\lambda(f,f')=ff'f^{-1}$ for any $(f,f')\in\mathrm{Aut}_{\mathcal{C}}(A)$;
- for any $g\in G$, $M\in{}_A\mathcal{C}_g$,

$$\rho_M\circ C_{M,A}\circ C_{A,M}=\rho_M\circ(\upsilon_{\chi(g)}^{-1}\otimes\mathrm{id}_M)\ . \tag{4.10}$$

Proposition 4.1 ([GRW26, Thm 5.12 & Prop. 5.13])**.** *The category*

$${}_A\mathcal{C}^{\mathrm{gr}}:=\bigoplus_{g\in G}{}_A\mathcal{C}_g \tag{4.11}$$

admits a structure of χ-crossed braided category. The H-crossed action on ${}_A\mathcal{C}^{\mathrm{gr}}$ is given by the strictly monoidal functor

$$\psi:\underline{H}\to\mathrm{Aut}_{\otimes}({}_A\mathcal{C})\ ,\qquad h\mapsto {}^h(-) \tag{4.12}$$

sending each group element h to a monoidal autoequivalence ${}^h(-)$ as follows: for $M,N\in{}_A\mathcal{C}$ and a morphism $f\colon M\to N$,

- *${}^hM:=M$ as the same object in $\mathcal{C}$, but with the left A-action twisted by υ_h^{-1}:*

$$\rho_{{}^hM}=\rho_M\circ(\upsilon_h^{-1}\otimes\mathrm{id}_M)\ , \tag{4.13}$$

- *${}^hf:=f$,*

and the monoidal structure morphisms of ${}^h(-)$ are given by:

- *the isomorphism* $h_{M,N}\colon {}^hM \otimes_A {}^hN \to {}^h(M \otimes_A N)$ *is the unique isomorphism making the following diagram in* $\mathcal{C}$ *commute:*

$$\begin{array}{ccc} {}^hM \otimes {}^hN & \xrightarrow{\pi_{{}^hM,{}^hN}} & {}^hM \otimes_A {}^hN \\ \downarrow{\scriptstyle \mathrm{id}_{M\otimes N}} & & \exists!\, h_{M,N} \downarrow \\ {}^h(M \otimes N) & \xrightarrow{{}^h\pi_{M,N}} & {}^h(M \otimes_A N) \end{array} \tag{4.14}$$

- *the isomorphism* $h_A\colon A \to {}^hA$ *is given by*

$$\upsilon_h^{-1}\colon A \to A\ . \tag{4.15}$$

For $M, N \in {}_A\mathcal{C}^{\mathrm{gr}}$ *and* M *being homogeneous of degree* g*, the* χ*-crossed braiding*

$$\widetilde{C}_{M,N}\colon M \otimes_A N \to {}^{\chi_g}N \otimes_A M \tag{4.16}$$

is the unique isomorphism making the following diagram in $\mathcal{C}$ *commute:*

$$\begin{array}{ccc} M \otimes N & \xrightarrow{\pi_{M,N}} & M \otimes_A N \\ \downarrow{\scriptstyle C_{M,N}} & & \exists!\, \widetilde{C}_{M,N} \downarrow \\ N \otimes M & \xrightarrow{\pi_{{}^{\chi_g}N,M}} & {}^{\chi_g}N \otimes_A M \end{array} \tag{4.17}$$

If $\mathcal{C}$ *is further equipped with a balancing isomorphism* ϑ *such that* $\vartheta_A = \mathrm{id}_A$*, then* ${}_A\mathcal{C}^{\mathrm{gr}}$ *is a* χ*-crossed balanced category whose* χ*-crossed balancing* $\widetilde{\vartheta}$ *has underlying morphism* ϑ*.*

Consider the induction functor

$$A \otimes -\colon \mathcal{C} \longrightarrow {}_A\mathcal{C}\ , \qquad X \mapsto A \otimes X\ , \qquad f \mapsto \mathrm{id}_A \otimes f\ , \tag{4.18}$$

where the A-action on the free module $A \otimes M$ is given by algebra multiplication. Recall that the monoidal structure on ${}_A\mathcal{C}$ is given as in (4.4). The induction functor is monoidal [FS01, Prop. 5.11]. For any $X, Y \in \mathcal{C}$ the monoidal structure morphism

$$a_{X,Y}\colon (A \otimes X) \otimes_A (A \otimes Y) \to A \otimes X \otimes Y \tag{4.19}$$

is induced by:

$$\begin{array}{ccc} (A \otimes X) \otimes (A \otimes Y) & \xrightarrow{\pi_{A\otimes X, A\otimes Y}} & (A \otimes X) \otimes_A (A \otimes Y) \\ & {\scriptstyle (\mu\otimes \mathrm{id}_{X\otimes Y})(\mathrm{id}_A \otimes C^{-1}_{A,X} \otimes \mathrm{id}_Y)} \searrow & \downarrow{\scriptstyle a_{X,Y}} \\ & & A \otimes X \otimes Y \end{array} \tag{4.20}$$

with the inverse given by

$$A \otimes X \otimes Y \xrightarrow{\mathrm{id}_{A\otimes X} \otimes \eta \otimes \mathrm{id}_Y} A \otimes X \otimes A \otimes Y \xrightarrow{\pi_{A\otimes X, A\otimes Y}} (A \otimes X) \otimes_A (A \otimes Y)\ . \tag{4.21}$$

A left A-module M in ${}_A\mathcal{C}$ is called *finitely generated* if there exists a dualisable object $X \in \mathcal{C}$, together with an epimorphism

$$A \otimes X \twoheadrightarrow M \tag{4.22}$$

in ${}_A\mathcal{C}$. We denote the full subcategory of finitely generated A-modules by ${}_A\mathcal{C}^{\mathrm{f.g.}}$.

Remark 4.2. ${}_A\mathcal{C}^{\mathrm{f.g.}}$ is in general not abelian. For instance, let $\mathcal{C}$ be the category of $\mathbb{K}$-vector spaces and let $A = \mathbb{K}[X_1, X_2, ...]$ be the polynomial ring with countably many generators. A is finitely generated as an A-module, but the kernel of

$$f : A \to \mathbb{K}\ , \qquad X_i \mapsto 0\ ,\ 1 \mapsto 1 \tag{4.23}$$

is not a finitely generated ideal and hence not a finitely generated module over A.

Lemma 4.3 ([CGR17, Lemma 2.8])**.** *If the tensor product in* $\mathcal{C}$ *is right exact, then* ${}_A\mathcal{C}^{\mathrm{f.g.}}$ *is a monoidal subcategory of* ${}_A\mathcal{C}$*.*

Corollary 4.4. *If the category ${}_A\mathcal{C}^{\mathrm{gr}}$ admits a χ-crossed braided/balanced structure, then so does its full subcategory of finitely generated modules.*

Remark 4.5. If $\mathcal{C}$ is rigid, then ${}_A\mathcal{C} = {}_A\mathcal{C}^{\mathrm{f.g.}}$ because every $M \in {}_A\mathcal{C}$ is a retract of the free module $A \otimes M$; however, this does not guarantee the rigidity of ${}_A\mathcal{C}$. For example, let $\mathcal{C}$ be the category of finite-dimensional $\mathbb{K}$-vector spaces and let $A = \mathbb{K}[X]/(X^d)$ for some positive integer d. Then the one-dimensional A-module $\mathbb{K}$, on which X acts as 0, admits no dual. If $\mathcal{C}$ is rigid and A is separable, we showed in [GRW26, Prop. 5.16] that ${}_A\mathcal{C}$ is rigid, and that ${}_A\mathcal{C}^{\mathrm{gr}}$ as defined in (4.11) is a rigid G-graded monoidal category. However, these assumptions fail for our main example in the next subsection, where the ambient category is not rigid and the algebra is not separable.

4.2. **An example from unrolled quantum $\mathfrak{sl}_2$.** Fix a primitive $2p$-th root of unity $q = e^{\frac{\pi i}{p}}$ in $\mathbb{C}$, where $p \geq 2$ is an integer. The *unrolled quantum* $\mathfrak{sl}_2$, denoted by $\overline{U}_q^H\mathfrak{sl}_2$, is the Hopf algebra generated by

$$\{E, F, H, K^{\pm 1}\} \ , \tag{4.24}$$

with the relations

$$\begin{gathered} KEK^{-1} = q^2E \ , \qquad KFK^{-1} = q^{-2}F \ , \qquad [E,F] = \frac{K-K^{-1}}{q-q^{-1}} \ , \\ [H,E] = 2E \ , \qquad [H,F] = -2F \ , \qquad [H,K] = 0 \ , \\ E^p = F^p = 0 \ , \end{gathered} \tag{4.25}$$

where the coproduct, counit, and antipode are defined via

$$\begin{aligned} \Delta(E) &= E \otimes K + 1 \otimes E \ , & \varepsilon(E) &= 0 \ , & S(E) &= -EK^{-1} \ , \\ \Delta(F) &= K^{-1} \otimes F + F \otimes 1 \ , & \varepsilon(F) &= 0 \ , & S(F) &= -KF \ , \\ \Delta(K^{\pm 1}) &= K^{\pm 1} \otimes K^{\pm 1} \ , & \varepsilon(K^{\pm 1}) &= 1 \ , & S(K^{\pm 1}) &= K^{\mp 1} \ , \\ \Delta(H) &= H \otimes 1 + 1 \otimes H \ , & \varepsilon(H) &= 0 \ , & S(H) &= -H \ . \end{aligned} \tag{4.26}$$

For more details about the unrolled quantum $\mathfrak{sl}_2$, see e.g. [CGP14].

Definition 4.6. A $\overline{U}_q^H\mathfrak{sl}_2$-module is of *weight type* if the H-action on it is diagonalisable, and K acts by $q^H := e^{\pi i H/p}$.

To simplify our notation, we denote the category of finite-dimensional weight modules of $\overline{U}_q^H\mathfrak{sl}_2$ by

$$\mathcal{C}^{\mathrm{f.d.}} := \mathrm{Rep}^{\mathrm{f.d.}}_{\mathrm{wt}}\overline{U}_q^H\mathfrak{sl}_2 \ . \tag{4.27}$$

$\mathcal{C}^{\mathrm{f.d.}}$ is an abelian ribbon category [CGP14, Sec. 2.2]: using notations in (1.4), for any $M, N \in \mathcal{C}$,

- the braiding is given by
$$C_{M,N} = \tau \circ (R\cdot) \ , \tag{4.28}$$
where τ is permutation, and
$$R = q^{\frac{H\otimes H}{2}} \sum_{n=0}^{p-1} \frac{(q-q^{-1})^n}{[n]!} q^{\frac{1}{2}n(n-1)} E^n \otimes F^n \ ; \tag{4.29}$$
- the ribbon twist is given by
$$\vartheta_M = (v^{-1}\cdot) \ , \tag{4.30}$$
where
$$v = K^{p-1} \sum_{n=0}^{p-1} \frac{(q-q^{-1})^n}{[n]!} q^{\frac{1}{2}n(n-1)} S(F^n) q^{-\frac{1}{2}H^2} E^n \ . \tag{4.31}$$

Note that $q^{\frac{H\otimes H}{2}}$ and $q^{-\frac{1}{2}H^2}$ are not elements in $\overline{U}_q^H\mathfrak{sl}_2$. They are defined via their actions on H-eigenvectors. For example, if $Hx = \mu x$, $Hy = \nu y$, then $q^{\frac{H\otimes H}{2}}(x\otimes y) = q^{\mu\nu/2}x\otimes y$, $q^{-\frac{1}{2}H^2}x = q^{-\frac{\mu^2}{2}}x$.

Denote by

$$\mathcal{C} \coloneqq \mathrm{Ind}(\mathcal{C}^{\mathrm{f.d.}}) \tag{4.32}$$

the *Ind-completion* of $\mathcal{C}^{\mathrm{f.d.}}$, which can be identified with the category of arbitrary-dimensional weight modules. Indeed, every weight module is the filtered sum of its finite-dimensional weight submodules, and conversely, filtered colimits of finite-dimensional weight modules are again weight modules.

Remark 4.7. The compact objects of $\mathcal{C}$ are precisely the objects of $\mathcal{C}^{\mathrm{f.d.}}$. Since $\mathcal{C}^{\mathrm{f.d.}}$ is rigid, every compact object of $\mathcal{C}$ is dualisable, hence $\mathcal{C}$ is compact-rigid [BJS18, Rem. 1.6].

For each $k\in\mathbb{Z}$, let $\mathbb{C}_k$ denote the one-dimensional $\overline{U}_q^H\mathfrak{sl}2$-module on which H acts by the scalar $2pk$ and both E and F act by 0. We fix a basis vector $\mathbf{1}_k\in\mathbb{C}_k$. Each $\mathbb{C}_k$ is invertible in $\mathcal{C}$ with inverse $\mathbb{C}_{-k}$, and for any $k,l\in\mathbb{Z}$,

$$\mathbb{C}_k\otimes\mathbb{C}_l \cong \mathbb{C}_{k+l}\ . \tag{4.33}$$

Consider the algebra object

$$A = \bigoplus_{k\in\mathbb{Z}}\mathbb{C}_k\ , \tag{4.34}$$

in $\mathcal{C}$, with multiplication and unit given by

$$\begin{aligned} \mu\colon A\otimes A\to A\ , \qquad & \eta\colon \mathbb{C}\to A\ , \\ \mathbf{1}_k\otimes\mathbf{1}_l\mapsto\mathbf{1}_{k+l}\ , \qquad & 1\mapsto\mathbf{1}_0\ . \end{aligned} \tag{4.35}$$

Thus A is a *simple-current algebra* in the sense of [SY26, Def. 2.8]. Moreover, equation (4.29) implies that the self-braiding of A is just permutation. It follows that A is commutative. We may therefore consider the monoidal category ${}_A\mathcal{C}$ of left A-modules, with tensor product given by the relative tensor product $\otimes_A$.

Proposition 4.8 ([SY26, Thm. C])**.** *The category ${}_A\mathcal{C}^{\mathrm{f.g.}}$ of finitely generated A-modules in $\mathcal{C}$ is a tensor category. More precisely, it is a locally finite $\mathbb{C}$-linear abelian rigid monoidal category, and its tensor unit, the regular module A, is simple.*

Proposition 4.9. *An A-module is finitely generated if and only if all of its H-eigenspaces are finite-dimensional.*

Proof. Suppose that M is finitely generated, then there exists a finite-dimensional $\overline{U}_q^H\mathfrak{sl}_2$-module X with a surjection

$$f\colon A\otimes X\twoheadrightarrow M\ . \tag{4.36}$$

Since f intertwines with H-action, it does not mix H-eigenspaces corresponding to different H-eigenvalues. Therefore, for each H-eigenspace of M, there is a finite-dimensional H-eigenspace of $A\otimes X$ with the same eigenvalue mapped onto it by f, which implies that every H-eigenspace of M should be finite-dimensional as well.

Conversely, without loss of generality we may assume that M is homogeneous of degree α and that all of its H-eigenspaces are finite-dimensional. Let $\overline{\alpha}\in\mathbb{C}$ be an arbitrary representative of α. Then we can write $M=\bigoplus_{k\in\mathbb{Z}}M_k$, where M_k is the finite-dimensional H-eigenspace of eigenvalue $2k+\overline{\alpha}$. Let $N=\bigoplus_{k=0}^{p-1}M_k\subset M$. It's clear that N is finite-dimensional and we claim that the A-intertwiner $\rho_{M_{|N}}\colon A\otimes N\to M$ is bijecive. Indeed, for any $l,k\in\mathbb{Z}$, the map

$$\rho_{\mathbf{1}_k}\colon M_l\to M_{l+2pk}\ , \qquad m\mapsto\rho_M(\mathbf{1}_k\otimes m) \tag{4.37}$$

is an isomorphism. Since H- and K-actions are diagonalisable on N, and both E- and F-actions are nilpotent, the $\overline{U}_q^H\mathfrak{sl}_2$-module generated by N is also finite-dimensional, which we denote by $\langle N\rangle$. Then $\rho_{M_{|\langle N\rangle}}: A\otimes\langle N\rangle\to M$ is a surjective morphism in ${}_A\mathcal{C}$. □

Denote by $\widetilde{A}$ the commutative algebra in $Z(\mathcal{C})$ given by A together with the permutation half-braiding

$$\tau_X: X\otimes A\to A\otimes X\ ,\qquad x\otimes a\mapsto a\otimes x\ . \tag{4.38}$$

This is indeed an intertwiner because K acts on A by identity. Consider the crossed module $\pi:\mathbb{C}/2\mathbb{Z}\to\mathbb{C}/\mathbb{Z}$, where π is the projection map, and $\mathbb{C}/\mathbb{Z}$ acts on $\mathbb{C}/2\mathbb{Z}$ trivially. We define

- a $\mathbb{C}/2\mathbb{Z}$-action on A:

$$\varphi:\mathbb{C}/2\mathbb{Z}\to\mathrm{Aut}_{\mathcal{C}}(A)\ ,\qquad \alpha\mapsto\varphi_\alpha\ , \tag{4.39}$$

  where for each $\alpha\in\mathbb{C}/2\mathbb{Z}$, φ_α is given by

$$\varphi_\alpha: A\to A\ ,\qquad \mathbf{1}_k\mapsto q^{-pk\alpha}\mathbf{1}_k\ ; \tag{4.40}$$

- a $\mathbb{C}/\mathbb{Z}$-action on the algebra A:

$$\upsilon:\mathbb{C}/\mathbb{Z}\to\mathrm{Aut}_{\mathcal{C}}(A)\ ,\qquad h\mapsto\upsilon_h\ , \tag{4.41}$$

  where for each $h\in\mathbb{C}/\mathbb{Z}$, υ_h is given by

$$\upsilon_h: A\to A\ ,\qquad \mathbf{1}_k\mapsto q^{-2pkh}\mathbf{1}_k\ . \tag{4.42}$$

Note that $q^{-pk\alpha}$ and q^{-2pkh} are well-defined independently of the choice of representatives, so this abuse of notation is harmless. It is immediate to see that both φ and υ are actually algebra automorphisms, and both lift to automorphisms of $\widetilde{A}$ in $Z(\mathcal{C})$.

In this setting, equations (4.28) and (4.29) imply that

$$\begin{gathered}\textit{an } A\textit{-module is } \alpha\textit{-local for some } \alpha\in\mathbb{C}/2\mathbb{Z} \textit{ in the sense of } (4.8)\\ \textit{if and only if all of its } H\textit{-eigenvalues are representatives of } \alpha \textit{ in } \mathbb{C}.\end{gathered} \tag{4.43}$$

Furthermore, direct computation verifies that conditions (4.9) and (4.10) hold.

Theorem 4.10. *${}_A\mathcal{C}^{\mathrm{f.g.}}$ is a π-crossed ribbon category.*

Proof. As we see above, the data $(A,\tau,\varphi,\upsilon)$ satisfy all the conditions for Proposition 4.1. Moreover, since $\mathcal{C}$ is the category of weight modules, we have

$${}_A\mathcal{C}={}_A\mathcal{C}^{\mathrm{gr}}=\bigoplus_{\alpha\in\mathbb{C}/2\mathbb{Z}}{}_A\mathcal{C}_\alpha\ . \tag{4.44}$$

Therefore, ${}_A\mathcal{C}$ is a π-crossed balanced category. By Proposition 4.8, ${}_A\mathcal{C}^{\mathrm{f.g.}}$ is a tensor subcategory of ${}_A\mathcal{C}$, so it inherits the π-crossed balanced structure. By [GRW26, Prop. 5.16], the rigidity of ${}_A\mathcal{C}^{\mathrm{f.g.}}$ is compatible with the G-grading, in the sense that the dual of a g-local module is g^{-1}-local.

It remains to verify condition (RB1). For any finitely generated g-local module M, this reduces to the commutativity of the following diagram:

$$\begin{array}{ccc}\chi_g(M^\vee) & \xrightarrow{\chi_g\widetilde{\vartheta}_{M^\vee}} & M^\vee\\ {\scriptstyle\nu_{\chi_g,M}}\downarrow & \nearrow_{(\widetilde{\vartheta}_M)^\vee} & \\ (\chi_g M)^\vee & & \end{array} \tag{4.45}$$

where $M^\vee$ denotes the left dual of M, and $\nu_{\chi_g,M}$ is as defined in Remark 2.2. In our setting, both $\mathcal{C}$ and ${}_A\mathcal{C}$ admit right *internal Homs*, which we denote by $[-,-]$ and $\underline{\mathrm{Hom}}_A(-,-)$, respectively. Since both $-\otimes_A M^\vee$ and $\underline{\mathrm{Hom}}_A(M,-)$ are right adjoint to $-\otimes_A M$, we may then set

$$M^\vee=\underline{\mathrm{Hom}}_A(M,A)\ . \tag{4.46}$$

Our strategy is to apply the theory in [SY26] to identify $M^\vee$ with $[M,\mathbb{1}]$, thereby simplifying the morphisms in (4.45).

Recall that for any objects $X,Y,Z\in\mathcal{C}$ and $M,N,L\in{}_A\mathcal{C}$, the internal homs are characterised by adjunctions:

$$\mathrm{Hom}_{\mathcal{C}}(X\otimes Y,Z)\cong\mathrm{Hom}_{\mathcal{C}}(X,[Y,Z])\ , \tag{4.47a}$$

$$\mathrm{Hom}_A(M\otimes_A N,L)\cong\mathrm{Hom}_A\big(M,\underline{\mathrm{Hom}}_A(N,L)\big)\ . \tag{4.47b}$$

Denote by

$$\mathrm{ev}_X\colon[X,\mathbb{1}]\otimes X\to\mathbb{1}\ ,\qquad \mathrm{ev}^A_M\colon\underline{\mathrm{Hom}}_A(M,A)\otimes_A M\to A \tag{4.48}$$

the corresponding counits, evaluated at the tensor units $\mathbb{1}$ and A, respectively.

In our example, the algebra A is *Frobenius* in the sense of [SY26, Def. 3.9], with Frobenius form

$$\delta\colon A=\bigoplus_{k\in\mathbb{Z}}\mathbb{C}_k\longrightarrow\mathbb{1} \tag{4.49}$$

given by the projection onto the summand $\mathbb{C}_0=\mathbb{1}$. This yields an isomorphism

$$\delta_A\colon A\longrightarrow[A,\mathbb{1}] \tag{4.50}$$

given by the adjoint of $\delta\circ\mu\colon A\otimes A\to\mathbb{1}$ [SY26, Thm. 4.15]. By a left-module analogue of [SY26, Lem. 3.11], this induces an isomorphism

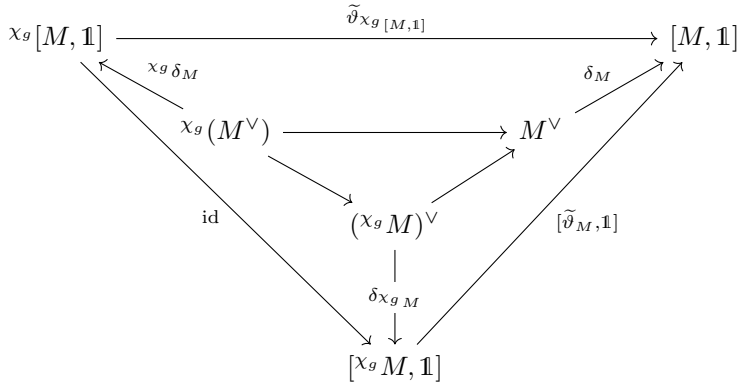
$$\delta_M\colon M^\vee=\underline{\mathrm{Hom}}_A(M,A)\xrightarrow{\sim}[M,\mathbb{1}]\ , \tag{4.51}$$

which is contravariantly natural in M, and whose adjoint is the composite

$$M^\vee\otimes M\xrightarrow{\pi_{M^\vee,M}}M^\vee\otimes_A M\xrightarrow{\mathrm{ev}^A_M}A\xrightarrow{\delta}\mathbb{1}\ . \tag{4.52}$$

Here $[M,\mathbb{1}]$ is equipped with the A-module structure induced by the left-module analogue of [SY26, Lem. 3.11], with respect to which δ_M is A-linear.

To prove the commutativity of (4.45), we establish the following commutative diagram, where the omitted labels on the inner triangle are the same as those in (4.45):



(4.53)

It suffices to show it in the underlying category $\mathcal{C}$. By construction, the underlying morphism of $\widetilde{\vartheta}$ is just ϑ in $\mathcal{C}$. Hence, by [SY26, Lem. 3.14],

$$[\widetilde{\vartheta}_M,\mathbb{1}]=\widetilde{\vartheta}_{\chi_g[M,\mathbb{1}]} \tag{4.54}$$

as underlying morphisms in $\mathcal{C}$. Thus it remains to verify the commutativity of the three outer regions. The upper region commutes by (BL1), applied to the A-module isomorphism δ_M. The

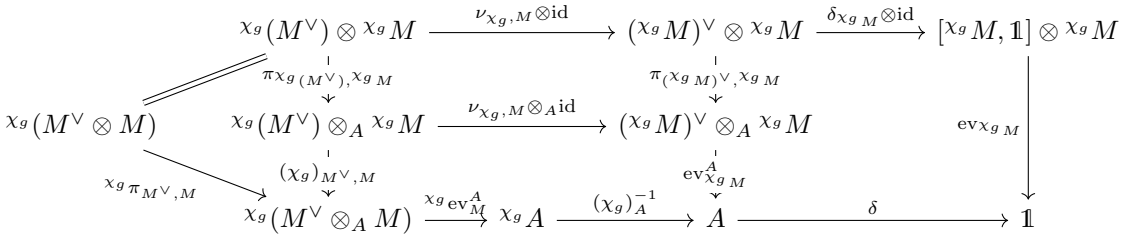
right-hand region commutes by the naturality of $\{\delta_M\}_{M\in {}_A\mathcal{C}}$, applied to the A-module morphism $\widetilde{\vartheta}_M$. It is left to check the left-hand region:

$$\delta_{{}^{\chi_g}M}\circ \nu_{\chi_g,M} = {}^{\chi_g}\delta_M \ . \tag{4.55}$$

By adjunction

$$\mathrm{Hom}_{\mathcal{C}}\big({}^{\chi_g}(M^\vee),\, [{}^{\chi_g}M, \mathbb{1}]\big) \cong \mathrm{Hom}_{\mathcal{C}}\big({}^{\chi_g}(M^\vee)\otimes {}^{\chi_g}M,\, \mathbb{1}\big) \ , \tag{4.56}$$

we can alternatively compare the adjoints of the two sides. For the left-hand side, consider the following commutative diagram:



where the left triangle commutes by the defining property of the structure morphism $(\chi_g)_{M^\vee,M}$ as in (4.14); the upper rectangle uses the definition of $\otimes_A$ for morphisms; the lower rectangle is the defining characterisation of $\nu_{\chi_g,M}$; and the right rectangle follows from (4.52). Consequently, the adjoint of the left-hand side of (4.55) is the composite

$$\delta\circ \upsilon_{\chi_g}\circ \mathrm{ev}^A_M\circ \pi_{M^\vee,M} \ , \tag{4.57}$$

where we use $(\chi_g)_A = \upsilon_{\chi_g}^{-1}$ as in (4.15). Similarly, the adjoint of the right-hand side of (4.55) is given by

$$\delta\circ \mathrm{ev}^A_M\circ \pi_{M^\vee,M} \ . \tag{4.58}$$

These two agree because $\delta\circ\upsilon_h = \delta$ for all $h\in H$. Indeed, υ_h preserves each summand $\mathbb{C}_k$ and acts trivially on $\mathbb{C}_0$, while δ vanishes on $\mathbb{C}_k$ for $k\neq 0$.

□

Remark 4.11. In [CGR17] the authors considered finitely generated local modules of A in $\mathcal{C}$, which correspond to ${}_A\mathcal{C}_0^{\mathrm{f.g.}}\oplus {}_A\mathcal{C}_1^{\mathrm{f.g.}}$ in our setting.

5. π-Crossed ribbon quasi-Hopf algebras from quantum $\mathfrak{sl}_2$

As an explicit example, we construct in this section a non-semisimple $(\pi : \mathbb{C}/2\mathbb{Z}\to\mathbb{C}/\mathbb{Z})$-crossed ribbon quasi-Hopf algebra ${}^{\sigma'}\overline{\mathfrak{U}}^{\sigma}$, whose category of finite-dimensional representations is π-crossed ribbon equivalent to ${}_A\mathcal{C}^{\mathrm{f.g.}}$ (Theorems 5.6, 5.9, 5.10). Each graded component of ${}^{\sigma'}\overline{\mathfrak{U}}^{\sigma}$ is finite-dimensional, hence each homogeneous component of its category of finite-dimensional representations is a finite abelian category, in contrast to the category of finite-dimensional weight modules of the unrolled quantum $\mathfrak{sl}_2$.

5.1. **A family of algebras.** Consider the crossed module $\pi\colon \mathbb{C}/2\mathbb{Z}\to\mathbb{C}/\mathbb{Z}$ where π is the projection map and $\mathbb{C}/\mathbb{Z}$ acts on $\mathbb{C}/2\mathbb{Z}$ trivially. Fix a primitive $2p$th root of unity $q = e^{\frac{\pi i}{p}}$, where $p\geq 2$ is an integer. Recall the notations in (1.4). For every $\alpha\in\mathbb{C}/2\mathbb{Z}$, we define an algebra $\overline{\mathfrak{U}}_\alpha$ generated by $\{E,F,K^{\pm 1}\}$ with the relations:

$$\begin{aligned} E^p = F^p = K^p - q^{p\overline{\alpha}} = 0 \ , \qquad KEK^{-1} = q^2E \ , \\ KFK^{-1} = q^{-2}F \ , \qquad [E,F] = \frac{K-K^{-1}}{q-q^{-1}} \ , \end{aligned} \tag{5.1}$$

where $\overline{\alpha}$ is an arbitrary representative for α in $\mathbb{C}$.

Let $\mathcal{D}_\alpha$ denote the category of finite-dimensional representations of $\overline{\mathfrak{U}}_\alpha$. Note that for each $\alpha \in \mathbb{C}/2\mathbb{Z}$, the action of K on any $\overline{\mathfrak{U}}_\alpha$-module is diagonalisable because K^p acts as a nonzero scalar. The direct sum

$$\mathcal{D} = \bigoplus_{\alpha\in\mathbb{C}/2\mathbb{Z}} \mathcal{D}_\alpha \tag{5.2}$$

is a $\mathbb{C}/2\mathbb{Z}$-graded $\mathbb{C}$-linear abelian category. Our goal is to equip

$$\overline{\mathfrak{U}} = \{\overline{\mathfrak{U}}_\alpha\}_{\alpha\in\mathbb{C}/2\mathbb{Z}} \tag{5.3}$$

with the structure of a π-crossed ribbon quasi-Hopf algebra such that

$$\mathcal{D} \cong {}_A\mathcal{C}^{\mathrm{f.g.}} \tag{5.4}$$

as π-crossed ribbon categories, where ${}_A\mathcal{C}^{\mathrm{f.g.}}$ is as in Theorem 4.10.

Let $\pi_1\colon \mathbb{C}/2p\mathbb{Z} \to \mathbb{C}/2\mathbb{Z}$ be the projection homomorphism. For each $a \in \mathbb{C}/2p\mathbb{Z}$ such that $\pi_1(a) = \alpha$, there is an idempotent $e_a \in \overline{\mathfrak{U}}_\alpha$ defined via

$$e_a = \frac{1}{p}\sum_{l=0}^{p-1} q^{-al}K^l \ . \tag{5.5}$$

Note that q^{-al} makes sense since its value does not depend on the choice of representatives.

Lemma 5.1. *For every $\alpha \in \mathbb{C}/2\mathbb{Z}$, $0 \le l \le p-1$, we have $K^l = \sum_{\pi_1(a)=\alpha} q^{al}e_a$ in $\overline{\mathfrak{U}}_\alpha$.*

Proof.

$$\sum_{\pi_1(a)=\alpha} q^{al}e_a = \frac{1}{p}\sum_a q^{al}(\sum_{i=0}^{p-1} q^{-ai}K^i) = \frac{1}{p}\sum_i K^i(\sum_a q^{a(l-i)}) = \sum_i K^i\delta_{l,i} = K^l \ .$$

□

Lemma 5.2. *In each $\overline{\mathfrak{U}}_\alpha$, the idempotents defined above satisfy:*

$$\begin{aligned} Ke_a = q^a e_a = e_a K \ , \qquad Ee_a = e_{a+2}E \ , \qquad Fe_a = e_{a-2}F \ , \\ K^{-1}e_a = q^{-a}e_a = e_aK^{-1} \ , \qquad e_ae_b = \delta_{a,b}e_a \ , \qquad \sum_{\pi_1(a)=\alpha} e_a = \mathbb{1}_\alpha \ . \end{aligned} \tag{5.6}$$

Proof. The proof is by straightforward computation. Here we just show the last two:

$$e_ae_b = \frac{1}{p}\sum_{l=0}^{p-1} q^{-al}K^le_b = (\frac{1}{p}\sum_{l=0}^{p-1} q^{(b-a)l})e_b = \delta_{a,b}e_b \ , \tag{5.7}$$

$$\sum_{\pi_1(a)=\alpha} e_a = \frac{1}{p}\sum_{a,l} q^{-al}K^l = \frac{1}{p}\sum_l K^l(\sum_a q^{-al}) = \sum_l K^l\delta_{l,0} = \mathbb{1}_\alpha \ . \tag{5.8}$$

□

Let $\pi_2\colon \mathbb{C} \to \mathbb{C}/2p\mathbb{Z}$ be the projection. Fix a set-theoretic section $\sigma' : \mathbb{C}/2p\mathbb{Z} \to \mathbb{C}$ of π_2 such that $\sigma'(0) = 0$. Recall that $\mathcal{C} = \mathrm{Ind}(\mathrm{Rep}^{\mathrm{f.d.}}_{\mathrm{wt}}\overline{U}_q^H\mathfrak{sl}_2)$ as in (4.32), and the algebra $A \in \mathcal{C}$ is given as in (4.34). Following the idea in [CGR17, Sec. 6.2], we define a $\mathbb{C}$-linear functor

$$F_{\sigma'}\colon \mathcal{D} \to {}_A\mathcal{C}^{\mathrm{f.g.}} \ , \qquad M \mapsto A\underline{\otimes}M \ , \qquad f \mapsto \mathrm{id}_A\underline{\otimes}f \ , \tag{5.9}$$

where

- $\underline{\otimes}$ denotes the tensor product for underlying vector spaces instead of the tensor product in $\mathcal{C}$;

- for any $\alpha \in \mathbb{C}/2\mathbb{Z}$ and any $M \in \mathcal{D}_\alpha$, the $\overline{U}_q^H\mathfrak{sl}_2$-action on $A\underline{\otimes}M$ is defined by:

$$K \cdot (\mathbf{1}_k \underline{\otimes} e_a m) = q^a (\mathbf{1}_k \underline{\otimes} e_a m) \ , \tag{5.10a}$$
$$H \cdot (\mathbf{1}_k \underline{\otimes} e_a m) = (2pk + \sigma'(a))(\mathbf{1}_k \underline{\otimes} e_a m) \ , \tag{5.10b}$$
$$E \cdot (\mathbf{1}_k \underline{\otimes} e_a m) = \mathbf{1}_{k+\kappa_+(a)} \underline{\otimes} e_{a+2} E m \ , \tag{5.10c}$$
$$F \cdot (\mathbf{1}_k \underline{\otimes} e_a m) = \mathbf{1}_{k+\kappa_-(a)} \underline{\otimes} e_{a-2} F m \ , \tag{5.10d}$$

  for all $a \in \mathbb{C}/2p\mathbb{Z}$ with $\pi_1(a) = \alpha$, where $\kappa_\pm(a) = \frac{\sigma'(a)\pm 2 - \sigma'(a\pm 2)}{2p}$ are integers;
- the A-action $\rho_{A\underline{\otimes}M} \coloneqq A \otimes A\underline{\otimes}M \to A\underline{\otimes}M$ is given by left multiplication on A.

Proposition 5.3. *$F_{\sigma'}$ is a well-defined $\mathbb{C}/2\mathbb{Z}$-graded $\mathbb{C}$-linear functor.*

Proof. We need to show:

- the $\overline{U}_q^H\mathfrak{sl}_2$-action defined above satisfies the algebra relations in $\overline{U}_q^H\mathfrak{sl}_2$;
- $\rho_{A\underline{\otimes}M}$ intertwines with $\overline{U}_q^H\mathfrak{sl}_2$-action;
- if $M \in \mathcal{D}_\alpha$, then $F_{\sigma'}(M) \in {}_A\mathcal{C}_\alpha^{\mathrm{f.g.}}$.

We show for instance $E^p(\mathbf{1}_k \underline{\otimes} e_a m) = 0$. Indeed,

$$E^p(\mathbf{1}_k \underline{\otimes} e_a m) = \mathbf{1}_{k+\kappa_+(a)+\ldots+\kappa_+(a+2p-2)} \underline{\otimes} e_a E^p m = 0 \ . \tag{5.11}$$

To show that $\rho_{A\underline{\otimes}M}$ is indeed a morphism in $\mathcal{C}$, we just need to check that the following diagram commutes for any $x \in \overline{U}_q^H\mathfrak{sl}_2$:

$$\begin{array}{ccc} A \otimes A\underline{\otimes}M & \xrightarrow{\rho_{A\underline{\otimes}M}} & A\underline{\otimes}M \\ \downarrow{\scriptstyle x\cdot} & & {\scriptstyle x\cdot}\downarrow \\ A \otimes A\underline{\otimes}M & \xrightarrow{\rho_{A\underline{\otimes}M}} & A\underline{\otimes}M \end{array} \tag{5.12}$$

It suffices to check this for the generators. For instance, recalling the Hopf algebra structure (4.26) on $\overline{U}_q^H\mathfrak{sl}_2$, for $x = E$, $\mathbf{1}_k \otimes (\mathbf{1}_l \underline{\otimes} e_a m) \in A \otimes A\underline{\otimes}M$, we have

$$E \cdot \big(\rho_{A\underline{\otimes}M}(\mathbf{1}_k \otimes (\mathbf{1}_l \underline{\otimes} e_a m))\big) = \mathbf{1}_{k+l+\kappa_+(a)} \underline{\otimes} e_{a+2} E m \ , \tag{5.13a}$$
$$\begin{aligned} \rho_{A\underline{\otimes}M}\big(E \cdot (\mathbf{1}_k \otimes (\mathbf{1}_l \underline{\otimes} e_a m))\big) &= \rho_{A\underline{\otimes}M}\big((1 \otimes E + E \otimes K) \cdot (\mathbf{1}_k \otimes (\mathbf{1}_l \underline{\otimes} e_a m))\big) \\ &= \rho_{A\underline{\otimes}M}\big(\mathbf{1}_k \otimes (\mathbf{1}_{l+\kappa_+(a)} \underline{\otimes} e_{a+2} E m)\big) \\ &= \mathbf{1}_{k+l+\kappa_+(a)} \underline{\otimes} e_{a+2} E m \ . \end{aligned} \tag{5.13b}$$

Moreover, by our definition of H-action (5.10b), we can see that $A\underline{\otimes}M$ is α-local following (4.43). Finally, let $X \subset A\underline{\otimes}M$ be the finite-dimensional $\overline{U}_q^H\mathfrak{sl}_2$-submodule generated by

$$\mathbf{1}_0 \underline{\otimes} M \ . \tag{5.14}$$

It is finite-dimensional, hence dualisable, and the A-action gives a surjection

$$A \otimes X \twoheadrightarrow A\underline{\otimes}M \ , \tag{5.15}$$

because X contains $\mathbf{1}_0 \underline{\otimes} M$. Therefore, $A\underline{\otimes}M$ is indeed finitely generated.

□

Proposition 5.4. *$F_{\sigma'}$ is a $\mathbb{C}$-linear equivalence.*

Proof. The proof follows similar arguments as in [CGR17, Prop. 6.6]. Here we omit the details and only give the inverse explicitly.

For any $\alpha \in \mathbb{C}/2\mathbb{Z}$ and any $M \in {}_A\mathcal{C}_\alpha^{\text{f.g.}}$, we can decompose M as

$$M = \bigoplus_{k\in\mathbb{Z}} \bigoplus_{\substack{a\in\mathbb{C}/2p\mathbb{Z} \\ \pi_1(a)=\alpha}} M_{2pk+\sigma'(a)} \ , \tag{5.16}$$

where $M_{2pk+\sigma'(a)}$ is the subspace of M corresponding to H-eigenvalue $2pk+\sigma'(a)$. Let us denote

$$M_{(0)} = \bigoplus_{\substack{a\in\mathbb{C}/2p\mathbb{Z} \\ \pi_1(a)=\alpha}} M_{\sigma'(a)} \ . \tag{5.17}$$

We define a $\mathbb{C}/2\mathbb{Z}$-graded $\mathbb{C}$-linear functor $G_{\sigma'} \colon {}_A\mathcal{C}^{\text{f.g.}} \to \mathcal{D}$ degreewise by

$$(G_{\sigma'})_\alpha \colon {}_A\mathcal{C}_\alpha^{\text{f.g.}} \longrightarrow \mathcal{D}_\alpha \ , \qquad M \longmapsto M_{(0)} \ , \qquad (f\colon M \to N) \longmapsto f|_{M_{(0)}} \ . \tag{5.18}$$

Since M is finitely generated, each $M_{\sigma'(a)}$ is finite-dimensional by Proposition 4.9; hence, $M_{(0)}$ is a finite direct sum of finite-dimensional spaces, and is therefore finite-dimensional. For any $t \in \mathbb{Z}$, the action of $\mathbf{1}_t \in A$ on M, which we denote by $\mathbf{1}_t \bullet -$, induces an isomorphism

$$\mathbf{1}_t \bullet - \colon M_{\sigma'(a)} \xrightarrow{\sim} M_{2pt+\sigma'(a)} \ . \tag{5.19}$$

Using these isomorphisms, we define the $\overline{\mathfrak{U}}_\alpha$-action on $M_{(0)}$ as follows. On each summand $M_{\sigma'(a)}$, set

- $K_{\mathcal{D}} \coloneqq q^H$,
- $E_{\mathcal{D}} \coloneqq [M_{\sigma'(a)} \xrightarrow{E\cdot} M_{\sigma'(a)+2} \xrightarrow{\mathbf{1}_{-\kappa_+(a)}\bullet} M_{\sigma'(a+2)}]$,
- $F_{\mathcal{D}} \coloneqq [M_{\sigma'(a)} \xrightarrow{F\cdot} M_{\sigma'(a)-2} \xrightarrow{\mathbf{1}_{-\kappa_-(a)}\bullet} M_{\sigma'(a-2)}]$,

where subscript $\mathcal{D}$ distinguishes the generators of $\overline{\mathfrak{U}}_\alpha$ from those of $\overline{U}_q^H\mathfrak{sl}_2$. It is straightforward to check that these actions satisfy the defining relations of $\overline{\mathfrak{U}}_\alpha$ in (5.1). It remains to check that $G_{\sigma'}$ is well-defined on morphisms. For $f\colon M \to N$ in ${}_A\mathcal{C}_\alpha$, since f is in particular a morphism in $\mathcal{C}$, it commutes with H and hence preserves the H-weight spaces. Therefore

$$f(M_{(0)}) \subseteq N_{(0)} \ . \tag{5.20}$$

Moreover, combining the $\overline{U}_q^H\mathfrak{sl}_2$-linearity and A-linearity of f, we conclude that $f|_{M(0)}$ is a $\overline{\mathfrak{U}}_\alpha$-intertwiner.

Following the arguments in [CGR17, Prop. 6.6], one can check that $G_{\sigma'}$ indeed defines an inverse for $F_{\sigma'}$. □

Proposition 5.5. *Let σ_1' and σ_2' be two sections of $\pi_2\colon \mathbb{C} \to \mathbb{C}/2p\mathbb{Z}$, then $F_{\sigma_1'}$ is naturally isomorphic to $F_{\sigma_2'}$.*

Proof. For any $a \in \mathbb{C}/2p\mathbb{Z}$, $k \in \mathbb{Z}$, and any $M \in \mathcal{D}_{\pi_1(a)}$, we define an isomorphism

$$\eta_M^{\sigma_1',\sigma_2'} \colon F_{\sigma_1'}(M) \to F_{\sigma_2'}(M) \ , \qquad \mathbf{1}_k \underline{\otimes} e_a m \mapsto \mathbf{1}_{k+\frac{\sigma_1'(a)-\sigma_2'(a)}{2p}} \underline{\otimes} e_a m \ . \tag{5.21}$$

Verifying that this map is natural and intertwines both the A-action and the $\overline{U}_q^H\mathfrak{sl}_2$-action is straightforward and left to the reader. □

5.2. **Quasi-Hopf $\mathbb{C}/2\mathbb{Z}$-coalgebra structure.** Our next step is to equip $\mathcal{D}$ with a $\mathbb{C}/2\mathbb{Z}$-graded monoidal structure and to endow $F_{\sigma'}$ with the structure of a monoidal functor. Our strategy is to make a general ansatz for the structure morphisms of $F_{\sigma'}$, and then transport the monoidal structure on ${}_A\mathcal{C}^{\mathrm{f.g.}}$ to $\mathcal{D}$ to determine the counit, comultiplications, and coassociators on $\overline{\mathfrak{U}}$.

Denote by $\mathbb{1}_{\mathcal{D}}$ the (yet to be determined) tensor unit of $\mathcal{D}$, which should be a 1-dimensional $\overline{\mathfrak{U}}_0$-module. Fix a preferred basis vector $\mathbf{1} \in \mathbb{1}_{\mathcal{D}}$. The unital coherence map has the following general formula

$$\varphi_A \colon A \to A \underline{\otimes} \mathbb{1}_{\mathcal{D}} \ , \qquad \mathbf{1}_k \mapsto \zeta_k \mathbf{1}_{\xi(k)} \underline{\otimes} \mathbf{1} \ , \tag{5.22}$$

where $\zeta \colon \mathbb{Z} \to \mathbb{C}^\times$ is a $\mathbb{C}^\times$-valued function, and $\xi \colon \mathbb{Z} \to \mathbb{Z}$ is an integer-valued function.

- φ_A should intertwine the A-action, which means that for any $k, l \in \mathbb{Z}$,

$$\zeta_{k+l} \mathbf{1}_{\xi(k+l)} \underline{\otimes} \mathbf{1} = \zeta_k \mathbf{1}_{\xi(k)+l} \underline{\otimes} \mathbf{1} \ . \tag{5.23}$$

  Therefore, we obtain $\xi(k) = k + \xi(0)$, and we can assume $\zeta_k = 1$ for any $k \in \mathbb{Z}$. φ_A is then indeed an isomorphism.
- We next determine the counit $\varepsilon \colon \overline{\mathfrak{U}}_0 \longrightarrow \mathbb{C}$, or equivalently the $\overline{\mathfrak{U}}_0$-module structure on $\mathbb{1}_{\mathcal{D}}$, by requiring φ_A to intertwine the $\overline{U}_q^H \mathfrak{sl}_2$-action. By (5.10a), compatibility with the K-action gives

$$\varepsilon(K) = 1 \ . \tag{5.24}$$

  Similarly, compatibility with the H-action, using (5.10b), gives

$$\xi(0) = -\frac{\sigma'(0)}{2p} = 0 \ , \tag{5.25}$$

  where the last equality follows from our normalisation $\sigma'(0) = 0$. Finally, (5.10c) and (5.10d) imply

$$\varepsilon(E) = \varepsilon(F) = 0 \ . \tag{5.26}$$

Next, for any $a, b \in \mathbb{C}/2p\mathbb{Z}$, $\alpha = \pi_1(a)$, $\beta = \pi_1(b)$, $k, l \in \mathbb{Z}$, and $M \in \mathcal{D}_\alpha$, $N \in \mathcal{D}_\beta$, we make an ansatz for the structure morphism

$$\begin{aligned} &\varphi_{M,N} \colon F_{\sigma'}(M) \otimes_A F_{\sigma'}(N) \to F_{\sigma'}(M \otimes N) \ , \\ &\quad (\mathbf{1}_k \underline{\otimes} e_a m) \otimes_A (\mathbf{1}_l \underline{\otimes} e_b n) \mapsto c_{k,l,a,b} \mathbf{1}_{r_{k,l,a,b}} \underline{\otimes} (e_a m \otimes e_b n) \ , \end{aligned} \tag{5.27}$$

where $c_{k,l,a,b} \in \mathbb{C}^\times$ and $r_{k,l,a,b} \in \mathbb{Z}$. Note that the $\overline{\mathfrak{U}}_{\pi_1(a+b)}$-module structure on $M \otimes N$ is yet to be determined.

- φ is natural because morphisms between $\overline{\mathfrak{U}}$-modules can not mix K-eigenspaces.
- For $\varphi_{M,N}$ to be well defined over the relative tensor product and to intertwine the A-action, we must have, for every $t \in \mathbb{Z}$,

$$\begin{aligned} q^{pat} \zeta'_{k,l+t,a,b} \mathbf{1}_{\xi'(k,l+t,a,b)} \underline{\otimes} (e_a m \otimes e_b n) &= \zeta'_{k+t,l,a,b} \mathbf{1}_{\xi'(k+t,l,a,b)} \underline{\otimes} (e_a m \otimes e_b n) \\ &= \zeta'_{k,l,a,b} \mathbf{1}_{\xi'(k,l,a,b)+t} \underline{\otimes} (e_a m \otimes e_b n) \ . \end{aligned} \tag{5.28}$$

  Therefore, there exist functions $\zeta \colon (\mathbb{C}/2p\mathbb{Z})^2 \to \mathbb{C}^\times$ and $\xi \colon (\mathbb{C}/2p\mathbb{Z})^2 \to \mathbb{Z}$ so that

$$c_{k,l,a,b} = q^{-pal} \zeta_{a,b} \ , \qquad r_{k,l,a,b} = k + l + \xi(a,b) \ . \tag{5.29}$$

- $\varphi_{M,N}$ should intertwine the H-action. If we assume $\Delta_{\alpha,\beta}(K) = K \otimes K$, then following (5.10b) we have

$$2p(k+l) + \sigma'(a) + \sigma'(b) = 2p(k + l + \xi(a,b)) + \sigma'(a+b) \ . \tag{5.30}$$

Hence $\xi(a,b) = \frac{\sigma'(a)+\sigma'(b)-\sigma'(a+b)}{2p}$, which we have previously denoted by $\kappa(a,b)$. $\varphi_{M,N}$ is then an isomorphism with the inverse given by

$$\begin{aligned} \varphi^{-1}_{M,N} &: F_{\sigma'}(M \otimes N) \to F_{\sigma'}(M) \otimes_A F_{\sigma'}(N) \ , \\ \mathbf{1}_k \underline{\otimes} (e_a m \otimes e_b n) &\mapsto \zeta^{-1}_{a,b} \mathbf{1}_{k-\kappa(a,b)} \underline{\otimes} e_a m \otimes_A \mathbf{1}_0 \underline{\otimes} e_b n \ . \end{aligned} \tag{5.31}$$

- $\varphi_{M,N}$ should intertwine the E- and F-actions. For example, for the E-action we want the following identity

$$\varphi_{M,N} \circ E \cdot (\mathbf{1}_k \underline{\otimes} e_a m \otimes_A \mathbf{1}_l \underline{\otimes} e_b n) = q^{-pal} \zeta_{a,b} \mathbf{1}_{k+l+\kappa(a,b)+\kappa_+(a+b)} \underline{\otimes} \big(E_{\mathcal{D}} \cdot (e_a m \otimes e_b n)\big) \ . \tag{5.32}$$

The LHS is

$$\begin{aligned} & q^{-pa(l+\kappa_+(b))} \zeta_{a,b+2} \mathbf{1}_{k+l+\kappa_+(b)+\kappa(a,b+2)} \underline{\otimes} (e_a m \otimes e_{b+2} E n) \\ & + q^{b-p(a+2)l} \zeta_{a+2,b} \mathbf{1}_{k+l+\kappa_+(a)+\kappa(a+2,b)} \underline{\otimes} (e_{a+2} E m \otimes e_b n) \ . \end{aligned} \tag{5.33}$$

Note that $\kappa_+(b) + \kappa(a,b+2) = \kappa_+(a) + \kappa(a+2,b) = \kappa(a,b) + \kappa_+(a+b)$. By comparing coefficients we obtain

$$\Delta_{\alpha,\beta}(E) = \sum_{\substack{a,b \in \mathbb{C}/2p\mathbb{Z} \\ \pi_1(a)=\alpha \\ \pi_1(b)=\beta}} \Big(q^{-p\kappa_+(b)a} \frac{\zeta_{a,b+2}}{\zeta_{a,b}} e_a \otimes E e_b + q^b \frac{\zeta_{a+2,b}}{\zeta_{a,b}} E e_a \otimes e_b \Big) \ . \tag{5.34}$$

By similar computation we also obtain

$$\Delta_{\alpha,\beta}(F) = \sum_{\substack{a,b \in \mathbb{C}/2p\mathbb{Z} \\ \pi_1(a)=\alpha \\ \pi_1(b)=\beta}} \Big(q^{-a} q^{-p\kappa_-(b)a} \frac{\zeta_{a,b-2}}{\zeta_{a,b}} e_a \otimes F e_b + \frac{\zeta_{a-2,b}}{\zeta_{a,b}} F e_a \otimes e_b \Big) \ . \tag{5.35}$$

Together with the previous assumption that $\Delta_{\alpha,\beta}(K) = K \otimes K$, we have defined an algebra homomorphism $\Delta_{\alpha,\beta} \colon \overline{\mathfrak{U}}_{\alpha+\beta} \longrightarrow \overline{\mathfrak{U}}_\alpha \otimes \overline{\mathfrak{U}}_\beta$. Indeed, the transported action on $M \otimes N$ satisfies all defining relations inherited from $\overline{U}^H_q \mathfrak{sl}_2$, while the additional relation $K^p = q^{p(\alpha+\beta)}$ follows from the K-weights.

- φ should be compatible with φ_A, i.e. the following two diagrams should commute:

$$\begin{array}{ccc} A \otimes_A F_{\sigma'}(M) & \xrightarrow{\varphi_A \otimes_A \mathrm{id}} & F_{\sigma'}(\mathbb{1}_{\mathcal{D}}) \otimes_A F_{\sigma'}(M) \\ \big\downarrow{\scriptstyle l} & & \big\downarrow{\scriptstyle \varphi_{\mathbb{1}_{\mathcal{D}},M}} \\ F_{\sigma'}(M) & \xleftarrow{F_{\sigma'} l} & F_{\sigma'}(\mathbb{1}_{\mathcal{D}} \otimes M) \end{array} \tag{5.36a}$$

$$\begin{array}{ccc} F_{\sigma'}(M) \otimes_A A & \xrightarrow{\mathrm{id} \otimes_A \varphi_A} & F_{\sigma'}(M) \otimes_A F_{\sigma'}(\mathbb{1}_{\mathcal{D}}) \\ \big\downarrow{\scriptstyle r} & & \big\downarrow{\scriptstyle \varphi_{M,\mathbb{1}_{\mathcal{D}}}} \\ F_{\sigma'}(M) & \xleftarrow{F_{\sigma'} r} & F_{\sigma'}(M \otimes \mathbb{1}_{\mathcal{D}}) \end{array} \tag{5.36b}$$

where by abuse of notation l and r denote the left- and right unitors in the corresponding category, respectively. Note that the left- and right unitors in ${}_A\mathcal{C}$ are given as in (4.5) and (4.6). By straightforward computation, if we want the left and right unitors of $\mathcal{D}$ to be trivial, then ζ should be normalised in the sense that $\zeta_{a,0} = \zeta_{0,a} = 1$ for all $a \in \mathbb{C}/2p\mathbb{Z}$.

- Lastly, the structure morphisms $\varphi_{M,N}$ must satisfy the monoidal coherence condition

$$\begin{array}{ccc} F_{\sigma'}(M)\otimes_A\left(F_{\sigma'}(N)\otimes_A F_{\sigma'}(K)\right) & \xrightarrow{\mathfrak{a}} & \left(F_{\sigma'}(M)\otimes_A F_{\sigma'}(N)\right)\otimes_A F_{\sigma'}(K) \\ \downarrow{\scriptstyle \mathrm{id}_{F_{\sigma'}(M)}\otimes_A\varphi_{N,K}} & & \downarrow{\scriptstyle \varphi_{M,N}\otimes_A\mathrm{id}_{F_{\sigma'}(K)}} \\ F_{\sigma'}(M)\otimes_A F_{\sigma'}(N\otimes K) & & F_{\sigma'}(M\otimes N)\otimes_A F_{\sigma'}(K) \\ \downarrow{\scriptstyle \varphi_{M,N\otimes K}} & & \downarrow{\scriptstyle \varphi_{M\otimes N,K}} \\ F_{\sigma'}(M\otimes(N\otimes K)) & \xrightarrow{F_{\sigma'}(\mathfrak{a}_{\mathcal{D}})} & F_{\sigma'}((M\otimes N)\otimes K) \end{array} \tag{5.37}$$

where $\mathfrak{a}$ denotes the associator in ${}_A\mathcal{C}$, and $\mathfrak{a}_{\mathcal{D}}$ the associator in $\mathcal{D}$ yet to be determined. Recall that $\mathfrak{a}$ is given in (4.7). Since all the arrows in (5.37) are invertible, we define $\mathfrak{a}_{\mathcal{D}}$ by requiring the diagram to commute. For any $\alpha,\beta,\gamma\in\mathbb{C}/2\mathbb{Z}$, if we set $M=\overline{\mathfrak{U}}_\alpha$, $N=\overline{\mathfrak{U}}_\beta$, $K=\overline{\mathfrak{U}}_\gamma$ to be the regular representations, and insert

$$(\mathbf{1}_0\underline{\otimes}\mathbb{1}_\alpha)\otimes_A\left((\mathbf{1}_0\underline{\otimes}\mathbb{1}_\beta)\otimes_A(\mathbf{1}_0\underline{\otimes}\mathbb{1}_\gamma)\right)=\sum_{\substack{a,b,c\in\mathbb{C}/2p\mathbb{Z}\\ \pi_1(a)=\alpha\\ \pi_1(b)=\beta\\ \pi_1(c)=\gamma}}(\mathbf{1}_0\underline{\otimes}e_a)\otimes_A\left((\mathbf{1}_0\underline{\otimes}e_b)\otimes_A(\mathbf{1}_0\underline{\otimes}e_c)\right)$$

at the top-left corner of (5.37), then by computing the two paths of the diagram, we obtain the formula for coassociator

$$\Phi_{\alpha,\beta,\gamma}=\sum_{\substack{a,b,c\in\mathbb{C}/2p\mathbb{Z}\\ \pi_1(a)=\alpha\\ \pi_1(b)=\beta\\ \pi_1(c)=\gamma}}q^{pa\kappa(b,c)}\frac{\zeta_{a,b}\zeta_{a+b,c}}{\zeta_{b,c}\zeta_{a,b+c}}e_a\otimes e_b\otimes e_c\ . \tag{5.38}$$

Summarising these together, we conclude the following result:

Theorem 5.6. *Fixing a function $\zeta\colon\mathbb{C}/2p\mathbb{Z}\times\mathbb{C}/2p\mathbb{Z}\to\mathbb{C}^\times$ normalised such that $\zeta(x,0)=\zeta(0,x)=1$ for all x, and a set-theoretic section $\sigma'\colon\mathbb{C}/2p\mathbb{Z}\to\mathbb{C}$ normalised by $\sigma'(0)=0$, the algebra $\overline{\mathfrak{U}}$ can be equipped with the structure of a quasi-bialgebra with $\mathbb{C}/2\mathbb{Z}$-graded comultiplications:*

$$\varepsilon(E)=\varepsilon(F)=0\ ,\qquad \varepsilon(K^{\pm1})=1\ , \tag{5.39a}$$

$$\Delta_{\alpha,\beta}(K^{\pm1})=K^{\pm1}\otimes K^{\pm1}\ , \tag{5.40a}$$

$$\Delta_{\alpha,\beta}(E)=\sum_{\substack{a,b\in\mathbb{C}/2p\mathbb{Z}\\ \pi_1(a)=\alpha\\ \pi_1(b)=\beta}}q^{-p\kappa_+(b)a}\frac{\zeta_{a,b+2}}{\zeta_{a,b}}e_a\otimes Ee_b+q^b\frac{\zeta_{a+2,b}}{\zeta_{a,b}}Ee_a\otimes e_b\ , \tag{5.40b}$$

$$\Delta_{\alpha,\beta}(F)=\sum_{\substack{a,b\in\mathbb{C}/2p\mathbb{Z}\\ \pi_1(a)=\alpha\\ \pi_1(b)=\beta}}q^{-a}q^{-p\kappa_-(b)a}\frac{\zeta_{a,b-2}}{\zeta_{a,b}}e_a\otimes Fe_b+\frac{\zeta_{a-2,b}}{\zeta_{a,b}}Fe_a\otimes e_b\ , \tag{5.40c}$$

$$\Phi_{\alpha,\beta,\gamma}=\sum_{\substack{a,b,c\in\mathbb{C}/2p\mathbb{Z}\\ \pi_1(a)=\alpha\\ \pi_1(b)=\beta\\ \pi_1(c)=\gamma}}q^{pa\kappa(b,c)}\frac{\zeta_{a,b}\zeta_{a+b,c}}{\zeta_{b,c}\zeta_{a,b+c}}e_a\otimes e_b\otimes e_c\ . \tag{5.41}$$

In this case, $F_{\sigma'}\colon \mathcal{D} \to {}_A\mathcal{C}^{\mathrm{f.g.}}$ as given in (5.9) is a monoidal functor, whose monoidal structure is given by:

- *for any $\alpha, \beta \in \mathbb{C}/2\mathbb{Z}$, $\pi_1(a) = \alpha$, $\pi_1(b) = \beta$, $M \in \mathcal{D}_\alpha$, $N \in \mathcal{D}_\beta$, we have the structure isomorphism*

$$\begin{aligned} \varphi_{M,N}\colon F_{\sigma'}(M) \otimes_A F_{\sigma'}(N) &\to F_{\sigma'}(M \otimes N) \ , \\ \mathbf{1}_k \underline{\otimes} e_a m \otimes_A \mathbf{1}_l \underline{\otimes} e_b n &\mapsto q^{-pal} \zeta_{a,b} \mathbf{1}_{k+l+\kappa(a,b)} \underline{\otimes} (e_a m \otimes e_b n) \ , \end{aligned} \tag{5.42}$$

  where $\kappa(a,b) = \frac{\sigma'(a)+\sigma'(b)-\sigma'(a+b)}{2p}$;
- *for the tensor unit $\mathbb{C} \in \mathcal{D}_0$, we have*

$$\varphi_A\colon A \to A \underline{\otimes} \mathbb{C} \ , \qquad \mathbf{1}_k \mapsto \mathbf{1}_k \underline{\otimes} \mathbf{1} \ . \tag{5.43}$$

Proof. We have constructed a tensor product on $\mathcal{D}$, together with a non-trivial associator and trivial left and right unitors. It remains to show that they satisfy the coherence conditions so that we indeed defined a monoidal structure on $\mathcal{D}$. Since $F_{\sigma'}$ is an equivalence of categories, a diagram in $\mathcal{D}$ commutes if and only if its image under $F_{\sigma'}$ commutes in ${}_A\mathcal{C}^{\mathrm{f.g.}}$. Therefore, the coherence conditions in $\mathcal{D}$ follow from the coherence conditions in ${}_A\mathcal{C}^{\mathrm{f.g.}}$ and the commutativity of the diagrams (5.36) and (5.37). Following Proposition 2.4 we know that the coassociator Φ automatically satisfy (H2)–(H4). □

Let $\pi_3\colon \mathbb{C} \to \mathbb{C}/2\mathbb{Z}$ be the projection map and let $\sigma : \mathbb{C}/2\mathbb{Z} \to \mathbb{C}$ be a normalised set-theoretic section of π_3 with $\sigma(0) = 0$. From now on, for any $a, b \in \mathbb{C}/2p\mathbb{Z}$, we specify

$$\zeta_{a,b} = q^{-\frac{\sigma'(b)\sigma\pi_1(a)}{2}} \ . \tag{5.44}$$

We then obtain the following family of coproducts Δ^σ on $\overline{\mathfrak{U}} = \{\overline{\mathfrak{U}}_\alpha\}_{\alpha\in\mathbb{C}/2\mathbb{Z}}$ as in (1.10):

$$\begin{aligned} \Delta^\sigma_{\alpha,\beta}(E) &= q^{-\sigma(\alpha)} \mathbb{1}_\alpha \otimes E + E \otimes K \ , \\ \Delta^\sigma_{\alpha,\beta}(F) &= q^{\sigma(\alpha)} K^{-1} \otimes F + F \otimes \mathbb{1}_\beta \ , \\ \Delta^\sigma_{\alpha,\beta}(K^{\pm 1}) &= K^{\pm 1} \otimes K^{\pm 1} \ , \end{aligned} \tag{5.45}$$

with the coassociators in (1.11) and their inverses given by

$$\Phi^\sigma_{\alpha,\beta,\gamma} = \sum_{\substack{\pi_1(a)=\alpha \\ \pi_1(b)=\beta \\ \pi_1(c)=\gamma}} q^{c\iota_{\alpha,\beta}} e_a \otimes e_b \otimes e_c \ , \tag{5.46a}$$

$$(\Phi^\sigma_{\alpha,\beta,\gamma})^{-1} = \sum_{\substack{\pi_1(a)=\alpha \\ \pi_1(b)=\beta \\ \pi_1(c)=\gamma}} q^{-c\iota_{\alpha,\beta}} e_a \otimes e_b \otimes e_c \ , \tag{5.46b}$$

where $\iota_{\alpha,\beta} \coloneqq \frac{\sigma(\alpha)+\sigma(\beta)-\sigma(\alpha+\beta)}{2} \in \mathbb{Z}$ is a normalised 2-cocycle. By Lemmas 5.1 and 5.2, we can alternatively write $\Phi^\sigma_{\alpha,\beta,\gamma}$ as

$$\Phi^\sigma_{\alpha,\beta,\gamma} = \mathbb{1}_\alpha \otimes \mathbb{1}_\beta \otimes K^{\iota_{\alpha,\beta}} \ . \tag{5.47}$$

Different choices of σ yield different monoidal structures on $\mathcal{D}$, but they are all monoidally equivalent to ${}_A\mathcal{C}^{\mathrm{f.g.}}$.

Lemma 5.7. *For $a \in \mathbb{C}/2p\mathbb{Z}$ with $\pi_1(a) = \alpha + \beta$, it holds that*

$$\Delta^\sigma_{\alpha,\beta}(e_a) = \sum_{\substack{\pi_1(b)=\beta \\ c=a-b}} e_c \otimes e_b \ . \tag{5.48}$$

Proof.

$$\begin{aligned}\Delta^{\sigma}_{\alpha,\beta}(e_a) &= \frac{1}{p}\sum_{l=0}^{p-1} q^{-al} K^l \otimes K^l = \frac{1}{p}\sum_{l=0}^{p-1} q^{-al}(K^l \otimes \sum_{\pi_1(b)=\beta} q^{bl} e_b) \\ &= \sum_{\substack{\pi_1(b)=\beta \\ c=a-b}} (\frac{1}{p}\sum_{l}^{p-1} q^{-cl} K^l) \otimes e_b = \sum_{\substack{\pi_1(b)=\beta \\ c=a-b}} e_c \otimes e_b \ ,\end{aligned} \tag{5.49}$$

where in the third equality we applied Lemma 5.1. □

We can further define a family of antipodes S^σ as in (1.12) by

$$S^\sigma_\alpha(E) = -q^{-\sigma(\alpha)} E K^{-1} \ , \quad S^\sigma_\alpha(F) = -q^{\sigma(\alpha)} K F \ , \quad S^\sigma_\alpha(K^{\pm 1}) = K^{\mp 1} \ , \tag{5.50}$$

with the evaluation and coevaluation elements as in (1.13):

$$\boldsymbol{\alpha}^\sigma_\alpha = \mathbb{1}_\alpha \ , \qquad \boldsymbol{\beta}^\sigma_\alpha = K^{\frac{\sigma(\alpha)+\sigma(-\alpha)}{2}} \ . \tag{5.51}$$

Lemma 5.8. *For any $a \in \mathbb{C}/2p\mathbb{Z}$ and $\alpha \in \mathbb{C}/2\mathbb{Z}$ with $\pi_1(a) = \alpha$, we have*

$$S^\sigma_\alpha(e_a) = e_{-a} \ . \tag{5.52}$$

Proof.

$$S^\sigma_\alpha(e_a) = \frac{1}{p}\sum_{l=0}^{p-1} q^{-al} S^\sigma_\alpha(K^l) = \frac{1}{p}\sum_{l=0}^{p-1} q^{-al} K^{-l} = \frac{1}{p}\sum_{l=0}^{p-1} q^{-(-a)(p-l)} K^{p-l} = e_{-a} \ . \tag{5.53}$$

□

In this way, we have endowed $\overline{\mathfrak{U}}$ with the structure of a quasi-Hopf $\mathbb{C}/2\mathbb{Z}$-coalgebra, which we denote by $\overline{\mathfrak{U}}^\sigma$. We check in A.1 explicitly that $\overline{\mathfrak{U}}^\sigma$ satisfies (H1)–(H6). Consequently, $\mathcal{D}$ carries the structure of a rigid $\mathbb{C}/2\mathbb{Z}$-graded monoidal category.

5.3. **$\mathbb{C}/\mathbb{Z}$-Crossed structure.** We next define a $\mathbb{C}/\mathbb{Z}$-crossed structure on $\overline{\mathfrak{U}}^\sigma$, i.e. a family of algebra isomorphisms

$$\{\gamma_{h,\alpha} \colon \overline{\mathfrak{U}}^\sigma_\alpha \to \overline{\mathfrak{U}}^\sigma_\alpha\}_{(h,\alpha)\in\mathbb{C}/\mathbb{Z}\times\mathbb{C}/2\mathbb{Z}} \ , \tag{5.54}$$

and a family of invertible elements

$$\{\Omega^h_{\alpha,\beta} \in \overline{\mathfrak{U}}^\sigma_\alpha \otimes \overline{\mathfrak{U}}^\sigma_\beta\}_{(h,\alpha,\beta)\in\mathbb{C}/\mathbb{Z}\times\mathbb{C}/2\mathbb{Z}\times\mathbb{C}/2\mathbb{Z}} \ . \tag{5.55}$$

Our approach is to equip $F_{\sigma'}$ with the structure of a $\mathbb{C}/\mathbb{Z}$-equivariant $\mathbb{C}/2\mathbb{Z}$-functor and transport the $\mathbb{C}/\mathbb{Z}$-action on ${}_A\mathcal{C}^{\text{f.g.}}$ to $\mathcal{D}$. Concretely, we need a family of monoidal natural isomorphisms

$$\{\omega^h \colon {}^hF_{\sigma'}(-) \Longrightarrow F_{\sigma'}({}^h(-))\}_{h\in\mathbb{C}/\mathbb{Z}} \tag{5.56}$$

satisfying (2.10a), where the ${}^h(-)$ on the left hand side denotes the action of h on ${}_A\mathcal{C}^{\text{f.g.}}$ as in (4.12), and the ${}^h(-)$ on the right hand side denotes the (yet to be defined) action of h on $\mathcal{D}$ induced by γ_h.

Let us start with a general ansatz for ω. For any $h \in \mathbb{C}/\mathbb{Z}$, $a \in \mathbb{C}/2p\mathbb{Z}$, and $M \in \mathcal{D}_{\pi_1(a)}$, we define

$$\omega^h_M \colon {}^hF_{\sigma'}(M) \to F_{\sigma'}({}^hM) \ , \qquad \mathbf{1}_k \underline{\otimes} e_a m \mapsto \delta^h_{k,a} \mathbf{1}_k \underline{\otimes} \gamma_h^{-1}(e_a) m \ , \tag{5.57}$$

where $\delta_k, a^h \in \mathbb{C}^\times$, and γ_h is the to be determined algebra isomorphism. Recall the $\mathbb{C}/\mathbb{Z}$-action on A is given as in (4.42).

- It is easy to see that ω^h is indeed a natural isomorphism.

- In order to make ω^h intertwine the A-action, for any $t \in \mathbb{Z}$, we need to enforce
$$q^{2pth}\delta^h_{k+t,a} = \delta^h_{k,a} \ . \tag{5.58}$$
By abuse of notation, $\delta^h_{k,a}$ is of the form $q^{-2pkh}\delta^h_a$.
- To make ω^h_M commute with the H- and K-actions, within each $\overline{\mathfrak{U}}_\alpha$ we set
$$\gamma_{h,\alpha}(K^{\pm 1}) = K^{\pm 1} \ . \tag{5.59}$$
- ω^h_M should also intertwine the E- and F-actions. Applying ω^h_M after E yields
$$\omega^h_M\big(E \cdot (\mathbf{1}_k \underline{\otimes} e_a m)\big) = q^{-2p(k+\kappa_+(a))h}\delta^h_{a+2}\mathbf{1}_{k+\kappa_+(a)} \underline{\otimes} e_{a+2} E_{\mathcal{D}} m \ . \tag{5.60}$$
First applying ω^h_M and then E, we obtain
$$E \cdot \big(\omega^h_M(\mathbf{1}_k \underline{\otimes} e_a m)\big) = q^{-2pkh}\delta^h_a \mathbf{1}_{k+\kappa_+(a)} \underline{\otimes} e_{a+2}\gamma_h^{-1}(E_{\mathcal{D}})m \ . \tag{5.61}$$
By equating these two expressions, within each $\overline{\mathfrak{U}}_\alpha$, we obtain
$$\gamma_{h,\alpha}(E) = \sum_{\substack{a \in \mathbb{C}/2p\mathbb{Z} \\ \pi_1(a)=\alpha}} \frac{\delta^h_a}{\delta^h_{a+2}} q^{2p\kappa_+(a)h} E e_a \ . \tag{5.62}$$
Similarly, we also get
$$\gamma_{h,\alpha}(F) = \sum_{\substack{a \in \mathbb{C}/2p\mathbb{Z} \\ \pi_1(a)=\alpha}} \frac{\delta^h_a}{\delta^h_{a-2}} q^{2p\kappa_-(a)h} F e_a \ . \tag{5.63}$$
$\gamma_{h,\alpha}$ is indeed an algebra map: the additional relation $K^p = q^{p\alpha}$ is preserved since $\gamma_{h,\alpha}(K) = K$, while the remaining defining relations of $\overline{\mathfrak{U}}^\sigma_\alpha$ are inherited from $\overline{U}^H_q\mathfrak{sl}_2$.
- Since ${}^0(-) = \mathrm{id}$, we have
$$\delta^0_a = 1 \ , \qquad \forall\, a \in \mathbb{C}/2p\mathbb{Z} \ , \tag{5.64}$$
which implies (C1). Moreover, for general $h, h' \in \mathbb{C}/\mathbb{Z}$, condition (2.10a) implies
$$\delta^{h+h'}_a = \delta^h_a \delta^{h'}_a \ , \tag{5.65}$$
hence condition (C2) holds. As we want each h-action on $\mathcal{D}$ to be strictly unital, we further need to require
$$\delta^h_0 = 1 \ , \qquad \forall\, h \in \mathbb{C}/\mathbb{Z} \ , \tag{5.66}$$
which implies condition (C3).
- Lastly, ω^h should be a monoidal natural transformation. We construct the structure morphisms of ${}^h(-)$ on $\mathcal{D}$ by requiring the following diagram to commute:
$$\begin{array}{ccc}
F_{\sigma'}({}^hM) \otimes_A F_{\sigma'}({}^hN) & \xrightarrow{\varphi_{{}^hM,{}^hN}} & F_{\sigma'}({}^hM \otimes {}^hN) \\
\uparrow{\scriptstyle \omega^h_M \otimes_A \omega^h_N} & & {\scriptstyle F_{\sigma'}{}^h{}_{M,N}}\downarrow \\
{}^hF_{\sigma'}(M) \otimes_A {}^hF_{\sigma'}(N) & & F_{\sigma'}\big({}^h(M \otimes N)\big) \\
\downarrow{\scriptstyle {}^h{}_{F_{\sigma'}(M),F_{\sigma'}(N)}} & & {\scriptstyle \omega^h_{M\otimes N}}\uparrow \\
{}^h\big(F_{\sigma'}(M) \otimes_A F_{\sigma'}(N)\big) & \xrightarrow{{}^h\varphi_{M,N}} & {}^hF_{\sigma'}(M \otimes N)
\end{array} \tag{5.67}$$

where the structure morphisms of ${}^h(-)$ on ${}_A\mathcal{C}^{\mathrm{f.g.}}$ is given in (4.14). For any $\alpha,\beta\in\mathbb{C}/2\mathbb{Z}$, if we set $M=\overline{\mathfrak{U}}_\alpha$ and $N=\overline{\mathfrak{U}}_\beta$ to be the regular representations, and insert

$$(\mathbf{1}_0\underline{\otimes}\mathbb{1}_\alpha)\otimes_A(\mathbf{1}_0\underline{\otimes}\mathbb{1}_\beta)=\sum_{\substack{a,b\in\mathbb{C}/2p\mathbb{Z}\\ \pi_1(a)=\alpha\\ \pi_1(b)=\beta}}(\mathbf{1}_0\underline{\otimes}e_a)\otimes_A(\mathbf{1}_0\underline{\otimes}e_b)$$

at the middle of the left column, then by computing the two paths of the above diagram, we obtain a family of invertible elements $\{\Omega^h_{\alpha,\beta}\in\overline{\mathfrak{U}}^\sigma_\alpha\otimes\overline{\mathfrak{U}}^\sigma_\beta\}_{(h,\alpha,\beta)\in\mathbb{C}/\mathbb{Z}\times\mathbb{C}/2\mathbb{Z}\times\mathbb{C}/2\mathbb{Z}}$ given by:

$$\Omega^h_{\alpha,\beta}=\sum_{\substack{a,b\in\mathbb{C}/2p\mathbb{Z}\\ \pi_1(a)=\alpha\\ \pi_1(b)=\beta}}q^{-2p\kappa(a,b)h}\frac{\delta^h_{a+b}}{\delta^h_a\delta^h_b}e_a\otimes e_b\ . \tag{5.68}$$

Summarising these together, we have the following result:

Theorem 5.9. *Fix a normalised set-theoretic section*

$$\sigma'\colon\mathbb{C}/2p\mathbb{Z}\longrightarrow\mathbb{C}\ ,\qquad\sigma'(0)=0\ , \tag{5.69}$$

and a function

$$\delta\colon\mathbb{C}/\mathbb{Z}\times\mathbb{C}/2p\mathbb{Z}\longrightarrow\mathbb{C}^\times\ ,\qquad(h,a)\longmapsto\delta^h_a\ , \tag{5.70}$$

satisfying

$$\delta^0_a=1\ ,\qquad\delta^h_0=1\ ,\qquad\delta^{h+h'}_a=\delta^h_a\delta^{h'}_a\ , \tag{5.71}$$

for all $a\in\mathbb{C}/2p\mathbb{Z}$ and $h,h'\in\mathbb{C}/\mathbb{Z}$. Then the quasi-Hopf $\mathbb{C}/2\mathbb{Z}$-coalgebra $\overline{\mathfrak{U}}^\sigma$ damits a $\mathbb{C}/\mathbb{Z}$-crossed structure. The corresponding family of algebra isomorphisms $\{\gamma_{h,\alpha}\colon\overline{\mathfrak{U}}^\sigma_\alpha\xrightarrow{\sim}\overline{\mathfrak{U}}^\sigma_\alpha\}_{(h,\alpha)\in\mathbb{C}/\mathbb{Z}\times\mathbb{C}/2\mathbb{Z}}$ are given by:

$$\gamma_{h,\alpha}(K^{\pm1})=K^{\pm1}\ , \tag{5.72a}$$

$$\gamma_{h,\alpha}(E)=\sum_{\substack{a\in\mathbb{C}/2p\mathbb{Z}\\ \pi_1(a)=\alpha}}\frac{\delta^h_a}{\delta^h_{a+2}}q^{2p\kappa_+(a)h}Ee_a\ , \tag{5.72b}$$

$$\gamma_{h,\alpha}(F)=\sum_{\substack{a\in\mathbb{C}/2p\mathbb{Z}\\ \pi_1(a)=\alpha}}\frac{\delta^h_a}{\delta^h_{a-2}}q^{2p\kappa_-(a)h}Fe_a\ , \tag{5.72c}$$

and the corresponding family of invertible elements $\{\Omega^h_{\alpha,\beta}\in\overline{\mathfrak{U}}^\sigma_\alpha\otimes\overline{\mathfrak{U}}^\sigma_\beta\}_{(h,\alpha,\beta)\in\mathbb{C}/\mathbb{Z}\times\mathbb{C}/2\mathbb{Z}\times\mathbb{C}/2\mathbb{Z}}$ are given by:

$$\Omega^h_{\alpha,\beta}=\sum_{\substack{a,b\in\mathbb{C}/2p\mathbb{Z}\\ \pi_1(a)=\alpha\\ \pi_1(b)=\beta}}q^{-2p\kappa(a,b)h}\frac{\delta^h_{a+b}}{\delta^h_a\delta^h_b}e_a\otimes e_b\ . \tag{5.73}$$

Moreover, $(F_{\sigma'},\{\omega^h\}_{h\in\mathbb{C}/\mathbb{Z}})\colon\mathcal{D}\to{}_A\mathcal{C}^{\mathrm{f.g.}}$ is a $\mathbb{C}/\mathbb{Z}$-equivariant $\mathbb{C}/2\mathbb{Z}$-graded monoidal functor. For any $h\in\mathbb{C}/\mathbb{Z}$, $a\in\mathbb{C}/2p\mathbb{Z}$, $M\in\mathcal{D}_{\pi_a}$, ω^h_M is given by

$$\omega^h_M\big(\mathbf{1}_k\underline{\otimes}e_am\big)=q^{-2pkh}\delta^h_a\mathbf{1}_k\underline{\otimes}e_am\ . \tag{5.74}$$

Proof. By construction, the maps $\gamma_{h,\alpha}$ are algebra isomorphisms satisfying (C1)–(C3). It remains to verify that, for each $h\in\mathbb{C}/\mathbb{Z}$, the structure morphisms $h_{M,N}$ define a monoidal structure on ${}^h(-)$ in $\mathcal{D}$. Since $F_{\sigma'}$ is an equivalence, the coherence diagram for $h_{M,N}$ commutes in $\mathcal{D}$ if and only if its image under $F_{\sigma'}$ commutes in ${}_A\mathcal{C}^{\mathrm{f.g.}}$. By the commutativity of (5.67), this follows from

the monoidal coherence of the $\mathbb{C}/\mathbb{Z}$-action on ${}_A\mathcal{C}^{\mathrm{f.g.}\cdot}$. Hence each ${}^h(-)$ is a strictly unital monoidal functor. Therefore, by Proposition 2.9, the corresponding elements $\Omega^h_{\alpha,\beta}$ satisfy (C4)–(C7). □

From now on, we set $\delta \equiv 1$, we then obtain the formula given in (1.14) and (1.15). We denote the resulting $\mathbb{C}/\mathbb{Z}$-crossed quasi-Hopf $\mathbb{C}/2\mathbb{Z}$-coalgebra by ${}^{\sigma'}\overline{\mathfrak{U}}^{\sigma}$. In A.2 we conduct a direct verification that ${}^{\sigma'}\overline{\mathfrak{U}}^{\sigma}$ indeed carries a $\mathbb{C}/\mathbb{Z}$-crossed quasi-Hopf $\mathbb{C}/2\mathbb{Z}$-coalgebra structure.

5.4. **π-Crossed ribbon structure.** Our final step is to construct a π-crossed ribbon structure on ${}^{\sigma'}\overline{\mathfrak{U}}^{\sigma}$ and upgrade $F_{\sigma'}$ to a π-crossed ribbon equivalence.

We can construct a π-crossed braiding on $\mathcal{D}$ by solving the equation induced by the following commutative diagram for homogeneous objects M and N:

$$\begin{array}{ccccc} F_{\sigma'}(M\otimes N) & \xrightarrow{\varphi^{-1}_{M,N}} & F_{\sigma'}(M)\otimes_A F_{\sigma'}(N) & \xrightarrow{\widetilde{C}_{F_{\sigma'}(M),F_{\sigma'}(N)}} & {}^{\pi(|M|)}F_{\sigma'}(N)\otimes_A F_{\sigma'}(M) \\ \downarrow{\scriptstyle F_{\sigma'}C_{M,N}} & & & & \downarrow{\scriptstyle \omega_N^{\pi(|M|)}\otimes_A \mathrm{id}} \\ F_{\sigma'}({}^{\pi(|M|)}N\otimes M) & & \xleftarrow{\varphi_{\pi(|M|)N,M}} & & F_{\sigma'}({}^{\pi(|M|)}N)\otimes_A F_{\sigma'}(M) \end{array} \tag{5.75}$$

where $\widetilde{C}$ is the π-crossed braiding in ${}_A\mathcal{C}^{f.g.\cdot}$, and C is the yet to be determined π-crossed braiding in $\mathcal{D}$. For any $\alpha,\beta\in\mathbb{C}/2\mathbb{Z}$, if we set $M=\overline{\mathfrak{U}}_\alpha$ and $N=\overline{\mathfrak{U}}_\beta$ to be the regular representations, and insert

$$\mathbf{1}_0\underline{\otimes}(\mathbb{1}_\alpha\otimes\mathbb{1}_\beta)=\sum_{\substack{a,b\in\mathbb{C}/2p\mathbb{Z}\\ \pi_1(a)=\alpha\\ \pi_1(b)=\beta}}\mathbf{1}_0\underline{\otimes}(e_a\otimes e_b)$$

at the top-left corner, we then obtain

$$\mathbf{1}_0\underline{\otimes}\tau(R_{\alpha,\beta})=\varphi_{\pi(\alpha)\overline{\mathfrak{U}}_\beta,\overline{\mathfrak{U}}_\alpha}\circ(\omega^{\pi(\alpha)}_{\overline{\mathfrak{U}}_\beta}\otimes_A\mathrm{id}_{F_{\sigma'}(\overline{\mathfrak{U}}_\alpha)})\circ\widetilde{C}_{F_{\sigma'}(\overline{\mathfrak{U}}_\alpha),F_{\sigma'}(\overline{\mathfrak{U}}_\beta)}\circ\varphi^{-1}_{\overline{\mathfrak{U}}_\alpha,\overline{\mathfrak{U}}_\beta}\big(\mathbf{1}_0\underline{\otimes}(\mathbb{1}_\alpha\otimes\mathbb{1}_\beta)\big)\ , \tag{5.76}$$

where τ is the usual transposition. By a straightforward but lengthy computation we get

$$R_{\alpha,\beta}=\sum_{n=0}^{p-1}\sum_{\substack{a,b\in\mathbb{C}/2p\mathbb{Z}\\ \pi_1(a)=\alpha\\ \pi_1(b)=\beta}}\frac{(q-q^{-1})^n}{[n]!}q^{\frac{n(n-1)+\zeta_a^n(\zeta_b^0-2n)}{2}}q^{\frac{\sigma(\alpha)\sigma(\beta)}{2}}q^{(\zeta_b^{-n}+n)\sigma(\alpha)}(E^ne_a\otimes F^ne_b)\ , \tag{5.77}$$

where for any $c\in\mathbb{C}/2p\mathbb{Z}$ and any integer $1-p\le k\le p-1$, $\zeta_c^k:=\sigma'(c+2k)-\sigma(\pi_1(c))$ is an even integer depending on the choices of σ and σ'. The intermediate steps are referred to A.3.

Similarly, by solving the equation induced by the following commutative diagram for a homogeneous object M, we can construct a π-crossed ribbon structure on $\mathcal{D}$:

$$\begin{array}{ccc} {}^{\pi(|M|)}F_{\sigma'}(M) & \xrightarrow{\widetilde{\vartheta}^{-1}_{F_{\sigma'}(M)}} & F_{\sigma'}(M) \\ {\scriptstyle\omega_M^{\pi(|M|)}}\downarrow & \nearrow_{F_{\sigma'}(\vartheta_M^{-1})} & \\ F_{\sigma'}({}^{\pi(|M|)}M) & & \end{array} \tag{5.78}$$

where $\widetilde{\vartheta}$ is the π-crossed ribbon twist in ${}_A\mathcal{C}^{f.g.\cdot}$, and ϑ is the yet to be determined π-crossed ribbon twist in $\mathcal{D}$. For any $\alpha\in\mathbb{C}/2\mathbb{Z}$, if we set $M=\overline{\mathfrak{U}}_\alpha$, and put at the top-left corner

$$\sum_{\substack{a\in\mathbb{C}/2p\mathbb{Z}\\ \pi_1(a)=\alpha}}\mathbf{1}_0\underline{\otimes}e_a=\mathbf{1}_0\underline{\otimes}\mathbb{1}_\alpha\ , \tag{5.79}$$

we then obtain

$$F_{\sigma'}(\vartheta_{\underline{\mathfrak{U}}_\alpha}^{-1}) \circ \omega_{\underline{\mathfrak{U}}_\alpha}^{\pi(\alpha)}(\mathbf{1}_0 \underline{\otimes} \mathbb{1}_\alpha) = \widetilde{\vartheta}_{F_{\sigma'}(\underline{\mathfrak{U}}_\alpha)}^{-1}(\mathbf{1}_0 \underline{\otimes} \mathbb{1}_\alpha) \ . \tag{5.80}$$

By a straightforward computation we get

$$v_\alpha = \sum_{n=0}^{p-1} \sum_{\substack{a \in \mathbb{C}/2p\mathbb{Z} \\ \pi_1(a)=\alpha}} \frac{(q^{-1}-q)^n}{[n]!} q^{\frac{-n(n+3)}{2}} q^{-\frac{(\sigma'(a))^2}{2}} q^{a(p-n-1)} F^n E^n e_a \ . \tag{5.81}$$

The intermediate steps are referred to A.4.

Summarising these together, we conclude the following result

Theorem 5.10. *${}^{\sigma'}\overline{\mathfrak{U}}^{\sigma}$ can be endowed with the structure of a π-crossed ribbon quasi-Hopf algebra with the π-crossed R-matrices and π-crossed ribbon elements given by:*

$$R_{\alpha,\beta} = \sum_{n=0}^{p-1} \sum_{\substack{a,b \in \mathbb{C}/2p\mathbb{Z} \\ \pi_1(a)=\alpha \\ \pi_1(b)=\beta}} \frac{(q-q^{-1})^n}{[n]!} q^{\frac{n(n-1)+\zeta_a^n(\zeta_b^0-2n)}{2}} q^{\frac{\sigma(\alpha)\sigma(\beta)}{2}} q^{(\zeta_b^{-n}+n)\sigma(\alpha)} (E^n e_a \otimes F^n e_b) \ , \tag{5.82}$$

$$v_\alpha = \sum_{n=0}^{p-1} \sum_{\substack{a \in \mathbb{C}/2p\mathbb{Z} \\ \pi_1(a)=\alpha}} \frac{(q^{-1}-q)^n}{[n]!} q^{\frac{-n(n+3)}{2}} q^{-\frac{(\sigma'(a))^2}{2}} q^{a(p-n-1)} F^n E^n e_a \ , \tag{5.83}$$

for any $\alpha, \beta \in \mathbb{C}/2\mathbb{Z}$.

The functor $F_{\sigma'} : \mathcal{D} \to {}_A\mathcal{C}^{\mathrm{f.g.}}$ is a π-crossed ribbon equivalence.

Proof. We have to verify that we have indeed constructed a π-crossed ribbon structure on $\mathcal{D}$. Once this is established, Propositions 2.11 and 2.13 imply that $\{R_{\alpha,\beta}\}_{\alpha,\beta\in\mathbb{C}/2\mathbb{Z}}$ forms a family of π-crossed R-matrices, and that $\{v_\alpha\}_{\alpha\in\mathbb{C}/2\mathbb{Z}}$ forms a family of π-crossed ribbon elements.

Since $F_{\sigma'}$ is an equivalence, verifying conditions (B1)–(B4), (BL1)–(BL3), and (RB1) for $\{C_{M,N}\}_{M,N\in\mathcal{D}_{\mathrm{hom}}}$ and $\{\vartheta_M\}_{M\in\mathcal{D}_{\mathrm{hom}}}$ in $\mathcal{D}$ is equivalent to verifying them for $\{F_{\sigma'}(C_{M,N})\}_{M,N\in\mathcal{D}_{\mathrm{hom}}}$ and $\{F_{\sigma'}(\vartheta_M)\}_{M\in\mathcal{D}_{\mathrm{hom}}}$ in ${}_A\mathcal{C}^{\mathrm{f.g.}}$. These conditions all hold as consequence of the commutativity of diagrams (5.75) and (5.78), together with the π-crossed ribbon structure already established on ${}_A\mathcal{C}^{\mathrm{f.g.}}$. Finally, by our construction, $F_{\sigma'}$ is a π-crossed ribbon equivalence. □

5.5. **Graded coends in ${}_A\mathcal{C}^{\mathrm{f.g.}}$.** Recall the graded coends defined in (3.15). We now give explicit formulas for them in ${}_A\mathcal{C}^{\mathrm{f.g.}}$ as free A-modules.

By Proposition 3.9, the graded coends in $\mathcal{D}$ are

$$\mathcal{L}_\alpha := \overline{(\overline{\mathfrak{U}}_\alpha)^*} \in \mathcal{D}_0 \ , \qquad \alpha \in \mathbb{C}/2\mathbb{Z} \ . \tag{5.84}$$

We first lift these coadjoint representations to finite-dimensional weight modules of $\overline{U}_q^H\mathfrak{sl}_2$. Note that each $\overline{\mathfrak{U}}_\alpha$ admits a structure of graded algebra with

$$\deg(E) = 2 \ , \qquad \deg(F) = -2 \ , \qquad \deg(K^{\pm 1}) = 0 \ . \tag{5.85}$$

We write

$$\overline{\mathfrak{U}}_\alpha = \bigoplus_{d\in 2\mathbb{Z}} (\overline{\mathfrak{U}}_\alpha)_d \ , \tag{5.86}$$

where $(\overline{\mathfrak{U}}_\alpha)_d$ is homogeneous of degree d.

Lemma 5.11. *For every $\alpha \in \mathbb{C}/2\mathbb{Z}$, the coadjoint representation $\mathcal{L}_\alpha$ extends to a finite-dimensional weight module $\widetilde{\mathcal{L}}_\alpha$ of $\overline{U}_q^H \mathfrak{sl}_2$ by setting*

$$H \cdot \varphi = -d\,\varphi \ , \qquad \varphi \in \big((\overline{\mathfrak{U}}_\alpha)_d\big)^* \ . \tag{5.87}$$

Proof. The actions of $E, F, K^\pm$ on $\widetilde{\mathcal{L}}_\alpha$ are given by the coadjoint action on $\mathcal{L}_\alpha$. We just need to show that these actions respect the algebra relations in $\overline{U}_q^H \mathfrak{sl}_2$.

The relations involving only $E, F, K^{\pm 1}$ already hold because $\mathcal{L}_\alpha$ is a $\overline{\mathfrak{U}}_0$-module. It remains to check the relations involving H and the weight-module condition.

Using (3.78), the coproducts (1.10), and the antipodes (1.12), we obtain, for any $y \in \overline{\mathfrak{U}}_\alpha$,

$$\langle E \cdot \varphi, y\rangle = q^{-\sigma(-\alpha)}\big\langle \varphi,\, yE - EK^{-1}yK\big\rangle \ , \tag{5.88a}$$

$$\langle F \cdot \varphi, y\rangle = q^{\sigma(-\alpha)}\big\langle \varphi,\, KyF - KFy\big\rangle \ , \tag{5.88b}$$

$$\langle K \cdot \varphi, y\rangle = \langle \varphi, K^{-1}yK\rangle \ . \tag{5.88c}$$

Consequently,

$$E \cdot \big((\overline{\mathfrak{U}}_\alpha)_d\big)^* \subset \big((\overline{\mathfrak{U}}_\alpha)_{d-2}\big)^* \ , \qquad F \cdot \big((\overline{\mathfrak{U}}_\alpha)_d\big)^* \subset \big((\overline{\mathfrak{U}}_\alpha)_{d+2}\big)^* \ . \tag{5.89}$$

Together with H-action defined in (5.87), these give

$$[H, E] = 2E \ , \qquad [H, F] = -2F \ , \qquad [H, K] = 0 \ . \tag{5.90}$$

Moreover, by the algebra relations of $\overline{\mathfrak{U}}_\alpha$ in (5.1), for homogeneous $y \in (\overline{\mathfrak{U}}_\alpha)_d$, we have

$$K^{-1}yK = q^{-d}y \ . \tag{5.91}$$

Thus K acts on $\big((\overline{\mathfrak{U}}_\alpha)_d\big)^*$ by q^{-d}, which is exactly q^H, so this is a weight module. □

Recall the projection $\pi_2 \colon \mathbb{C} \to \mathbb{C}/2p\mathbb{Z}$ and its normalised set theoretic section $\sigma' \colon \mathbb{C}/2p\mathbb{Z} \to \mathbb{C}$, and set

$$n_{\sigma'}(\lambda) \coloneqq \frac{\sigma'(\pi_2(\lambda)) - \lambda}{2p} \in \mathbb{Z} \ , \qquad \lambda \in \mathbb{C} \ . \tag{5.92}$$

Write $\mathrm{Res}(X)$ for the object of $\mathcal{D}$ obtained from $X \in \mathcal{C}^{\mathrm{f.d.}}$ by forgetting the H-action.

Lemma 5.12. *For every $M \in \mathcal{C}^{\mathrm{f.d.}}$, there is a natural isomorphism*

$$\begin{aligned} \Xi_M \colon F_{\sigma'}(\mathrm{Res}(M)) &\xrightarrow{\sim} A \otimes M \ , \\ \mathbf{1}_k \underline{\otimes} m_\lambda &\longmapsto \mathbf{1}_{k+n_{\sigma'}(\lambda)} \otimes m_\lambda \ , \end{aligned} \tag{5.93}$$

where $Hm_\lambda = \lambda m_\lambda$.

Proof. The map Ξ_M is clearly A-linear and bijective. We check that it also intertwines the $\overline{U}_q^H \mathfrak{sl}_2$-actions, using the action on $F_{\sigma'}(\mathrm{Res}(M))$ defined in (5.10a)–(5.10d).

Write $a = \pi_2(\lambda)$. By (5.10b), Ξ_M intertwines the H-actions, and thus also the K-actions, because

$$2pk + \sigma'(a) = 2p\big(k + n_{\sigma'}(\lambda)\big) + \lambda \ . \tag{5.94}$$

For the E-action, note that Em_λ has weight $\lambda + 2$ and that

$$\kappa_+(a) + n_{\sigma'}(\lambda + 2) = n_{\sigma'}(\lambda) \ . \tag{5.95}$$

Using (5.10c), we obtain

$$\begin{aligned} &\Xi_M\big(E \cdot (\mathbf{1}_k \underline{\otimes} m_\lambda)\big) = \Xi_M\big(\mathbf{1}_{k+\kappa_+(a)} \underline{\otimes} Em_\lambda\big) \\ &= \mathbf{1}_{k+\kappa_+(a)+n_{\sigma'}(\lambda+2)} \otimes Em_\lambda = \mathbf{1}_{k+n_{\sigma'}(\lambda)} \otimes Em_\lambda \ . \end{aligned} \tag{5.96}$$

On the other hand, the coproduct $\Delta(E) = E \otimes K + 1 \otimes E$ and the trivial E-action on A give

$$E \cdot \Xi_M(\mathbf{1}_k \underline{\otimes} m_\lambda) = E \cdot \big(\mathbf{1}_{k+n_{\sigma'}(\lambda)} \otimes m_\lambda\big) = \mathbf{1}_{k+n_{\sigma'}(\lambda)} \otimes Em_\lambda \ . \tag{5.97}$$

Hence Ξ_M intertwines the E-actions. The verification for F is similar.

Finally, naturality holds because morphisms of weight modules preserve H-weight spaces. □

Proposition 5.13. *For every $\alpha \in \mathbb{C}/2\mathbb{Z}$, the graded coend $\mathcal{L}_\alpha^A := \int^{M\in {}_A\mathcal{C}_\alpha^{\mathrm{f.g.}}} M^\vee \otimes_A M$ in ${}_A\mathcal{C}^{\mathrm{f.g.}}$ exists and can be chosen as*

$$\mathcal{L}_\alpha^A = A \otimes \widetilde{\mathcal{L}}_\alpha \ . \tag{5.98}$$

Proof. Since F is a monoidal equivalence, it sends coends to coends. By Lemma 5.12, we have $F(\mathcal{L}_\alpha) \cong A \otimes \widetilde{\mathcal{L}}_\alpha$, where $\widetilde{\mathcal{L}}_\alpha$ are as in Lemma 5.11. □

Appendix A. π-Crossed ribbon quasi-Hopf algebra conditions for ${}^{\sigma'}\overline{\mathfrak{U}}^{\sigma}$

A.1. **Quasi-Hopf $\mathbb{C}/2\mathbb{Z}$-coalgebra conditions.** We first verify that $\overline{\mathfrak{U}}^\sigma$ presented in Section 1.1 indeed carries the structure of a quasi-Hopf $\mathbb{C}/2\mathbb{Z}$-coalgebra. By Proposition 2.4, it remains only to check (H5) and (H6). However, we provide a complete verification of all conditions here as an independent check. It is straightforward to check that the counit in (1.7) and the coproducts in (1.10) are algebra homomorphisms. For instance, for any $\alpha, \beta \in \mathbb{C}/2\mathbb{Z}$, we check that $\Delta^\sigma_{\alpha,\beta}(E)^p = 0$:

$$\begin{aligned}
\Delta^\sigma_{\alpha,\beta}(E)^p &= (q^{-\sigma(\alpha)}\mathbb{1}_\alpha \otimes E + E \otimes K)^p = \sum_{i=0}^{p} q^{-i\sigma(\alpha)} q^{i(p-i)} \begin{bmatrix} p \\ i \end{bmatrix} E^{p-i} \otimes E^i K^{p-i} \\
&= E^p \otimes K^p + \sum_{i=1}^{p-1} q^{-i\sigma(\alpha)} q^{i(p-i)} \begin{bmatrix} p \\ i \end{bmatrix} E^{p-i} \otimes E^i K^{p-i} + q^{-p\sigma(\alpha)} \mathbb{1}_\alpha \otimes E^p \\
&= 0 \ ,
\end{aligned} \tag{A.1}$$

where in the last step we use $E^p = 0$ and $\{p\} = 0$.

We check in the following that $\overline{\mathfrak{U}}^\sigma$ satisfies (H1)–(H6).

For (H1), here we only check the generator E, the rest can be done similarly:

$$(\varepsilon \otimes \mathrm{id}_\alpha) \circ \Delta^\sigma_{0,\alpha}(E) = \varepsilon(\mathbb{1}_0)E + \varepsilon(E)K = E = \varepsilon(E)q^{-\sigma(\alpha)}\mathbb{1}_\alpha + \varepsilon(K)E = (\mathrm{id}_\alpha \otimes \varepsilon) \circ \Delta^\sigma_{\alpha,0}(E) \ . \tag{A.2}$$

For (H2), we also only check the generator E here. Let's write $\Phi^\sigma_{\alpha,\beta,\gamma} = \sum_{a,b,c} \xi_{a,b,c} e_a \otimes e_b \otimes e_c$, where $\xi_{a,b,c} = q^{\frac{c(\sigma(\alpha)+\sigma(\beta)-\sigma(\alpha+\beta))}{2}}$. We then have

$$\begin{aligned}
&\Phi^\sigma_{\alpha,\beta,\gamma} \cdot (\mathrm{id}_\alpha \otimes \Delta^\sigma_{\beta,\gamma}) \circ \Delta^\sigma_{\alpha,\beta+\gamma}(E) \\
&= \sum_{a,b,c} \xi_{a,b,c} q^{-\sigma(\alpha)-\sigma(\beta)} e_a \otimes e_b \otimes E e_{c-2} + \sum_{a,b,c} \xi_{a,b,c} q^{-\sigma(\alpha)} e_a \otimes E e_{b-2} \otimes K e_c \\
&\quad + \sum_{a,b,c} \xi_{a,b,c} E e_{a-2} \otimes K e_b \otimes K e_c \\
&= \sum_{a,b,c-2} \xi_{a,b,c-2} q^{-\sigma(\alpha+\beta)} e_a \otimes e_b \otimes E e_{c-2} + \sum_{a,b-2,c} \xi_{a,b-2,c} q^{-\sigma(\alpha)} e_a \otimes E e_{b-2} \otimes K e_c \\
&\quad + \sum_{a-2,b,c} \xi_{a-2,b,c} E e_{a-2} \otimes K e_b \otimes K e_c \\
&= (\Delta^\sigma_{\alpha,\beta} \otimes \mathrm{id}_\gamma) \circ \Delta^\sigma_{\alpha+\beta,\gamma}(E) \cdot \Phi^\sigma_{\alpha,\beta,\gamma} \ .
\end{aligned} \tag{A.3}$$

We now check (H3). By applying Lemma 5.7 we have

$$\begin{aligned}&(\Phi^{\sigma}_{\alpha,\beta,\gamma}\otimes \mathbb{1}_{\zeta})((\mathrm{id}_{\alpha}\otimes\Delta^{\sigma}_{\beta,\gamma}\otimes\mathrm{id}_{\zeta})\Phi^{\sigma}_{\alpha,\beta+\gamma,\zeta})(\mathbb{1}_{\alpha}\otimes\Phi^{\sigma}_{\beta,\gamma,\zeta})\\ =&\sum_{\substack{\pi_1(a)=\alpha\\ \pi_1(b)=\beta\\ \pi_1(c)=\gamma\\ \pi_1(d)=\zeta}}\xi_{a,b,c}\xi_{a,b+c,d}\xi_{b,c,d}e_a\otimes e_b\otimes e_c\otimes e_d\ .\end{aligned}\tag{A.4}$$

Similarly,

$$\begin{aligned}&((\Delta^{\sigma}_{\alpha,\beta}\otimes\mathrm{id}_{\gamma}\otimes\mathrm{id}_{\zeta})\Phi^{\sigma}_{\alpha+\beta,\gamma,\zeta})((\mathrm{id}_{\alpha}\otimes\mathrm{id}_{\beta}\otimes\Delta^{\sigma}_{\gamma,\zeta})\Phi^{\sigma}_{\alpha,\beta,\gamma+\zeta})\\ =&\sum_{\substack{\pi_1(a)=\alpha\\ \pi_1(b)=\beta\\ \pi_1(c)=\gamma\\ \pi_1(d)=\zeta}}\xi_{a+b,c,d}\xi_{a,b,c+d}e_a\otimes e_b\otimes e_c\otimes e_d\ .\end{aligned}\tag{A.5}$$

To see whether these two terms are equal, we need to compare the coefficients

$$\xi_{a,b,c}\xi_{a,b+c,d}\xi_{b,c,d}\overset{!}{=}\xi_{a+b,c,d}\xi_{a,b,c+d}\ ,\tag{A.6}$$

which is a straightforward computation that we leave to the reader.

For (H4) we have

$$\begin{aligned}&(\mathrm{id}_{\alpha}\otimes\varepsilon\otimes\mathrm{id}_{\gamma})\Phi^{\sigma}_{\alpha,0,\gamma}=\sum_{a,c}\sum_{b}\xi_{a,b,c}e_a\otimes\varepsilon(e_b)\otimes e_c\\ =&\sum_{a,c}\xi_{a,0,c}e_a\otimes e_c=\sum_{a,c}q^{\frac{c\sigma(0)}{2}}e_a\otimes e_c=\mathbb{1}_{\alpha}\otimes\mathbb{1}_{\gamma}\ ,\end{aligned}\tag{A.7}$$

where we use the assumption that $\sigma(0)=0$.

We next check the generator E for (H5). Using Sweedler notation we have:

$$S^{\sigma}_{-\alpha}(E_{(1,-\alpha)})\boldsymbol{\alpha}^{\sigma}_{\alpha}E_{(2,\alpha)}=q^{-\sigma(-\alpha)}\mathbb{1}_{\alpha}\mathbb{1}_{\alpha}E-q^{-\sigma(-\alpha)}EK^{-1}\mathbb{1}_{\alpha}K=0=\varepsilon(E)\boldsymbol{\alpha}^{\sigma}_{\alpha}\ ,\tag{A.8}$$

and

$$\begin{aligned}&E_{(1,\alpha)}\boldsymbol{\beta}^{\sigma}_{\alpha}S^{\sigma}_{-\alpha}(E_{(2,-\alpha)})=-q^{-\sigma(\alpha)}\boldsymbol{\beta}^{\sigma}_{\alpha}q^{-\sigma(-\alpha)}EK^{-1}+E\boldsymbol{\beta}^{\sigma}_{\alpha}K^{-1}\\ =&(E\boldsymbol{\beta}^{\sigma}_{\alpha}-q^{-\sigma(\alpha)-\sigma(-\alpha)}\boldsymbol{\beta}^{\sigma}_{\alpha}E)K^{-1}=0=\varepsilon(E)\boldsymbol{\beta}^{\sigma}_{\alpha}\ .\end{aligned}\tag{A.9}$$

Finally we check (H6). Writing $\Phi^{\sigma}_{\alpha,-\alpha,\alpha}=\sum_i X_i\otimes Y_i\otimes Z_i$, we have

$$\begin{aligned}&\sum_i S^{\sigma}_{\alpha}(X_i)\boldsymbol{\alpha}_{-\alpha}Y_i\boldsymbol{\beta}_{-\alpha}S^{\sigma}_{\alpha}(Z_i)=\sum_{\pi_1(a)=\alpha}q^{\frac{a(\sigma(\alpha)+\sigma(-\alpha))}{2}}e_{-a}K^{\frac{\sigma(\alpha)+\sigma(-\alpha)}{2}}\\ =&\Big(\sum_{\pi_1(-a)=-\alpha}q^{\frac{a(\sigma(\alpha)+\sigma(-\alpha))}{2}}e_{-a}\Big)K^{\frac{\sigma(\alpha)+\sigma(-\alpha)}{2}}=K^{-\frac{\sigma(\alpha)+\sigma(-\alpha)}{2}}K^{\frac{\sigma(\alpha)+\sigma(-\alpha)}{2}}=\mathbb{1}_{-\alpha}\ ,\end{aligned}\tag{A.10}$$

where in the second last step we applied Lemma 5.1. The other identity can be verified by similar computation.

A.2. $\mathbb{C}/\mathbb{Z}$**-Crossed conditions.** We next check the $\mathbb{C}/\mathbb{Z}$-crossed structure on $\overline{\mathfrak{U}}^{\sigma}$.

For any $h\in\mathbb{C}/\mathbb{Z}$, $\alpha\in\mathbb{C}/2\mathbb{Z}$, $\gamma_{h,\alpha}$ is indeed an algebra map. Here we check

$$\gamma_{h,\alpha}(EF-FE)=\gamma_{h,\alpha}(\frac{K-K^{-1}}{q-q^{-1}})$$

as an example:

$$\begin{aligned}
&\gamma_{h,\alpha}(EF-FE)\\
&=(\sum_a q^{2p\kappa_+(a)h}Ee_a)(\sum_b q^{2p\kappa_-(b)h}Fe_b)-(\sum_c q^{2p\kappa_-(c)h}Fe_c)(\sum_d q^{2p\kappa_+(d)h}Ee_d)\\
&=\sum_a q^{2p(\kappa_+(a)+\kappa_-(a+2))h}e_{a+2}EF-\sum_c q^{2p(\kappa_-(c)+\kappa_+(c-2))h}e_{c-2}FE\\
&=\sum_a e_a(EF-FE)=EF-FE=\frac{K-K^{-1}}{q-q^{-1}}=\gamma_{h,\alpha}(\frac{K-K^{-1}}{q-q^{-1}})\ ,
\end{aligned}\tag{A.11}$$

where in the second last line we use the fact that $\kappa_+(a)+\kappa_-(a+2)=0$.

We next check that ${}^{\sigma'}\overline{\mathfrak{U}}^{\sigma}$ is indeed a $\mathbb{C}/\mathbb{Z}$-crossed quasi-Hopf $\mathbb{C}/2\mathbb{Z}$-coalgebra with the $\mathbb{C}/\mathbb{Z}$-action and the invertible elements given in (1.14) and (1.15). It's immediate to see that (C1)–(C3) hold. By Proposition 2.9, conditions (C4)–(C7) hold automatically. Here we provide an independent check explicitly.

To check (C4), it suffices to show it for the generators. For example, we show it for F:

$$\begin{aligned}
&\Omega^h_{\alpha,\beta}\cdot(\gamma^{-1}_{h,\alpha}\otimes\gamma^{-1}_{h,\beta})\circ\Delta^{\sigma}_{\alpha,\beta}\circ\gamma_{h,\alpha+\beta}(F)\\
&=(\sum_{\substack{\pi_1(c)=\alpha\\ \pi_1(d)=\beta}} q^{-2p\kappa(c,d)h}e_c\otimes e_d)\big(\sum_{\substack{\pi_1(a)=\alpha\\ \pi_1(b)=\beta}}(q^{\sigma(\alpha)}K^{-1}e_a\otimes q^{-2p\kappa_-(b)h}Fe_b+q^{-2p\kappa_-(a)h}Fe_a\otimes e_b)\big)\\
&=\sum_{\substack{\pi_1(a)=\alpha\\ \pi_1(b)=\beta}}(q^{\sigma(\alpha)}q^{-2ph(\kappa(a,b-2)+\kappa_-(b)-\kappa_-(a+b))}K^{-1}e_a\otimes Fe_b+q^{-2ph(\kappa(a-2,b)+\kappa_-(a)-\kappa_-(a+b))}Fe_a\otimes e_b)\\
&=\sum_{\substack{\pi_1(a)=\alpha\\ \pi_1(b)=\beta}}(q^{\sigma(\alpha)}q^{-2p\kappa(a,b)h}K^{-1}e_a\otimes Fe_b+q^{-2p\kappa(a,b)h}Fe_a\otimes e_b)\\
&=(q^{\sigma(\alpha)}K^{-1}\otimes F+F\otimes\mathbb{1}_\beta)(\sum_{\substack{\pi_1(a)=\alpha\\ \pi_1(b)=\beta}} q^{-2p\kappa(a,b)h}e_a\otimes e_b)=\Delta^{\sigma}_{\alpha,\beta}(F)\cdot\Omega^h_{\alpha,\beta}\ .
\end{aligned}\tag{A.12}$$

Conditions (C5) and (C6) are easy. Finally, we check (C7): the LHS is

$$\begin{aligned}
&\Phi^{\sigma}_{\alpha,\beta,\gamma}\cdot(\mathrm{id}_\alpha\otimes\Delta^{\sigma}_{\beta,\gamma})\Omega^h_{\alpha,\beta+\gamma}\cdot(\mathbb{1}_\alpha\otimes\Omega^h_{\beta,\gamma})\\
&=(\sum_{\substack{\pi_1(a)=\alpha\\ \pi_1(b)=\beta\\ \pi_1(c)=\gamma}}\xi_{a,b,c}e_a\otimes e_b\otimes e_c)(\sum_{\substack{\pi_1(\widetilde{a})=\alpha\\ \pi_1(\widetilde{b})=\beta\\ \pi_1(\widetilde{a})=\gamma}} q^{-2p\kappa(\widetilde{a},\widetilde{b}+\widetilde{c})h}e_{\widetilde{a}}\otimes e_{\widetilde{b}}\otimes e_{\widetilde{c}})(\sum_{\substack{\pi_1(b')=\beta\\ \pi_1(c')=\gamma}} q^{-2p\kappa(b',c')h}\mathbb{1}_\alpha\otimes e_{b'}\otimes e_{c'})\\
&=\sum_{\substack{\pi_1(a)=\alpha\\ \pi_1(b)=\beta\\ \pi_1(c)=\gamma}}\xi_{a,b,c}q^{-2ph(\kappa(a,b+c)+\kappa(b,c))}e_a\otimes e_b\otimes e_c\ ,
\end{aligned}\tag{A.13}$$

and the RHS is

$$
\begin{aligned}
&(\Delta^{\sigma}_{\alpha,\beta}\otimes \mathrm{id}_\gamma)\Omega^h_{\alpha+\beta,\gamma}\cdot(\Omega^h_{\alpha,\beta}\otimes \mathbb{1}_\gamma)\cdot(\gamma^{-1}_{h,\alpha}\otimes\gamma^{-1}_{h,\beta}\otimes\gamma^{-1}_{h,\gamma})\Phi^{\sigma}_{\alpha,\beta,\gamma}\\
&=\Big(\sum_{\substack{\pi_1(a)=\alpha\\ \pi_1(b)=\beta\\ \pi_1(c)=\gamma}} q^{-2p\kappa(a+b,c)h}e_a\otimes e_b\otimes e_c\Big)\Big(\sum_{\substack{\pi_1(a')=\alpha\\ \pi_1(b')=\beta}} q^{-2p\kappa(a',b')h}e_{a'}\otimes e_{b'}\otimes \mathbb{1}_\gamma\Big)\Big(\sum_{\substack{\pi_1(\widetilde{a})=\alpha\\ \pi_1(\widetilde{b})=\beta\\ \pi_1(\widetilde{c})=\gamma}} \xi_{\widetilde{a},\widetilde{b},\widetilde{c}}e_{\widetilde{a}}\otimes e_{\widetilde{b}}\otimes e_{\widetilde{c}}\Big)\\
&=\sum_{\substack{\pi_1(a)=\alpha\\ \pi_1(b)=\beta\\ \pi_1(c)=\gamma}} \xi_{a,b,c}q^{-2ph(\kappa(a+b,c)+\kappa(a,b))}e_a\otimes e_b\otimes e_c\ .
\end{aligned}
\tag{A.14}
$$

Note that $\kappa(a,b+c)+\kappa(b,c)=\kappa(a+b,c)+\kappa(a,b)$, hence the LHS and RHS coincide.

A.3. **R-Matrices computation.** We compute the RHS of (5.76) step by step:

- $\varphi^{-1}_{\underline{\mathfrak{U}}_\alpha,\underline{\mathfrak{U}}_\beta}\big(\mathbf{1}_0\underline{\otimes}(\mathbb{1}_\alpha\otimes\mathbb{1}_\beta)\big)=$

$$
\sum_{a,b} q^{\frac{\sigma'(b)\sigma(\alpha)}{2}}\Big((\mathbf{1}_{-\kappa(a,b)}\underline{\otimes}e_a)\otimes_A(\mathbf{1}_0\underline{\otimes}e_b)\Big);
\tag{A.15}
$$

- $\widetilde{C}_{F_{\sigma'}(\overline{\mathfrak{U}}_\alpha),F_{\sigma'}(\overline{\mathfrak{U}}_\beta)}\circ\varphi^{-1}_{\underline{\mathfrak{U}}_\alpha,\underline{\mathfrak{U}}_\beta}\big(\mathbf{1}_0\underline{\otimes}(\mathbb{1}_\alpha\otimes\mathbb{1}_\beta)\big)=$

$$
\begin{aligned}
&\sum_{n=0}^{p-1}\sum_{a,b} q^{\frac{\sigma'(b)\sigma(\alpha)}{2}}\frac{(q-q^{-1})^n}{[n]!}q^{\frac{1}{2}n(n-1)}q^{\frac{H\otimes H}{2}}F^n\otimes E^n\Big((\mathbf{1}_0\underline{\otimes}e_b)\otimes_A(\mathbf{1}_{-\kappa(a,b)}\underline{\otimes}e_a)\Big)\\
&=\sum_{a,b,n}\frac{(q-q^{-1})^n}{[n]!}q^{\frac{n(n-1)+\sigma'(b)\sigma(\alpha)}{2}}q^{\frac{H\otimes H}{2}}\Big((\mathbf{1}_{\kappa^n_-(b)}\underline{\otimes}F^ne_b)\otimes_A(\mathbf{1}_{\kappa^n_+(a)-\kappa(a,b)}\underline{\otimes}E^ne_a)\Big)\\
&=\sum_{a,b,n}\frac{(q-q^{-1})^n}{[n]!}q^{\frac{n(n-1)+\sigma'(b)\sigma(\alpha)}{2}}q^{\frac{\left(\sigma'(a+b)-\sigma'(b)+2n\right)\left(\sigma'(b)-2n\right)}{2}}(\mathbf{1}_{\kappa^n_-(b)}\underline{\otimes}F^ne_b)\otimes_A(\mathbf{1}_{\kappa^n_+(a)-\kappa(a,b)}\underline{\otimes}E^ne_a)\ ,
\end{aligned}
\tag{A.16}
$$

  where $\kappa^n_+(a)=\frac{\sigma'(a)+2n-\sigma'(a+2n)}{2p}$, $\kappa^n_-(b)=\frac{\sigma'(b)-2n-\sigma'(b-2n)}{2p}$;

- $(\omega^{\pi(\alpha)}_{\overline{\mathfrak{U}}_\beta}\otimes_A\mathrm{id}_{F_{\sigma'}(\overline{\mathfrak{U}}_\alpha)})\circ\widetilde{C}_{F_{\sigma'}(\overline{\mathfrak{U}}_\alpha),F_{\sigma'}(\overline{\mathfrak{U}}_\beta)}\circ\varphi^{-1}_{\underline{\mathfrak{U}}_\alpha,\underline{\mathfrak{U}}_\beta}\big(\mathbf{1}_0\underline{\otimes}(\mathbb{1}_\alpha\otimes\mathbb{1}_\beta)\big)=$

$$
\begin{aligned}
&\sum_{a,b,n}\frac{(q-q^{-1})^n}{[n]!}q^{\frac{n(n-1)+\sigma'(b)\sigma(\alpha)}{2}}q^{\frac{\left(\sigma'(a+b)-\sigma'(b)+2n\right)\left(\sigma'(b)-2n\right)}{2}}q^{-2p\kappa^n_-(b)\alpha}\\
&\qquad(\mathbf{1}_{\kappa^n_-(b)}\underline{\otimes}F^ne_b)\otimes_A(\mathbf{1}_{\kappa^n_+(a)-\kappa(a,b)}\underline{\otimes}E^ne_a)\ ;
\end{aligned}
\tag{A.17}
$$

- $\varphi_{\pi(\alpha)\overline{\mathfrak{U}}_\beta,\overline{\mathfrak{U}}_\alpha}\circ(\omega^{\pi(\alpha)}_{\overline{\mathfrak{U}}_\beta}\otimes_A \mathrm{id}_{F_{\sigma'}(\overline{\mathfrak{U}}_\alpha)})\circ\widetilde{C}_{F_{\sigma'}(\overline{\mathfrak{U}}_\alpha),F_{\sigma'}(\overline{\mathfrak{U}}_\beta)}\circ\varphi^{-1}_{\overline{\mathfrak{U}}_\alpha,\overline{\mathfrak{U}}_\beta}\big(\mathbf{1}_0\underline{\otimes}(\mathbb{1}_\alpha\otimes\mathbb{1}_\beta)\big)=$

$$
\begin{aligned}
&\sum_{a,b,n}\frac{(q-q^{-1})^n}{[n]!}q^{\frac{n(n-1)+\sigma'(b)\sigma(\alpha)}{2}}q^{\frac{\left(\sigma'(a+b)-\sigma'(b)+2n\right)\left(\sigma'(b)-2n\right)}{2}}q^{-2p\kappa^n_-(b)\alpha}\\
&\qquad q^{-p(b-2n)(\kappa^n_+(a)-\kappa(a,b))}q^{-\frac{\sigma'(a+2n)\sigma(\beta)}{2}}\mathbf{1}_0\underline{\otimes}(F^ne_b\otimes E^ne_a)\\
=&\sum_{a,b,n}\frac{(q-q^{-1})^n}{[n]!}q^{\frac{n(n-1)+\sigma'(b)\sigma(\alpha)}{2}}q^{\frac{\left(\sigma'(a+b)-\sigma'(b)+2n\right)\left(\sigma'(b)-2n\right)}{2}}q^{\sigma(\alpha)(\sigma'(b-2n)+2n-\sigma'(b))}\\
&\qquad q^{\frac{(2n-b)(2n-\sigma'(a+2n)-\sigma'(b)+\sigma'(a+b))}{2}}q^{-\frac{\sigma'(a+2n)\sigma(\beta)}{2}}\mathbf{1}_0\underline{\otimes}(F^ne_b\otimes E^ne_a)\\
=&\sum_{a,b,n}\frac{(q-q^{-1})^n}{[n]!}q^{\frac{n(n-1)}{2}}q^{\frac{(2\sigma'(b-2n)+4n-\sigma'(b))\sigma(\alpha)}{2}}q^{\frac{\sigma'(a+2n)(\sigma'(b)-2n-\sigma(\beta))}{2}}\mathbf{1}_0\underline{\otimes}(F^ne_b\otimes E^ne_a)\\
=&\sum_{a,b,n}\frac{(q-q^{-1})^n}{[n]!}q^{\frac{n(n-1)+\zeta^n_a(\zeta^0_b-2n)}{2}}q^{\frac{\sigma(\alpha)\sigma(\beta)}{2}}q^{(\zeta_b^{-n}+n)\sigma(\alpha)}\mathbf{1}_0\underline{\otimes}(F^ne_b\otimes E^ne_a)\ ,
\end{aligned}
\tag{A.18}
$$

Therefore,

$$
R_{\alpha,\beta}=\sum_{a,b,n}\frac{(q-q^{-1})^n}{[n]!}q^{\frac{n(n-1)+\zeta^n_a(\zeta^0_b-2n)}{2}}q^{\frac{\sigma(\alpha)\sigma(\beta)}{2}}q^{(\zeta_b^{-n}+n)\sigma(\alpha)}(E^ne_a\otimes F^ne_b)\ . \tag{A.19}
$$

A.4. **Ribbon elements computation.** We compute the RHS of (5.80) step by step:

$$
\begin{aligned}
\widetilde{\vartheta}^{-1}_{F_{\sigma'}(\overline{\mathfrak{U}}_\alpha)}(\mathbf{1}_0\underline{\otimes}\mathbb{1}_\alpha)&=K^{p-1}\sum_{n=0}^{p-1}\frac{(q-q^{-1})^n}{[n]!}q^{\frac{n(n-1)}{2}}S(F^n)q^{-\frac{H^2}{2}}E^n\Big(\sum_{\substack{a\in\mathbb{C}/2p\mathbb{Z}\\ \pi_1(a)=\alpha}}\mathbf{1}_0\underline{\otimes}e_a\Big)\\
&=\sum_{a,n}\frac{(q-q^{-1})^n}{[n]!}q^{\frac{n(n-1)}{2}}K^{p-1}S(F^n)q^{-\frac{H^2}{2}}(\mathbf{1}_{\kappa^n_+(a)}\underline{\otimes}E^ne_a)\\
&=\sum_{a,n}\frac{(q-q^{-1})^n}{[n]!}q^{\frac{n(n-1)}{2}}q^{-2p\kappa^n_+(a)\alpha}q^{-\frac{(\sigma'(a+2n))^2}{2}}K^{p-1}S(F^n)(\mathbf{1}_{\kappa^n_+(a)}\underline{\otimes}E^ne_a)\\
&=\sum_{a,n}\frac{(q^{-1}-q)^n}{[n]!}q^{\frac{n(n-1)}{2}}q^{-2p\kappa^n_+(a)\alpha}q^{-\frac{(\sigma'(a+2n))^2}{2}}K^{p-1}(KF)^n(\mathbf{1}_{\kappa^n_+(a)}\underline{\otimes}E^ne_a)\\
&=\sum_{a,n}\frac{(q^{-1}-q)^n}{[n]!}q^{\frac{3n(n-1)}{2}}q^{-2p\kappa^n_+(a)\alpha}q^{-\frac{(\sigma'(a+2n))^2}{2}}K^{p-1+n}F^n(\mathbf{1}_{\kappa^n_+(a)}\underline{\otimes}E^ne_a)\\
&=\sum_{a,n}\frac{(q^{-1}-q)^n}{[n]!}q^{\frac{3n(n-1)}{2}}q^{-2p\kappa^n_+(a)\alpha}q^{-\frac{(\sigma'(a+2n))^2}{2}}q^{a(n+p-1)}(\mathbf{1}_0\underline{\otimes}F^nE^ne_a)\\
&=\sum_{a,n}\frac{(q^{-1}-q)^n}{[n]!}q^{\frac{3n(n-1)}{2}}q^{-\frac{(\sigma'(a)+2n)^2}{2}}q^{a(n+p-1)}(\mathbf{1}_0\underline{\otimes}F^nE^ne_a)\\
&=\sum_{a,n}\frac{(q^{-1}-q)^n}{[n]!}q^{\frac{-n(n+3)}{2}}q^{-\frac{(\sigma'(a))^2}{2}}q^{a(p-n-1)}(\mathbf{1}_0\underline{\otimes}F^nE^ne_a)\ .
\end{aligned}
\tag{A.20}
$$

Since $\omega^{\pi(\alpha)}_{\underline{\mathfrak{U}}_\alpha}(\mathbf{1}_0 \underline{\otimes} \mathbb{1}_\alpha) = \mathbf{1}_0 \underline{\otimes} \mathbb{1}_\alpha$, and $F_{\sigma'}(\vartheta^{-1}_{\underline{\mathfrak{U}}_\alpha})(\mathbf{1}_0 \underline{\otimes} \mathbb{1}_\alpha) = \mathbf{1}_0 \underline{\otimes} (v_\alpha \cdot \mathbb{1}_\alpha)$, we then have

$$v_\alpha = \sum_{a,n} \frac{(q^{-1}-q)^n}{[n]!} q^{\frac{-n(n+3)}{2}} q^{-\frac{(\sigma'(a))^2}{2}} q^{a(p-n-1)} F^n E^n e_a \ . \tag{A.21}$$

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